‡‡‡ Chapter 9 was amended by **Local Law 77 of 2023**. This law has an effective date of June 10, 2023.

CHAPTER 9

FIRE PROTECTION SYSTEMS

SECTION BC 901
GENERAL

901.1 Scope. The provisions of this chapter shall specify where fire protection systems are required and shall apply to the design, installation and operation of fire protection systems.

901.1.1 Referenced standards. Where this code makes reference to the nationally recognized standards NFPA 13, NFPA 13D, NFPA 13R, NFPA 14, NFPA 20, NFPA 72, or NFPA 92, such standard shall be as modified for New York City in accordance with Appendix Q of this code.

901.2 Fire protection systems. Fire protection systems shall be installed, repaired, operated and maintained in accordance with this code and the *New York City Fire Code*. Any fire protection system for which an exception or reduction to the provisions of this code has been granted shall be considered to be a required system.

Exception: Any fire protection system or portion thereof not required by this code shall be permitted to be installed provided that such system meets the requirements of this code.

901.2.1 Fire protection systems within a building. Fire protection systems shall be dedicated to one building only.

Exception: Upon review and approval by the commissioner and the Fire Department, multiple buildings may be served by one fire protection system.

901.3 Modifications. No person shall remove or modify any fire protection system installed or maintained under the provisions of this code or the *New York City Fire Code* without approval by the commissioner.

901.4 Threads. Threads provided for Fire Department connections to sprinkler systems, standpipes, yard hydrants or any other fire hose connection shall be compatible with the connections used by the Fire Department.

901.5 Acceptance tests. Fire protection systems shall be tested in accordance with the requirements of this code and the *New York City Fire Code*. When required by this chapter, the tests shall be conducted in the presence of the department or an approved agency. Tests required by this code, the *New York City Fire Code* and the standards listed in this code shall be conducted at the expense of the owner or the owner's representative. It shall be unlawful to occupy portions of a structure until the required fire protection systems within that portion of the structure have been tested and approved.

901.6 Supervisory service. Where required, fire protection systems shall be monitored by a central supervising station in accordance with NFPA 72.

901.6.1 Automatic sprinkler systems. Automatic sprinkler systems shall be monitored by a central supervising station.

Exceptions:

1. A central supervising station is not required for automatic sprinkler systems protecting one- and two-family dwellings.
2. Limited area sprinkler systems serving no more than 6 sprinkler heads.

901.6.2 Fire alarm systems. Fire alarm systems required by the provisions of Section 907.2 of this code and the *New York City Fire Code* shall be monitored by a central supervising station in accordance with Section 907.6.5.

Exceptions:

1. Single- and multiple-station smoke alarms and carbon monoxide alarms required by Section 907.2.11.

2. Smoke detectors in Group I-3 occupancies.
3. Supervisory service is not required for automatic sprinkler systems in one- and two-family dwellings.

901.6.3 Group H. Manual fire alarm, automatic fire-extinguishing and emergency alarm systems in Group H occupancies shall be monitored by an approved supervising station.

Exception: When approved by the Fire Department, on-site monitoring at a constantly attended location shall be permitted provided that notifications to the Fire Department will be equal to those provided by a central supervising station.

901.7 Fire areas. Where buildings, or portions thereof, are divided into fire areas so as not to exceed the limits established for requiring a fire protection system in accordance with this chapter, such fire areas shall be separated by fire barriers constructed in accordance with Section 707 or horizontal assemblies constructed in accordance with Section 711, or both, having a fire-resistance rating of not less than that determined in accordance with Section 707.3.10.

901.8 Construction documents. Construction documents shall comply with Chapter 1 of Title 28 of the *Administrative Code*, Section 107 and other applicable provisions of this code and its referenced standards.

901.9 Special provisions for prior code buildings. The provisions of this chapter shall apply to alterations and changes of use or occupancy to prior code buildings in accordance with Sections 901.9.1 through 901.9.6.

901.9.1 Additions, alterations or repairs. Additions, alterations, renovations or repairs to existing systems shall conform to that required for new systems without requiring the existing system to comply with all of the requirements of this code, except as otherwise required in Sections 901.9.2 through 901.9.6. Additions, alterations or repairs shall not cause an existing installation to become unsafe, hazardous or overloaded.

901.9.1.1 Minor additions, alterations, renovations and repairs. Minor additions, alterations, renovations and repairs to existing systems shall meet the provisions for new construction, unless such work is done in the same manner and arrangement as was in the existing system, is not hazardous and is approved.

901.9.2 Additional requirements based on change of occupancy or use. Fire protection systems governed by this chapter shall be provided:

1. To the entire building as if the building were hereafter erected, where a change is made in the main use or dominant occupancy of such building.
2. Throughout a space, where a change is made in the occupancy group classification or usage of the space.

901.9.3 Additional requirements for enlargements. Fire protection systems shall be provided in enlarged portions of a building and where this chapter would require such systems in new construction for a space or building.

Exception: Section 901.9.3 shall not require sprinklers to be installed in enlarged portions of unsprinklered buildings to be occupied exclusively as one- or two-family dwellings. This exception shall not apply where sprinklers are otherwise required by the provisions of Sections 901.9.2 or 901.9.4.

901.9.4 Additional requirements based on value of alterations. Fire protection systems shall be provided to buildings and spaces in accordance with the provisions of Sections 901.9.4.1 through 901.9.4.3.

901.9.4.1 Alterations requiring fire protection systems throughout a building. If the value of alterations to the building equals or exceeds 60 percent of the value of the existing building, or, in the case of a building containing 4 or more dwelling units, 50 percent of the value of the existing building, the entire building shall be made to comply with the fire protection requirements of this chapter as if it were hereafter erected.

901.9.4.2 Alterations requiring fire protection systems in the space being altered. If the value of alterations of a space is between 30 percent and 60 percent of the value of the existing building, or, in the case of a building containing 4 or more dwelling units, if the value of alterations of a space is between 30 percent and 50 percent of the value of the existing building, those portions of the building being altered shall be made to comply with the fire protection requirements of this chapter.

901.9.4.3 Additional requirements for buildings containing 4 or more dwelling units. For buildings containing 4 or more dwelling units, if the value of alterations to an existing space classified in Occupancy Group R-1 or R-2 exceeds 50 percent of the value of the space, such space shall be made to comply with the fire protection requirements of this chapter.

901.9.5 Additional provisions. In buildings or spaces not otherwise required to provide fire protection systems in accordance with this chapter, fire protection systems shall be provided for the types of alterations described in Sections 901.9.5.1 through 901.9.5.4.

901.9.5.1 Additional requirements for providing smoke and carbon monoxide alarms in dwelling units during alterations involving removal of existing interior finishes. Smoke and/or carbon monoxide alarms complying with the location, interconnection and power source requirements of this chapter shall be provided throughout a dwelling unit when alteration work results in the removal of existing and/or installation of new interior wall or ceiling finishes permitting the installation of concealed wiring for all the required alarms throughout the dwelling unit.

901.9.5.2 Additional requirements for providing standpipes in newly constructed stair shafts. Where an alteration includes the addition or replacement of an entire exit stair shaft that is a required means of egress, the entire shaft shall be equipped with a standpipe in accordance with Section 905.

901.9.5.3 Additional requirements for enlargements of buildings with existing standpipe systems. Where the alteration involves the addition of stories to a building with an existing standpipe system, and one or more stair shafts are not currently equipped with standpipes, standpipes shall be provided to all stair shafts in accordance with this chapter.

Exception: Additional standpipes are not required where:

1. The alteration involves the addition of only one story;
2. Existing standpipes in existing stair shafts are extended in accordance with this chapter;
3. Standpipe hose connections are provided in compliance with Section 905.4, Item 6; and
4. The demand on the standpipe system, including any additional demand, with respect to flow and pressure does not exceed the capacity of the existing approved system.

901.9.5.4 Additional requirements for enlargements of buildings with no existing standpipe systems. Where the alteration involves the addition of stories to a building with no existing standpipe system, standpipes shall be provided to all stair shafts in accordance with this chapter.

Exception: Standpipes are not required where:

1. the alteration involves the addition of only one story;
2. the completed building does not exceed 7 stories; and
3. the completed building does not exceed 85 feet (25.9 m) in height.

901.9.6 Seismic supports. The determination as to whether seismic requirements apply to an alteration shall be made in accordance with the *1968 Building Code* and interpretations by the department relating to such determinations. Any applicable seismic loads and requirements shall be permitted to be determined in accordance with Chapter 16 of this code or the *1968 Building Code* and Reference Standard RS 9-6 of such code.

SECTION BC 902
DEFINITIONS

902.1 Definitions. The following terms are defined in Chapter 2:

ALARM NOTIFICATION APPLIANCE.

ALARM SIGNAL.

ALARM VERIFICATION FEATURE.

ANIMAL SERVICE FACILITY.

ANNUNCIATOR.

AUDIBLE ALARM NOTIFICATION APPLIANCE.

AUTOMATIC.

AUTOMATIC FIRE-EXTINGUISHING SYSTEM.

AUTOMATIC SMOKE DETECTION SYSTEM.

AUTOMATIC SPRINKLER SYSTEM.

AUTOMATIC WATER MIST SYSTEM.

AVERAGE AMBIENT SOUND LEVEL.

CARBON DIOXIDE EXTINGUISHING SYSTEMS.

CARBON MONOXIDE ALARM.

CARBON MONOXIDE DETECTOR.

CARBON MONOXIDE-PRODUCING EQUIPMENT.

CEILING LIMIT.

CLEAN AGENT.

COMMERCIAL COOKING SYSTEM.

COMMERCIAL MOTOR VEHICLE.

CONSTANTLY ATTENDED LOCATION.

DELUGE SPRINKLER SYSTEM.

DETECTOR, HEAT.

DRY-CHEMICAL EXTINGUISHING AGENT.

ELECTRICAL CIRCUIT PROTECTIVE SYSTEM.

EMERGENCY ALARM SYSTEM.

EMERGENCY VOICE/ALARM COMMUNICATIONS.

FIRE ALARM BOX, MANUAL.

FIRE ALARM CONTROL UNIT.

FIRE ALARM SIGNAL.

FIRE ALARM SYSTEM.

FIRE AREA.

FIRE COMMAND CENTER.

FIRE DETECTOR, AUTOMATIC.

FIRE PROTECTION SYSTEM.

FIRE PUMP.

Fire pump, automatic standpipe.

Fire pump, foam.

Fire pump, limited service.

Fire pump, special service.

Fire pump, sprinkler booster pump.

Fire pump, water mist system.

FIRE SAFETY FUNCTIONS.

FOAM-EXTINGUISHING SYSTEM.

HALOGENATED EXTINGUISHING SYSTEM.

INITIATING DEVICE.

LIMITED AREA SPRINKLER SYSTEM.

LISTED.

MANUAL FIRE ALARM BOX.

MULTIPLE-STATION ALARM DEVICE.

MULTIPLE-STATION SMOKE ALARM.

***‡NATURAL GAS ALARM.**

**Section 902.1 was added by Local Law 157 of 2016. This law has an effective date of May 1, 2025.*

NOTIFICATION ZONE.

POST-FIRE SMOKE PURGE SYSTEM.

PRESIGNAL SYSTEM.

RECORD DRAWINGS.

SINGLE-STATION SMOKE ALARM.

SMOKE ALARM.

SMOKE DETECTOR.

SMOKEPROOF ENCLOSURE.

STANDPIPE SYSTEM.

STANDPIPE, TYPES OF.

Automatic dry.

Automatic wet.

Manual dry.

Manual wet.

Semiautomatic dry.

STANDPIPE SYSTEM, CLASSES OF.

Class I system.

Class II system.

Class III system.

SUPERVISING STATION.

Supervising station, central.

Supervising station, proprietary.

Supervising station, remote.

SUPERVISORY SERVICE.

SUPERVISORY SIGNAL.

SUPERVISORY SIGNAL-INITIATING DEVICE.

TIRES, BULK STORAGE OF.

TROUBLE SIGNAL.

VALUE (OF ALTERATIONS TO DETERMINE REQUIRED FIRE PROTECTION).

VALUE (OF EXISTING BUILDING OR SPACE).

VISIBLE ALARM NOTIFICATION APPLIANCE.

WET CHEMICAL EXTINGUISHING AGENT.

WIRELESS PROTECTION SYSTEM.

ZONE.

ZONE, NOTIFICATION.

SECTION BC 903
AUTOMATIC SPRINKLER SYSTEMS

903.1 General. Automatic sprinkler systems shall comply with this section. Installation of automatic sprinkler systems shall comply with the special inspection requirements of Chapter 17.

903.1.1 Alternative protection. Where permitted by the *New York City Fire Code*, the Fire Department may approve the installation of alternative automatic fire-extinguishing systems complying with this code and the *New York City Fire Code* in lieu of automatic sprinkler protection.

903.1.2 Construction documents. Construction documents for automatic sprinkler systems shall contain plans that include the following data and information:

1. The location and size of water supplies and the location, number, and type of sprinkler heads to be used, with approximate location and size of all feed mains, valves and other essential features of the system. For hydraulically calculated systems, hydraulic data substantiating pipe sizes shown shall be submitted and hydraulic reference points and areas must be indicated on the plan. If any other methods are utilized to size the sprinkler system and its components, as allowed by NFPA 13, supporting documentation shall be submitted to the department.
2. A diagram showing the proposed sprinkler system in relation to principal construction features of the building, such as its size, walls, columns, and partitions; and such other information as may be necessary for the evaluation of the system.
3. The location, number, and type of any electrical or automatic devices or alarms to be used in the system.
4. In buildings where a new separate fire sprinkler system is required, the available water pressure at the top and bottom floors of each zone shall be shown on the riser diagram.
5. For street pressure-fed systems and fire pumps, a statement from the New York City Department of Environmental Protection, giving the minimum water pressure in the main serving the building.

903.2 Where required. Approved automatic sprinkler systems in new buildings and structures shall be provided in the locations described in Sections 903.2.1 through 903.2.13.

Exceptions:

1. Sprinklers shall not be required in electrical equipment rooms where all of the following conditions are met:

1.1 The room is dedicated to electrical equipment only.

1.2 Only dry-type electrical equipment is used.

1.3 Equipment is installed in a 2-hour fire-rated enclosure including protection for penetrations.

1.4 No combustible storage is permitted to be stored in the room.

2. Sprinklers shall not be permitted in elevator machine rooms and elevator machinery spaces.
3. Sprinklers shall not be required in rooms and spaces protected by an alternative fire suppression system in accordance with the *New York City Fire Code* and Section 904 of this code.

903.2.1 Group A. An automatic sprinkler system shall be provided throughout buildings and portions thereof used as Group A occupancies as provided in this section. For Group A-1, A-2, A-3 and A-4 occupancies, the automatic sprinkler system shall be provided throughout the story where the fire area containing the Group A-1, A-2, A-3 or A-4 occupancy is located, and throughout all stories from the Group A occupancy to, and including, the levels of exit discharge serving the Group A occupancy. For Group A-5 occupancies, the automatic sprinkler system shall be provided in the spaces indicated in Section 903.2.1.5. In all Group A occupancies providing live entertainment, dressing rooms and property rooms used in conjunction with such assembly occupancy shall be provided with an automatic sprinkler system. Stages shall comply with Section 410.7.

903.2.1.1 Group A-1. An automatic sprinkler system shall be provided for fire areas containing Group A-1 occupancies and intervening floors of the building where one of the following conditions exists:

1. The fire area exceeds 12,000 square feet (1114.8 m^2).
2. The fire area has an occupant load of 300 or more.
3. The fire area is located on a floor other than a level of exit discharge serving such occupancies.
4. The fire area contains a multitheater complex.

903.2.1.2 Group A-2. An automatic sprinkler system shall be provided for Group A-2 occupancies where any one of the following conditions exists:

1. The fire area exceeds 5,000 square feet (464.5 m^2).
2. The fire area has an occupant load of 300 or more.
3. The aggregate occupant load of all fire areas occupied by Group A, located on any given floor other than the level of exit discharge, is 300 or more.
4. The A-2 occupancy is used as a cabaret.

903.2.1.3 Group A-3. An automatic sprinkler system shall be provided for fire areas containing Group A-3 occupancies and intervening floors of the building where one of the following conditions exists:

1. The fire area exceeds 12,000 square feet (1114.8 m^2).
2. The fire area has an occupant load of 300 or more.
3. The aggregate occupant load of all fire areas occupied by Group A, located on any given floor other than the level of exit discharge, is 300 or more.

    Exception: Areas used exclusively as participant sports areas where the main floor area is located at the same level as the level of exit discharge of the main entrance and exit.

903.2.1.4 Group A-4. An automatic sprinkler system shall be provided for fire areas containing Group A-4 occupancies and intervening floors of the building where one of the following conditions exists:

1. The fire area exceeds 12,000 square feet (1114.8 m^2).
2. The fire area has an occupant load of 300 or more.
3. The aggregate occupant load of all fire areas occupied by Group A, located on any given floor other than the level of exit discharge, is 300 or more.

    Exception: Areas used exclusively as participant sports areas where the main floor area is located at the same level as the level of exit discharge of the main entrance and exit.

903.2.1.5 Group A-5. An automatic sprinkler system shall be provided in all enclosed areas of the structure, including but not limited to the concession concourse, concession stands, retail areas, press boxes and other accessory occupancies, in excess of 1,000 square feet (92.9 m^2).

903.2.1.6 Assembly occupancies on roofs. Where an occupied roof has an assembly occupancy with an occupant load exceeding 100 for Group A-2 and 300 for other Group A occupancies, all floors between the occupied roof and the level of exit discharge shall be equipped with an automatic sprinkler system in accordance with Section 903.3.1.1 or 903.3.1.2.

Exception: Open parking garages of Type I or Type II construction.

903.2.1.7 Multiple fire areas. An automatic sprinkler system shall be provided where multiple fire areas of Group A-1, A-2, A-3 or A-4 occupancies share exit or exit access components and the combined occupant load of these‡ fire areas is 300 or more.

903.2.2 Group B. An automatic sprinkler system shall be installed for Group B occupancies as provided in Sections 903.2.2.1 and 903.2.2.2.

903.2.2.1 Ambulatory health care facilities. An automatic sprinkler system shall be installed throughout all fire areas containing a Group B ambulatory health care facility occupancy. In buildings where ambulatory care is provided on levels other than the level of exit discharge, an automatic sprinkler system shall be installed throughout the entire floor where such care is provided as well as all floors below, and all floors between the level of ambulatory care and the nearest level of exit discharge, including the level of exit discharge.

903.2.2.2 Animal service facilities. An automatic sprinkler system shall be provided for animal service facilities.

Exceptions:

1. Animal service facilities that provide 24 hour in-person supervision of animals sheltered therein and are equipped with smoke alarms.
2. Animal service facilities that were in operation on or before December 31, 2016, and are equipped with an automatic smoke detection system.

903.2.3 Group E. An automatic sprinkler system shall be provided for Group E occupancies as follows:

1. Throughout all Group E fire areas greater than 12,000 square feet (1114.8 m^2) in area.
2. Throughout every portion of educational buildings below the level of exit discharge.

   Exception: An automatic sprinkler system is not required in any fire area or area below the level of exit discharge where every classroom throughout the building has not fewer than one exterior exit door at ground level without intervening corridors, passageways, interior exit stairways or ramps or exit passageways.

903.2.4 Group F. An automatic sprinkler system shall be provided throughout all buildings containing a Group F occupancy where any one of the following conditions exists:

1. Where a Group F-1 fire area exceeds 12,000 square feet (1114.8 m^2);
2. Where a Group F-1 fire area is located more than three stories above grade plane;
3. Where the combined area of all Group F-1 fire areas on all floors, including any mezzanines, exceeds 24,000 square feet (2229.7 m^2);
4. Where required by Section 280 of the *New York State Labor Law* for “factory buildings” defined in Section 2 of such law; or
5. Where a Group F-1 occupancy used for the manufacture of upholstered furniture or mattresses exceeds 2,500 square feet (232.3 m^2).

903.2.4.1 Woodworking operations. An automatic sprinkler system shall be provided throughout all Group F-1 occupancy fire areas that contain woodworking operations in excess of 2,500 square feet (232.3 m^2) in area that generate finely divided combustible waste or use finely divided combustible materials.

903.2.4.2 Repair garages. An automatic sprinkler system shall be provided throughout all buildings used as repair garages in accordance with Section 406, as follows:

1. Buildings two or more stories in height, including basements, with a fire area containing a repair garage exceeding 10,000 square feet (929 m^2).
2. One-story buildings with a fire area containing a repair garage exceeding 12,000 square feet (1114.8 m^2).
3. A Group F-1 fire area used for the repair of commercial trucks or buses where the fire area exceeds 5,000 square feet (464.5 m^2).
4. Buildings with a repair garage servicing vehicles parked in the basement.

903.2.4.3 Group F-1 fire areas. An automatic sprinkler system shall be provided throughout any Group F-1 occupancy fire area where any one of the following conditions exists:

1. The fire area exceeds 7,500 square feet (696.8 m^2).
2. The fire area of any size is located more than three stories above grade.

903.2.5 Group H. Automatic sprinkler systems shall be provided in high-hazard occupancies as required in the *New York City Fire Code* and Sections 903.2.5.1 through 903.2.5.3 of this code.

903.2.5.1 General. An automatic sprinkler system shall be installed in Group H occupancies. An automatic sprinkler system shall be installed throughout buildings with a main use or dominant occupancy of Group H.

903.2.5.2 Group H-5 occupancies. An automatic sprinkler system shall be installed throughout buildings containing Group H-5 occupancies. The design of the sprinkler system shall be not less than that required by this code for the occupancy hazard classifications in accordance with Table 903.2.5.2. Where the design area of the sprinkler system consists of a corridor protected by one row of sprinklers, the maximum number of sprinklers required to be calculated is 13.

TABLE 903.2.5.2
GROUP H-5 SPRINKLER DESIGN CRITERIA

LOCATION	OCCUPANCY HAZARD CLASSIFICATION
Fabrication areas	Ordinary Hazard Group 2
Service corridors	Ordinary Hazard Group 2
Storage rooms without dispensing	Ordinary Hazard Group 2
Storage rooms with dispensing	Extra Hazard Group 2
Corridors	Ordinary Hazard Group 2

903.2.5.3 Pyroxylin plastics. An automatic sprinkler system shall be provided in buildings, or portions thereof, where cellulose nitrate film or pyroxylin plastics are manufactured, stored or handled in quantities exceeding 100 pounds (45.4 kg).

903.2.6 Group I. An automatic sprinkler system shall be provided in Group I occupancies. An automatic sprinkler system shall be installed throughout buildings with a main use or dominant occupancy of Group I.

Exceptions:

1. An automatic sprinkler system installed in accordance with Section 903.3.1.2 or 903.3.1.3 shall be allowed in Group I-1 facilities if located in an I-1 occupancy building or a residential building, provided such building is six stories or less in height.
2. An automatic sprinkler system is not required where Group I-4 day care facilities are at the level of exit discharge and where every room where care is provided has not fewer than one exterior exit door.
3. In buildings where Group I-4 day care is provided on levels other than the level of exit discharge, an automatic sprinkler system in accordance with Section 903.3.1.1 shall be installed on the entire floor where care is provided, all floors between the level of care and the level of exit discharge, and all floors below the level of exit discharge other than areas classified as an open parking garage.

903.2.7 Group M. An automatic sprinkler system shall be provided throughout buildings containing a Group M occupancy where one of the following conditions exists:

1. A Group M fire area exceeds 12,000 square feet (1114.8 m^2).
2. A Group M fire area is located more than three stories above grade plane.
3. The combined area of all Group M fire areas on all floors, including any mezzanines, exceeds 24,000 square feet (2229.7 m^2).
4. A Group M occupancy used for the display and sale of upholstered furniture or mattresses exceeds 5,000 square feet (464.5 m^2).

903.2.7.1 High-piled storage. An automatic sprinkler system shall be provided in accordance with the *New York City Fire Code* in all buildings of Group M where storage of merchandise is in high-piled or rack storage arrays.

‡‡‡ **903.2.7.2 Group M fire areas.** An automatic sprinkler system shall be provided throughout any Group M occupancy fire area where any one of the following conditions exists:

1. The fire area exceeds 7,500 square feet (696.8 m^2).
2. The fire area of any size contains an unenclosed stair or escalator connecting two or more floors.

903.2.8 Group R. An automatic sprinkler system shall be installed in Group R fire areas. An automatic sprinkler system shall be installed throughout buildings with a main use or dominant occupancy of Group R.

Exception: An automatic sprinkler system shall not be required in detached one- and two-family dwellings and multiple single-family dwellings (townhouses), provided that such structures are not more than three stories above grade plane in height and have separate means of egress.

903.2.9 Group S-1. An automatic sprinkler system shall be provided throughout all buildings containing a Group S-1 occupancy where any one of the following conditions exists:

1. A Group S-1 fire area exceeds 12,000 square feet (1114.8 m^2).
2. The building is greater than 1,000 square feet (92.9 m^2) in area and the main use or dominant occupancy is Group S-1.
3. A Group S-1 fire area is located more than three stories above grade plane.
4. The combined area of all Group S-1 fire areas on all floors, including any mezzanines, exceeds 24,000 square feet (2229.7 m^2).
5. A Group S-1 fire area used for the storage of commercial motor vehicles where the fire area exceeds 5,000 square feet (464.5 m^2).
6. A Group S-1 occupancy used for the storage of upholstered furniture or mattresses exceeds 2,500 square feet (232.3 m^2).

903.2.9.1 Group S-1 fire areas. An automatic sprinkler system shall be provided throughout any Group S-1 occupancy fire area where the fire area exceeds 500 square feet (46.5 m^2).

903.2.9.2 Bulk storage of tires. Buildings and structures where the area for the storage of tires exceeds 500 square feet (46.5 m^2) or 7,500 cubic feet (212.3 m^3) shall be equipped throughout with an automatic sprinkler system in accordance with Section 903.3.1.1.

903.2.9.3 High-piled storage. An automatic sprinkler system shall be provided in accordance with the *New York City Fire Code* in all buildings or portions thereof in Group S-1 occupancies where the storage of merchandise is in high-piled or rack storage arrays.

903.2.10 Group S-2. An automatic sprinkler system shall be installed throughout buildings greater than 5,000 square feet (464.5 m^2) in area where the main use or dominant occupancy is Group S-2.

903.2.10.1 Commercial parking garages. An automatic sprinkler system shall be provided throughout buildings used for storage of commercial trucks, buses or other commercial motor vehicles where the fire area exceeds 5,000 square feet (464.5 m^2).

903.2.10.2 Group S-2 fire areas. An automatic sprinkler system shall be provided throughout any Group S-2 occupancy fire area greater than 5,000 square feet (464.5 m^2).

903.2.10.3 Parking garages. An automatic sprinkler system shall be provided throughout buildings classified as enclosed parking garages in accordance with Section 406.4 or where an open or enclosed parking garage is located beneath other occupancy groups.

Exception: Parking garages located beneath Group R-3 occupancies.

903.2.10.4 High-piled storage. An automatic sprinkler system shall be provided in all buildings or portions thereof of Group S-2 occupancies in accordance with the *New York City Fire Code*.

903.2.11 Specific building areas and hazards. An automatic sprinkler system shall be installed for building design or hazards in the locations set forth in Sections 903.2.11.1 through 903.2.11.13.

903.2.11.1 Above- or below-grade stories. An automatic sprinkler system shall be installed throughout every above- or below-grade story of all buildings where the floor area exceeds 1,500 square feet (139.4 m^2) and where not fewer than one of the following types of exterior wall openings is provided:

1. Openings below grade that lead directly to ground level by an exterior stairway complying with Section 1011 or an outside ramp complying with Section 1012. Openings shall be located in each 50 linear feet (15 240 mm), or fraction thereof, of exterior walls facing onto a street, public way or frontage space, in the story on at least one side. The required openings shall be distributed such that the lineal distance between adjacent openings does not exceed 50 feet (15 240 mm).
2. Openings entirely above the adjoining ground level totaling not less than 20 square feet (1.86 m^2) in each 50 linear feet (15 240 mm), or fraction thereof, of exterior walls facing onto a street, public way or frontage space, in the story on at least one side. The required openings shall be distributed such that the lineal distance between adjacent openings does not exceed 50 feet (15 240 mm). The bottom of the clear opening shall not exceed 36 inches (914.4 mm) measured from the floor.

903.2.11.1.1 Opening dimensions and access. Such openings shall have a minimum dimension of not less than 30 inches (762 mm). Such openings shall be accessible to the Fire Department from the exterior and shall not be obstructed in a manner that firefighting or rescue cannot be accomplished from the exterior.

903.2.11.1.2 Openings on one side only. Where such openings in a story are provided on only one side and the opposite wall of such story is more than 100 feet (30 480 mm) from such openings, the story shall be equipped throughout with an approved automatic sprinkler system, or openings as specified above shall be provided on not fewer than two sides of the story.

903.2.11.1.3 Below-grade stories. Where any portion of a below-grade story is located more than 75 feet (22 860 mm) from openings required by Section 903.2.11.1, or where walls, partitions, or other obstructions are installed that restrict the application of water from hose streams, the below-grade story shall be equipped throughout with an approved automatic sprinkler system.

903.2.11.2 Other above-grade stories. An automatic sprinkler system shall be installed throughout every above-grade story of buildings below a height of 100 feet (30 480 mm), other than the first story or ground floor, on which access is not provided directly from the outdoors by at least one window or readily identifiable access panel within each 50 feet (15 240 mm) or fraction thereof of horizontal length of every wall that fronts on a street or frontage space required pursuant to Section 501.3.1.

903.2.11.2.1 Opening dimensions and access. Such windows shall be openable from the inside or breakable from both the inside and the outside, and shall have a size when open of at least 24 inches by 36 inches (609.6 mm by 914.4 mm). Such panels shall be openable from both the inside and outside and shall have a height when open of 48 inches (1220 mm) and a width of at least 32 inches (812.8 mm). The sill of the window or panel shall not be higher than 36 inches (914.4 mm) above the inside floor. Where not all of the windows are openable or breakable, the windows intended to satisfy the requirements of this section shall be readily identifiable.

903.2.11.3 Other below-grade stories. An automatic sprinkler system shall be installed throughout every first basement or cellar story below grade of buildings on which access is not provided directly from the outdoors within each 100 feet (30 480 mm) or fraction thereof of horizontal length of every wall that fronts on a street or frontage space required pursuant to Section 501.3.1.

Exceptions:

1. One- and two-family dwellings need not provide direct access.
2. Any building classified in Occupancy Group R-2 not more than three stories in height and with not more than two dwelling units on any story need not provide direct access when such first basement or cellar story is used for dwelling units or for uses accessory to the residential use in the building.
3. Except as provided in Exception 2, above, for Group R-1 and R-2 occupancies, only one direct access from the outdoors to the first basement or cellar story consisting of a stair or door shall be required when such story is used for dwelling units or for uses accessory to the residential use in the building.

903.2.11.3.1 Opening dimensions and access. Such access shall be by stairs, doors, windows or other means that provide an opening 48 inches (1220 mm) high and 32 inches (812.8 mm) wide, the sill of which shall not be higher than 36 inches (914.4 mm) above the inside floor. If an areaway is used to provide below grade access, the minimum horizontal dimension shall be at least one-third the depth of the areaway or 6 feet (1828.8 mm), whichever is less.

903.2.11.4 Signs obstructing openings. Where wall signs are erected to cover doors or windows of existing buildings, access panels shall be provided as necessary to comply with the requirements of Section 903.2.11.

903.2.11.5 Compliance with the *New York State Multiple Dwelling Law*. Nothing in Section 903.2.11 shall be construed so as to supersede any applicable provisions of Section 54 of the *New York State Multiple Dwelling Law* relating to access to cellars or basements in multiple dwellings.

903.2.11.6 Rubbish and linen chutes. An automatic sprinkler system shall be installed at the top of rubbish and linen chutes, in chute access rooms, and in their terminal rooms. Chutes shall have additional sprinkler heads installed at alternate floors and at the lowest intake. Where a rubbish chute extends through a building more than one floor below the lowest intake, or through areas with no openings, such an extension shall have sprinklers installed that are recessed from the drop area of the chute and protected from freezing in accordance with Section 903.3.1.1. Such sprinklers shall be installed at alternate floors, beginning with the second level below the last intake and ending with the floor above the discharge. All chute sprinklers shall be accessible for servicing and shall not obstruct the vertical path for rubbish or linens.

903.2.11.7 Buildings 55 feet (16 764 mm) or more in height. An automatic sprinkler system shall be installed throughout buildings that have one or more stories with an occupant load of 30 or more located 55 feet (16 764 mm) or more above the lowest level of Fire Department vehicle access, measured to the finished floor.

Exceptions:

1. Open parking structures without any other occupancy groups, unless otherwise required.
2. Occupancies in Group F-2, unless required by the New York State Department of Labor.

903.2.11.8 Ducts conveying hazardous exhausts. Where required by the *New York City Mechanical Code*, automatic sprinklers shall be provided in ducts conveying hazardous exhaust or flammable or combustible materials.

Exception: Ducts where the largest cross-sectional diameter of the duct is less than 10 inches (254 mm).

903.2.11.9 Commercial cooking operations. An automatic sprinkler system shall not be installed in a commercial kitchen exhaust hood and duct system. Fire-extinguishing systems shall be installed in commercial cooking systems in accordance with this code and the *New York City Fire Code*.

903.2.11.10 Other buildings, occupancies and areas. In addition to the requirements of Section 903.2, the provisions indicated in Table 903.2.11.10 also require the installation of a suppression system for certain buildings and areas. Suppression systems shall also be required as provided for in other sections of this code, the *New York City Fuel Gas Code*, and the *New York City Mechanical Code*.

‡‡‡ TABLE 903.2.11.10
ADDITIONAL REQUIRED SUPPRESSION SYSTEMS

OCCUPANCY GROUP, SPECIFIED USE, MATERIALS OR EQUIPMENT (in alphabetical order)	CODE SECTION

Aerosol warehouses	FC 2804.4.1
Aircraft hangers	BC 412.4.6, BC 412.4.6.1, BC 412.6.5
Airport traffic control towers	BC 412.3.6
Atriums	BC 404.3
Automated storage; buildings with	FC 3209.2
Children’s play structures	BC 424.3
Chutes; refuse and laundry	BC 713.13
Chute vestibules	BC Appendix Q 22.15.2.2.1
Cold storage buildings: ice plants, food plants and food processing rooms with foam insulation up to 10 inches in thickness	BC 2603.3
Combustible fibers; storage at waterfront structures	FC 3706.6
Combustible fibers, loose; storage of more than 1,000 sq ft of	FC 3704.5
Commercial cooking systems	BC 904.12 FC 904.11
Commercial cooking systems with solid fuel storage	FC 609.5
Commercial cooking system with Type I hood	BC 904.2.1 MC 509.1
Covered mall and open mall buildings	BC 402.5
Dead-end public streets; buildings on	FC 503.3.1
Dip tank rooms	FC 2405.1
Dip tanks	FC 2405.3.4.1
Dry cleaning machines	FC 1208.3
Dry cleaning plants	FC 1208.2
Drying rooms	BC 417.4
Elevator lobbies	BC 3006.1.1
Exhausted enclosures	FC 5003.8.5
Extra-high-rack combustible storage; buildings with	FC 3208.5.1
Flammable and combustible liquid in Group H-2 or H-3 areas	FC 5705.3.7.3
Flammable and combustible liquid storage rooms	FC 5704.5.7.5.1
Flammable and combustible liquid storage warehouses	FC 3404.3.8.4
Flammable finishes	BC 416.5
Fuel-oil tanks and fuel-oil burning equipment; rooms containing	MC 1305.13.3
Furnaces: Class A and B	FC 3006.1
Furnaces: Class C and D	FC 3006.2
Gas rooms	FC 5003.8.4

Glazing in smoke partition	BC 710.2
Group H-2	BC 415.9.1.3
Group H-5, including but not limited to: workstations, gas cabinet, exhausted enclosures, pass-throughs in exit access corridors and exhaust ducts	BC 415.11
Group I-2	BC 407.6
Hardening and tempering tanks	FC 2405.4.4
Hazardous exhaust system ducts	MC 510.8
Hazardous materials; indoor handling or use of	FC 5005.1.8
Hazardous materials; indoor storage of	FC 5004.5
Hazardous Production Material ("HPM") corridors	FC 2703.10.3
Hazardous Production Material ("HPM") exhaust ducts	FC 2703.10.4
Hazardous Production Material ("HPM") facilities	FC 2703.10
Hazardous Production Material ("HPM") gas cabinets	FC 2703.10.2
Hazardous Production Material ("HPM") work station exhaust	FC 2703.10.1.1

continued

TABLE 903.2.11.10—continued
ADDITIONAL REQUIRED SUPPRESSION SYSTEMS

OCCUPANCY GROUP, SPECIFIED USE, MATERIALS OR EQUIPMENT (in alphabetical order)	CODE SECTION
High Pressure Gas Installations; buildings with	FGC G.2.3
Highly toxic and toxic compressed gases; exhausted enclosures for	FC 6004.1.3
Highly toxic and toxic compressed gases; gas cabinets containing	FC 6004.1.2
Highly toxic and toxic compressed gases; gas rooms utilizing	FC 6004.2.2.6
Highly toxic and toxic compressed gases; outdoor storage of	FC 6004.3.3
High-rise buildings	BC 403.3
Incidental uses	BC 509.4.2
Equipment platforms	BC 505.3.2
Kiosks in covered mall buildings	BC 402.6.2
Kiosks, displays, booths, or concession stands; covered	FC 314.5.1
Laboratory units; non-production	BC 427.6.1
Liquefied petroleum gas ("LPG") within buildings accessible to the public; storage of	FC 6109.9
Liquids, Class II and III, below grade storage of	FC 5704.3.5.1
Liquids, Class II and III, below grade storage of, accessory to retail	BC 414.2.5.1

Medical gas; storage of	FC 5306.2.1
Organic coatings; manufacturing of	BC 418.1
Oxidizer, solid and liquid; storage areas	FC 6304.1.4
Plastic light diffusing system	BC 2606.7.4
Pyroxylin plastic; areas with	FC 6504.1.1
Pyroxylin plastic; storage and manufacturing	FC 6504.2
Pyroxylin plastic; storage vaults	FC 6504.1.3
Rack storage	FC 3208.3
Radioactive materials and radiation-producing equipment; uses and occupancies involving	BC 428.3.4
Resin application areas	FC 2409.2.1
Silane gas; exhausted enclosures or gas cabinets for	FC 6404
Small arms ammunition and primers, black powder or smokeless propellant; storage of	FC 5606.7
Solid-piled and shelf storage	FC 3207.2
Smoke-protected assembly seating	BC 1029.6.2.3
Special amusement buildings	BC 411.4
Spray booths and rooms	FC 2404.3.3
Spray booths involving the use of organic peroxide coatings	FC 2408.2.2
Spray finishing in Group A, E, I or R	FC 2404.1
Stages	BC 410.7
Sterilization systems; rooms with	FC 5806.3.2
Storage	FC Table 3206.2 FC 3206.4
Substandard width public streets; buildings on	FC 503.8.2
Textile ceiling finish	BC 803
Textile wall coverings	BC 803
Underground buildings and spaces	BC 405.3
Unlimited area buildings	BC 507

903.2.11.11 Steel-plated and vault-like occupancies. An automatic sprinkler system shall be installed throughout all steel-plated or similarly reinforced or secured vault-like occupancies regardless of area.

903.2.11.12 Refuse collection and disposal areas. An automatic sprinkler system shall be installed throughout all areas used for the storage and sorting of refuse and recyclables.

903.2.11.13 Laundry drying areas. An automatic sprinkler system shall be installed in spaces in which two or more clothes drying machines are installed. Sprinkler heads shall be spaced to cover the areas 5 feet (1524 mm) on all sides of the drying machines.

903.2.12 During construction. Automatic sprinkler systems required during construction, alteration and demolition operations shall be provided in accordance with the *New York City Fire Code* and Chapter 33 of this code.

903.2.13 Type IV construction with cross-laminated timber (CLT) or structural composite lumber (SCL). Automatic sprinkler systems in accordance with NFPA 13 shall be required throughout buildings utilizing Type IV construction with CLT or SCL as follows:

1. In all occupancies where the building is more than three stories above grade plane.
2. In Group B occupancies, where a floor exceeds 28,500 square feet (2647.7 m^2).

903.3 Installation requirements. Automatic sprinkler systems shall be designed and installed in accordance with Sections 903.3.1 through 903.3.8.

903.3.1 Standards. Sprinkler systems shall be designed and installed in accordance with Section 903.3.1.1, 903.3.1.2 or 903.3.1.3.

‡‡‡ **903.3.1.1 NFPA 13 sprinkler systems.** Where the provisions of this code require that a building or portion thereof be equipped throughout with an automatic sprinkler system in accordance with this section, sprinklers shall be installed throughout in accordance with NFPA 13 except as provided in Sections 903.3.1.1.1 and 903.2 of this code.

903.3.1.1.1 Exempt locations protected by other means. When approved by the Fire Department, automatic sprinklers shall not be required in the following:

1. In rooms or areas protected with an approved automatic fire detection system in accordance with Section 907.2 that will respond to visible or invisible particles of combustion, and an alternative automatic fire-extinguishing system in accordance with this code and the *New York City Fire Code*. Sprinklers shall not be omitted from any room merely because it is of fire-resistance-rated construction or contains electrical equipment. This exemption shall not apply to a generator or transformer room unless, in addition to the above requirements, such room is separated from the remainder of the building by walls and floor/ceiling or roof/ceiling assemblies having a fire-resistance rating of not less than 2 hours, and the generator in such room shall not use high pressure flammable gas in excess of 15 psig (103.4 kPa gauge).
2. Machine rooms, machinery spaces, control rooms and control spaces associated with occupant evacuation elevators designed in accordance with Section 3008.

903.3.1.2 NFPA 13R sprinkler systems. Where allowed in buildings of Group R, up to and including six stories in height, automatic sprinkler systems shall be installed throughout in accordance with NFPA 13R.

The number of stories of Group R occupancies constructed in accordance with Sections 510.2 and 510.4 shall be measured from the horizontal assembly creating separate buildings.

903.3.1.2.1 Balconies and decks. Sprinkler protection shall be provided for exterior balconies, decks and ground-floor patios of dwelling units where the building is of Type V construction and automatic sprinkler protection is required for the Group R occupancy. Side wall sprinklers that are used to protect such areas shall be permitted to be located such that their deflectors are within 1 inch (25.4 mm) to 6 inches (152.4 mm) below the structural members, and a maximum distance of 14 inches (355.6 mm) below the deck of the exterior balconies and decks that are constructed of open wood joist construction.

903.3.1.2.2 Open-ended corridors. Sprinkler protection and freeze protection shall be provided in open-ended corridors and associated exterior stairways and ramps as specified in Section 1027.6, Exception 3.

903.3.1.3 NFPA 13D sprinkler systems. Where allowed, automatic sprinkler systems in one- and two-family dwellings and townhouses, up to and including six stories in height, shall be installed throughout in accordance with NFPA 13D.

‡‡‡ **903.3.2 Quick-response and residential sprinklers.** Where automatic sprinkler systems are required by this code, quick-response or residential automatic sprinklers shall be installed in all of the following areas in accordance with Section 903.3.1 and their listings:

1. Throughout all spaces within a smoke compartment containing care recipient sleeping units in Group I-2 in accordance with this code.
2. Throughout all spaces within a smoke compartment containing treatment rooms in ambulatory care facilities.

3. Dwelling units and sleeping units in Group I-1 and R occupancies.
4. Light-hazard occupancies as defined in NFPA 13.

903.3.3 Obstructed locations. Automatic sprinklers shall be installed with due regard to obstructions that will delay activation or obstruct the water distribution pattern. Automatic sprinklers shall be installed in or under covered kiosks, displays, booths, concession stands, or equipment that exceeds 4 feet (1220 mm) in width. Not less than a 3-foot (914.4 mm) clearance shall be maintained between automatic sprinklers and the top of piles of combustible fibers.

Exception: Kitchen equipment under exhaust hoods protected with a fire-extinguishing system in accordance with this code and the *New York City Fire Code*.

903.3.4 Actuation. Automatic sprinkler systems shall be automatically actuated unless specifically provided for in this code.

903.3.5 Water supplies. Water supplies for automatic sprinkler systems shall comply with this section and the standards referenced in Section 903.3.1. The potable water supply shall be protected against back flow in accordance with the requirements of this section, the *New York City Plumbing Code*, and Rules of the New York City Department of Environmental Protection.

903.3.5.1 Domestic services. Where the domestic service provides the water supply for the automatic sprinkler system, the supply shall be in accordance with NFPA 13, 13R and 13D.

903.3.5.1.1 Residential combination services. A single combination water supply shall be permitted in accordance with NFPA 13R.

903.3.5.2 Secondary on-site water supply. A secondary on-site water supply for high-rise buildings shall be provided in accordance with Section 403.3.3.

903.3.6 Hose threads. Fire hose threads and fittings used in connection with automatic sprinkler systems shall be approved and compatible with Fire Department hose threads.

903.3.7 Fire Department connections. The location of Fire Department connections for automatic sprinkler systems shall be installed in accordance with Sections 905 and 912.

903.3.8 Limited area sprinkler systems. Limited area sprinkler systems shall be in accordance with the standards listed in Section 903.3.1 except as provided in Sections 903.3.8.1 through 903.3.8.5.

903.3.8.1 Number of sprinklers. Limited area sprinkler systems shall not exceed six sprinklers in any single fire area.

903.3.8.2 Occupancy hazard classification. Only areas classified by NFPA 13 as Light Hazard or Ordinary Hazard Group 1 shall be permitted to be protected by limited area sprinkler systems.

903.3.8.3 Piping arrangement. Where a limited area sprinkler system is installed in a building with an automatic wet standpipe system, sprinklers shall be supplied by the standpipe system. Where a limited area sprinkler system is installed in a building without an automatic wet standpipe system, water shall be permitted to be supplied by the plumbing system provided that the plumbing system is capable of simultaneously supplying domestic and sprinkler demands.

903.3.8.4 Supervision. Control valves shall not be installed between the water supply and sprinklers unless the valves are of an approved indicating type that are supervised or secured in the open position.

903.3.8.5 Calculations. Hydraulic calculations in accordance with NFPA 13 shall be provided to demonstrate that the available water flow and pressure are adequate to supply all sprinklers installed in any single fire area with discharge densities corresponding to the hazard classification.

903.4 Sprinkler system supervision and alarms. Valves controlling the water supply for automatic sprinkler systems, pumps, tanks, water levels and temperatures, critical air pressures and water-flow switches on all sprinkler systems shall be electrically supervised by the fire alarm system where such fire alarm system is required by Section 907.

Exceptions:

1. Automatic sprinkler systems protecting one- and two-family dwellings.
2. Limited area sprinkler systems in accordance with Section 903.3.8.

3. Automatic sprinkler systems installed in accordance with NFPA 13R where a common supply main is used to supply both domestic water and the automatic sprinkler system, and a separate shutoff valve for the automatic sprinkler system is not provided.

4. Jockey pump control valves that are sealed or locked in the open position.

5. Control valves to commercial kitchen hoods, paint spray booths or dip tanks that are sealed or locked in the open position.

6. Valves controlling the fuel supply to fire pump engines that are sealed or locked in the open position.

7. Trim valves to pressure switches in dry, preaction and deluge sprinkler systems that are sealed or locked in the open position.

903.4.1 Signals and monitoring. Alarm, supervisory and trouble signals shall be distinctly different and shall be automatically transmitted to a central supervising station or when approved by the Fire Department, shall sound an audible signal at a constantly attended location.

Exceptions:

1. Underground key or hub valves in roadway boxes provided by the city or a public utility are not required to be monitored.
2. Backflow prevention device test valves, located in limited area sprinkler system supply piping, shall be locked in the open position. In occupancies required to be equipped with a fire alarm system, the backflow preventer valves shall be electrically supervised by a tamper switch installed in accordance with NFPA 72 and separately annunciated.

903.4.2 Alarms. An approved audible device, located on the exterior of the building in an approved location, shall be connected to each automatic sprinkler system. Such sprinkler water-flow alarm devices shall be activated by water flow equivalent to the flow of a single sprinkler of the smallest orifice size installed in the system. Alarm devices shall be provided on the exterior of the building in an approved location or in a location approved by the Fire Department, except in buildings equipped with a fire alarm system. Where a fire alarm system is installed, actuation of the automatic sprinkler system shall actuate the building fire alarm system.

903.4.3 Floor control valves. Approved supervised indicating control valves shall be provided at the point of connection to the riser on each floor in high-rise buildings.

903.5 Testing and maintenance. Sprinkler systems shall be tested and maintained in accordance with the *New York City Fire Code*.

903.6 Painting of dedicated sprinkler piping and valve handles. Dedicated sprinkler piping shall be painted and such painting certified in accordance with Sections 903.6.1 through 903.6.5 of this code. In addition to painting, sprinkler piping may also be identified by lettered legend in accordance with ANSI A13.1. Where the piping is required to be listed and labeled, such painting shall not obscure such labeling.

Exceptions:

1. Attachments, gauges, valves and operable parts of sprinkler systems other than valve handles.

2. Horizontal branch lines.

3. Where different color coding may be required by Section 3406 of the *New York City Fire Code* for facilities storing, handling, and using flammable and combustible liquids in connection with special operations.

903.6.1 New buildings. Cross connections and risers in new buildings, including buildings constructed pursuant to Section 28-101.4.2 of the *Administrative Code*, shall be painted red and the handles of valves serving dedicated sprinklers shall be painted green prior to the hydrostatic pressure test regardless of whether they will be enclosed at a later point in time.

Exception: Where a standpipe system is used as a combination standpipe and sprinkler system, the sprinkler risers and cross connections that are also used for the standpipe system shall be painted red and the handles of valves serving such combination system shall be painted yellow.

903.6.2 Alterations. Cross connections and risers for independent (stand-alone) existing sprinkler systems that are exposed during alterations, including alterations pursuant to Section 28-101.4.2 of the *Administrative Code*, shall be painted red and the handles of valves serving such existing sprinkler systems shall be painted green. Where the alteration requires a hydrostatic pressure test such painting shall be completed prior to such test.

Exception: Where a standpipe system is used as a combination standpipe and sprinkler system, the sprinkler risers and cross connections that are also used for the standpipe system shall be painted red and the handles of valves serving such combination system shall be painted yellow.

903.6.3 Retroactive requirement for completed buildings. Notwithstanding any other provision of law, all exposed risers and cross connections of completed buildings in existence on March 2, 2010, shall be painted red and all handles of valves serving such sprinkler system shall be painted green.

Exception: Where a standpipe system is used as a combination standpipe and sprinkler system, the sprinkler risers and cross connections that are also used for the standpipe system shall be painted red and the handles of valves serving such combination system shall be painted yellow.

903.6.4 Buildings under construction on March 2, 2010. Notwithstanding any other provision of law, where construction documents were approved and permits issued for the construction of a new building or alteration of an existing building prior to March 2, 2010, and the work is not signed off by the department prior to such date, all exposed cross connections and risers in any such building shall be painted red prior to the hydrostatic pressure test, including cross connections and risers that will be enclosed at a later point in time, and handles of valves serving such sprinkler system shall be painted green.

Exceptions:

1. Where a standpipe system is used as a combination standpipe and sprinkler system, the sprinkler risers and cross connections that are also used for the standpipe system shall be painted red and the handles of valves serving such combination system shall be painted yellow.
2. Cross connections and risers enclosed prior to March 2, 2010, need not be painted.

903.6.5 Certification of completion of system painting. For all buildings where sprinkler and combination sprinkler and standpipe systems are not subject to a special inspection pursuant to Section 1705.2.9 of this code, a licensed master plumber, licensed master fire suppression piping contractor, registered design professional or an individual holding an appropriate certificate of fitness from the Fire Department for the operation and/or maintenance of such system shall certify on forms provided by the department that all required painting has been completed in accordance with Section 903.6. Such certification shall be maintained on the premises and made available for inspection by the department and the Fire Department.

SECTION BC 904
ALTERNATIVE AUTOMATIC
FIRE-EXTINGUISHING SYSTEMS

904.1 General. Automatic fire-extinguishing systems, other than automatic sprinkler systems, shall be designed, installed, inspected, tested and maintained in accordance with the provisions of this section, the *New York City Fire Code*, and the applicable referenced standards.

904.1.1 Construction documents. Construction documents for alternative automatic fire-extinguishing systems shall be approved by the Fire Department and shall contain plans that include at least the following data and information:

1. Commercial kitchen suppression systems:
   - 1.1. Location of all surface, plenum and duct nozzles; surface dimensions and location of all cooking appliances; the location of automatic fuel shutoff and statement as to type (gas or electric); location and distance of the remote control or manual pull station;
   - 1.2. Identification of the grease filters to be used in any kitchen hood; the dimensions of all hoods and all related ducts, including termination of duct at the exterior of the building;

1.3. Identification of the fire suppression piping system; the make and model of the system; the type of extinguishing agent and number and size of agent containers; size, length, and type of all piping that will be used; the number and location of all fusible links or detectors and the temperature setting; any surface, plenum and duct nozzles.

2. For extinguishing agent systems, the plan should also include type and concentration of the extinguishing agent, the method of providing power supply to smoke or heat detectors, fire rating of partitions, location of all audible/visible alarms within and outside the location involved and the details of construction of the room to contain the extinguishing agent. If the area is not sprinklered, the following information is required:

   2.1. The size and location of the reserve supply, and

   2.2 Information as to why it has been determined that water is not effective as an extinguishing agent for the fire hazard in such location.

3. The plans must note whether the proposed system is connected to the building's fire alarm system.

   Exception: For that portion of a fire suppression piping system within an approved preengineered system, a schematic isometric diagram shall be acceptable in lieu of full plans, provided that the location and method of pressure relief must be indicated with areas and volumes to where said relief is taken.

904.2 Where required. Automatic fire-extinguishing systems installed as an alternative to the required automatic sprinkler systems of Section 903 shall be approved by the Fire Department. Automatic fire-extinguishing systems shall not be considered alternatives for the purposes of exceptions or reductions allowed by other requirements of this code.

904.2.1 Hood system suppression. Each required commercial kitchen exhaust hood and duct system required by Chapter 5 of the *New York City Mechanical Code* to have a Type I hood shall be protected with an automatic fire-extinguishing system installed in accordance with this code and the *New York City Fire Code*.

904.3 Installation. Automatic fire-extinguishing systems shall be installed in accordance with this section.

904.3.1 Electrical wiring. Electrical wiring shall be in accordance with the *New York City Electrical Code*.

904.3.2 Actuation. Automatic fire-extinguishing systems shall be automatically actuated and provided with a manual means of actuation in accordance with Section 904.12.1. Where more than one hazard could be simultaneously involved in fire due to their proximity, all hazards shall be protected by a single system designed to protect all hazards that could become involved.

Exception: Multiple systems shall be permitted to be installed if they are designed to operate simultaneously.

904.3.3 System interlocking. Automatic equipment interlocks with fuel shutoffs, ventilation controls, door closers, window shutters, conveyor openings, smoke and heat vents and other features necessary for proper operation of the fire-extinguishing system shall be provided as required by the design and installation standard utilized for the hazard.

904.3.4 Alarms and warning signs. Where alarms are required to indicate the operation of automatic fire-extinguishing systems, distinctive audible and visible alarms and warning signs shall be provided to warn of pending agent discharge. Where exposure to automatic-extinguishing agents poses a hazard to persons and a delay is required to ensure the evacuation of occupants before agent discharge, a separate warning signal shall be provided to alert occupants once agent discharge has begun. Audible signals shall be in accordance with Section 907.5.2.

904.3.5 Monitoring. Where a fire alarm system is installed, automatic fire-extinguishing systems shall be monitored by the fire alarm system in accordance with NFPA 72.

904.4 Inspection and testing. Automatic fire-extinguishing systems shall be inspected and tested in accordance with the provisions of this section and the *New York City Fire Code* prior to acceptance.

904.4.1 Inspection. Prior to conducting final acceptance tests, all of the following items shall be inspected:

1. Hazard specification for consistency with design hazard.

2. Type, location and spacing of automatic- and manual initiating devices.

3. Size, placement and position of nozzles or discharge orifices.

4. Location and identification of audible and visible alarm devices.
5. Identification of devices with proper designations.
6. Operating instructions.

904.4.2 Alarm testing. Notification appliances, connections to fire alarm systems and connections to central supervising stations shall be tested in accordance with this section and Section 907 to verify proper operation.

904.4.2.1 Audible and visible signals. The audibility and visibility of notification appliances signaling agent discharge or system operation, where required, shall be verified.

904.4.3 Monitor testing. Connections to protected premises and supervising station fire alarm systems shall be tested to verify proper identification and retransmission of alarms from automatic fire-extinguishing systems.

904.5 Wet-chemical systems. Wet-chemical extinguishing systems shall be installed, maintained, periodically inspected and tested in accordance with the *New York City Fire Code*.

904.6 Dry-chemical systems. Dry-chemical extinguishing systems shall be installed, maintained, periodically inspected and tested in accordance with *New York City Fire Code*. New dry-chemical extinguishing systems are not permitted for the protection of kitchen equipment.

904.7 Foam systems. Foam-extinguishing systems shall be installed, maintained, periodically inspected and tested in accordance with the *New York City Fire Code*.

904.8 Carbon dioxide systems. Carbon dioxide extinguishing systems shall be installed, maintained, periodically inspected and tested in accordance with the *New York City Fire Code*.

904.9 Halon systems. Halogenated extinguishing systems shall not be permitted. However, existing systems shall be maintained, periodically inspected and tested in accordance with the *New York City Fire Code*.

904.10 Clean-agent systems. Clean-agent fire-extinguishing systems shall be installed, maintained, periodically inspected and tested in accordance with the *New York City Fire Code*.

904.11 Automatic water mist systems. Automatic water mist systems shall be permitted in applications that are consistent with the applicable listing or approvals and shall comply with Sections 904.11.1 through 904.11.3.

904.11.1 Design and installation requirements. Automatic water mist systems shall be designed and installed in accordance with Sections 904.11.1.1 through 904.11.1.4.

904.11.1.1 General. Automatic water mist systems shall be designed and installed in accordance with the manufacturer's instructions and the *New York City Fire Code*.

904.11.1.2 Actuation. Automatic water mist systems shall be automatically actuated.

904.11.1.3 Water supply protection. Connections to a potable water supply shall be protected against backflow in accordance with the *New York City Plumbing Code*.

904.11.1.4 Secondary water supply. Where a secondary water supply is required for an automatic sprinkler system, an automatic water mist system shall be provided with an approved secondary water supply in accordance with the *New York City Fire Code*.

904.11.2 Water mist system supervision and alarms. Supervision and alarms shall be provided as required for automatic sprinkler systems in accordance with Section 903.4.

904.11.2.1 Monitoring. Monitoring shall be provided as required for automatic sprinkler systems in accordance with Section 903.4.1.

904.11.2.2 Alarms. Alarms shall be provided as required for automatic sprinkler systems in accordance with Section 903.4.2.

904.11.2.3 Floor control valves. Floor control valves shall be provided as required for automatic sprinkler systems in accordance with Section 903.4.3.

904.11.3 Testing, operation and maintenance. Automatic water mist systems shall be tested, operated and maintained in accordance with the *New York City Fire Code*.

904.12 Commercial cooking systems. The automatic fire-extinguishing system for commercial cooking systems shall be of a type recognized for protection of commercial cooking equipment and exhaust systems of the type and arrangement protected. Preengineered automatic wet-chemical extinguishing systems shall be approved by the Fire Commissioner, tested in accordance with UL 300, and listed and labeled for the intended application. The protected area shall include the area under the hood and over the cooking equipment, the area above or behind the filters and the opening of the hood into the branch duct. Where a preengineered system is installed and the size of the protected area exceeds that allowed for a single preengineered system, additional preengineered systems arranged for simultaneous operation shall be provided. Other types of automatic fire-extinguishing systems shall be listed and labeled for specific use as protection for commercial cooking operations. The system shall be installed in accordance with this code, its listing and the manufacturer's instructions. Only automatic fire-extinguishing systems of the following types shall be installed in accordance with the *New York City Fire Code*:

1. Foam water sprinkler system or foam water spray systems.
2. Wet-chemical extinguishing systems.

Automatic sprinkler systems, dry-chemical fire-extinguishing systems, and carbon dioxide fire-extinguishing systems shall not be installed to protect commercial cooking equipment and exhaust systems.

904.12.1 Manual system operation. A manual actuation device shall be located at or near a means of egress from the cooking area not less than 10 feet (3048 mm) and not more than 20 feet (6096 mm), from the kitchen exhaust system. The manual actuation device shall be installed not more than 48 inches (1220 mm) or less than 42 inches (1066.8 mm) above the finished floor and shall clearly identify the hazard protected. The manual actuation device shall require a maximum force of 40 pounds (177.9 N) and a maximum movement of 14 inches (355.6 mm) to actuate the fire suppression system.

Exception: Automatic sprinkler systems shall not be required to be equipped with manual actuation means.

904.12.2 System interconnection. The actuation of the fire suppression system shall automatically shut down the fuel or electrical power supply to the cooking equipment. The fuel and electrical supply reset shall be manual.

904.13 Domestic cooking systems in Group I-2. In Group I-2 occupancies where cooking facilities are installed in accordance with Section 407.2.6 of this code, the domestic cooking hood provided over the cooktop or range shall be equipped with an automatic fire-extinguishing system of a type recognized for protection of domestic cooking equipment. Preengineered automatic extinguishing systems shall be tested in accordance with UL 300A and listed and labeled for the intended application. The system shall be installed in accordance with this code, its listing and the manufacturer's instructions.

904.13.1 Manual system operation and interconnection. Manual actuation and system interconnection for the hood suppression system shall be installed in accordance with Sections 904.12.1 and 904.12.2, respectively.

904.13.2 Portable fire extinguishers for domestic cooking equipment in Group I-2. A portable fire extinguisher complying with Section 906 shall be installed within a 30-foot (9144 mm) distance of travel from domestic cooking appliances.

SECTION BC 905
STANDPIPE SYSTEMS

905.1 General. Standpipe systems shall be provided in buildings and structures in accordance with this section. Fire hose threads used in connection with standpipe systems shall be approved by the Fire Commissioner. Standpipe systems in buildings used for high-piled combustible storage shall be in accordance with the *New York City Fire Code*. Installation of standpipe systems shall comply with the special inspection requirements of Chapter 17 of this code.

Any space or room that contains equipment of such nature that the use of water would be ineffective in fighting a fire therein, or would be otherwise hazardous, shall have a conspicuous sign on each door opening on such space or room stating the nature of the use and the warning: "IN CASE OF FIRE, USE NO WATER."

905.1.1 Construction documents. Construction documents for standpipe systems shall contain plans that include, at a minimum, the following data and information:

1. The locations and sizes of all risers, cross-connections, hose racks, valves, Department connections, sources of water supply, piping, and other essential features of the system;
2. A floor plan for each group of floors that have typical riser locations and no special features within such group of floor levels, with the indication in title block of such plan indicating clearly the floors to which the arrangement is applicable;
3. A riser diagram showing the essential features of the system, including the risers, cross-connections, valves, Fire Department connections, tanks, pumps, sources of water supply, pipe sizes, capacities, floor heights, zone pressures, and other essential data and features of the system;
4. The available water pressure at the top and bottom floors of each zone, and at each floor where the pressure rating of piping, fittings, couplings, and valves are required to change, shall be shown on the riser diagram; and
5. For street pressure-fed systems and fire pumps, a statement from the New York City Department of Environmental Protection, giving the minimum water pressure in the main serving the building.

905.2 Installation standards. Standpipe systems shall be installed in accordance with this section and NFPA 14.

905.3 Required installations. Standpipe systems shall be installed where required by Sections 905.3.1 through 905.3.9 and in the locations indicated in Sections 905.4, 905.5 and 905.6. Standpipe systems are allowed to be combined with automatic sprinkler systems.

Exception: Standpipe systems are not required in buildings occupied entirely by Group R-3.

905.3.1 Applicability. Class III standpipe systems shall be installed throughout the following buildings:

1. In buildings two stories or more in height with floor area of 10,000 square feet (929 m^2) or greater on any story;
2. In buildings three stories or more in height with floor area of 7,500 square feet (696.8 m^2) or greater on any story;
3. In buildings of any area with a floor level having an occupant load of 30 or more that is located 55 feet (16 764 mm) or more above the lowest level of Fire Department vehicle access; and
4. In buildings of any area, constructed in accordance with Section 403, with occupied floors located 75 feet (22 60 mm) or more above the lowest level of Fire Department vehicle access.

Exceptions: The following exceptions are allowed as an alternative to the requirement of a Class III standpipe system:

1. Class I standpipes are allowed in buildings equipped throughout with an automatic sprinkler system in accordance with Section 903.3.1.1 or 903.3.1.2 provided that the following additional requirements are met:

1.1. A locked noncombustible storage cabinet shall be provided on the main entrance floor. One additional locked storage cabinet shall be provided on every tenth floor above the main entrance floor, such that no occupant on any floor would have to travel more than five floors to reach a cabinet in a location within 15 feet (4572 mm) of each standpipe riser. Where one standpipe riser is installed in the building, such cabinet shall contain at least one fog nozzle, one $1^1/_2$-inch (38.1 mm) spanner wrench, one $2^1/_2$-inch (63.5 mm) spanner wrench, one $2^1/_2$-inch (63.5 mm) by $1^1/_2$ inch (38.1 mm) nonswivel reducing coupling, and 125 feet (38.1 m) of inch (38.1 mm) hose.

1.1.1. The cabinet shall be kept locked, openable by a Fire Department city-wide standard key.

1.1.2. The cabinet shall be labeled, "FOR FIRE DEPARTMENT ONLY."

1.1.3. A metal sign stating clearly where the storage cabinet is located shall be placed in each stair enclosure on the main entrance floor and on each floor where the cabinet is located.

1.2. Hose valves are capped with a hose valve cap fastened to the valve with a chain.

2. Class I manual standpipes are allowed in open parking garages where the highest floor is located not more than 150 feet (45 720 mm) above the lowest level of Fire Department vehicle access.
3. Class I manual dry standpipes are allowed in open parking garages that are subject to freezing temperatures, provided that the hose connections are located as required for Class III standpipes in accordance with Section 905.6.
4. Class I standpipes are allowed in below-grade stories equipped throughout with an automatic sprinkler system.
5. Standpipe outlets may be omitted in portions of first floors or basements that are completely separated from the entrance hall or enclosed stairways leading to the upper floors, provided that portable fire extinguishers are installed, subject to the approval of the Fire Commissioner.

905.3.2 Group A. Class I automatic wet standpipes shall be provided in nonsprinklered Group A buildings having an occupant load exceeding 1,000 persons.

Exceptions:

1. Open-air-seating spaces without enclosed spaces.
2. Class I automatic dry and semiautomatic dry standpipes or manual wet standpipes are allowed in buildings where the highest floor surface used for human occupancy is not more than 75 feet (22 860 mm) above the lowest level of Fire Department vehicle access. Dry standpipes are permitted only where subject to freezing temperatures.

905.3.3 Covered mall buildings. Covered mall buildings and buildings connected thereto shall be equipped throughout with a Class I automatic wet standpipe system, except as permitted by Sections 905.3.3.1 through 905.3.3.3.

905.3.3.1 Covered-mall building height. Covered-mall buildings where the highest occupied floor level is located not more than 30 feet (9144 mm) above the lowest level of the Fire Department vehicle access shall be permitted to be provided with Class I hose connections connected to the mall sprinkler system in accordance with Section 8.17.5.2 of NFPA 13 regarding hose connections for Fire Department use and under the following conditions:

1. Any individual outlet shall be capable of delivering water flow at a rate of 250 gallons per minute (946.4 L/m) while concurrently supplying the mall sprinkler demand; and
2. Each of the two most hydraulically remote outlets shall be capable of concurrently delivering 250 gallons per minute (946.4 L/m) at a pressure of 100 pounds per square inch (689.4 kPa) with no mall sprinkler demand, based on a supply pressure at the system Fire Department connection of not more than 175 pounds per square inch (1206.6 kPa). Adequacy of the water supply available to the Fire Department shall be demonstrated through hydraulic calculations provided by the registered design professional.

905.3.3.2 Location of hose connections. Hose connections shall be provided in accordance with Section 905.4 and at each of the following locations:

1. Within the mall at the entrance to each exit passageway or exit.
2. At each floor-level landing within interior exit stairways opening directly on the mall.
3. At exterior public entrances to the mall of a covered mall building.
4. At public entrances at the perimeter line of an open mall building.
5. At other locations as necessary so that the distance to reach all portions of a tenant space does not exceed 150 feet (45 720 mm) from a hose connection.

905.3.3.3 Installation standard. Except as provided in Sections 905.3.3.1 and 905.3.3.2, the Class I hose connections and Fire Department connections shall be designed in conformance with NFPA 14.

905.3.4 Stages. Stages greater than 1,000 square feet in area (92.9 m^2), or any assembly occupancy having an occupant load of 1,000 or more with a stage of any size, shall be equipped with a Class III wet standpipe system with a hose station on each side of the stage. Such hose stations shall comply with Section 905.6 and shall have sufficient hose to provide protection for the entire stage area from either standpipe location. The exceptions allowed under Section 905.3.1 shall not apply to these hose stations.

905.3.5 Underground buildings. Underground buildings shall be equipped throughout with a Class I automatic wet or manual wet standpipe system.

905.3.6 Helistops and heliports. Buildings with a helistop or heliport that are equipped with a standpipe shall extend the standpipe to the roof level on which the helistop or heliport is located in accordance with the *New York City Fire Code*. All portions of the helistop and heliport area shall be within 150 feet (45 720 mm) of a $2^1/_2$-inch (63.5 mm) outlet on a Class I or III standpipe, in accordance with the *New York City Fire Code*.

905.3.7 Marinas and boatyards. Standpipes in marine terminals, piers, wharves, marinas and boatyards shall comply with the *New York City Fire Code* or other requirements of the Fire Department.

905.3.8 Rooftop gardens, landscaped roofs and green roofs. Buildings with a rooftop garden, landscaped roof, green roof, or roof used for any purpose other than weather protection or maintenance that are equipped with a standpipe system shall extend the standpipe system to the roof level on which the rooftop garden, landscaped roof, green roof, or roof used for any purpose other than weather protection or maintenance is located.

905.3.9 High-piled stock or rack storage. Where exit passageways are required in accordance with Chapter 10 of this code, a standpipe system shall be provided in accordance with the *New York City Fire Code* in all buildings containing high-piled stock or rack storage.

905.4 Location of Class I standpipe hose connections. Class I standpipe hose connections shall be provided in all of the following locations:

1. In every required stairway, a hose connection shall be provided for each floor level above or below grade. Hose connections shall be readily accessible and located at the riser on each floor-level landing and on the entrance floor above the standpipe riser control valve. Nonrequired enclosed stairways that do not serve as a means of egress are not required to have hose connections. Stairways without hose connections shall have a sign on the door to the stairway stating, “No standpipe connections in stairway.”
2. On each side of the wall adjacent to the exit opening of a horizontal exit.

   Exception: Where floor areas adjacent to a horizontal exit are reachable from exit stairway hose connections by a 30-foot (9144 mm) hose stream from a nozzle attached to 100 feet (30 480 mm) of hose, a hose connection shall not be required at the horizontal exit.
3. In every exit passageway at the entrance from the exit passageway to the other areas of a building.

   Exception: Where floor areas adjacent to an exit passageway are reachable from exit stairway hose connections by a 30-foot (9144 mm) hose stream from a nozzle attached to 100 feet (30 480 mm) of hose, a hose connection shall not be required at the entrance from the exit passageway to other areas of the building.
4. In covered mall buildings, in accordance with Section 905.3.3.2.
5. Where the roof has a slope of less than four units vertical in 12 units horizontal (33.3-percent slope), each standpipe shall be provided with a hose connection located either on the roof or at the highest landing of stairways with stair access to the roof. An additional hose connection shall be provided at the top of the most hydraulically remote standpipe for testing purposes. This additional hose connection shall not be required when a roof manifold is installed in accordance with NFPA 14.
6. Where the most remote portion of a floor or story is more than 150 feet (45 720 mm) from a hose connection, additional hose connections shall be provided in approved locations. For the purposes of this section, a penthouse with an occupant load greater than 10 shall be considered a story.
7. In any staircase where the change in elevation between floor landings is more than 25 feet (7620 mm), in addition to the hose connections required by Item 1, a hose connection shall be installed at the first intermediate stair landing below the higher floor level.

905.4.1 Protection. Risers and laterals of Class I standpipe systems not located within an enclosed stairway or pressurized enclosure shall be protected by a degree of fire resistance equal to that required for vertical enclosures in the building in which they are located. No standpipe riser shall be placed in any shaft containing a gas or fuel pipeline.

Exception: In buildings equipped throughout with an approved automatic sprinkler system, laterals that are not located within an enclosed stairway or pressurized enclosure are not required to be enclosed in fire resistance rated construction.

905.4.2 Interconnection. In buildings where more than one standpipe is provided, the standpipes shall be interconnected in accordance with NFPA 14.

905.5 Location of Class II standpipe hose connections. Class II standpipe hose connections shall be prohibited.

905.5.1 Reserved.

905.6 Location of Class III standpipe hose connections. Class III standpipe systems shall have $2^1/_2$-inch (63.5 mm) hose connections located as required for Class I standpipes in Section 905.4. At each hose connection, there shall be a hose station. The hose stations shall be equipped with a minimum of 125 feet (38 100 mm) but not more than a maximum of 150 feet (45 720 mm) of $1^1/_2$-inch (38.1 mm) fire hose connected to an adjustable fog nozzle. The hose shall be attached to the $2^1/_2$-inch (63.5 mm) hose connection by a $2^1/_2$-inch (63.5 mm) by $1^1/_2$-inch (38.1 mm) non-swivel reducing coupling. The hose shall be mounted on a rack and may be located in a cabinet, in accordance with Section 905.7. A pressure restricting device shall be installed when required by NFPA 14. Such pressure restricting device and reducing coupling shall be installed in such a manner that they are readily removable by the Fire Department.

905.6.1 Protection. Risers and laterals of Class III standpipe systems shall be protected as required for Class I systems in accordance with Section 905.4.1.

905.6.2 Interconnection. In buildings where more than one standpipe is provided, the standpipes shall be interconnected in accordance with NFPA 14.

905.7 Cabinets. Cabinets containing firefighting equipment such as standpipes, fire hoses, fire extinguishers or Fire Department valves shall not be blocked from use or obscured from view.

905.7.1 Cabinet equipment identification. Cabinets shall be identified in an approved manner by a permanently attached sign with white letters not less than 2 inches (50.8 mm) high and a red background color, indicating the equipment contained therein.

Exceptions:

1. Doors not large enough to accommodate a written sign with 2-inch lettering shall be marked with a permanently attached pictogram indicating the equipment contained therein, in addition to corresponding smaller white lettering on a red background adjacent to such pictogram.
2. Doors that have either an approved visual identification clear glass panel or a complete glass door panel are not required to be marked.

905.7.2 Locking cabinet doors. Cabinets shall be unlocked.

Exceptions:

1. Visual identification panels of glass or other approved transparent frangible material that is easily broken and allows access.
2. Approved locking arrangements.
3. Locking of cabinets shall be permitted in Group I-3.

905.8 Dry standpipes. Dry standpipes shall not be installed.

Exception: Where subject to freezing and in accordance with NFPA 14.

905.9 Valve supervision. Valves controlling water supplies shall be supervised in the open position so that a change in the normal position of the valve will generate a supervisory signal at the central supervising station required by Section 903.4. Where a fire alarm system is provided, a signal shall also be transmitted to the fire alarm system.

Exceptions:

1. Valves to underground key or hub valves in roadway boxes provided by the city or a public utility do not require supervision.

2. Valves locked in the normal position and inspected as provided in this code in buildings not equipped with a fire alarm system.

905.10 During construction. Standpipe systems required during construction, alteration and demolition operations shall be provided in accordance with Section 3303.8.

905.11 Painting of dedicated standpipes. Dedicated standpipes and the handles of valves serving standpipes shall be painted and such painting certified in accordance with Sections 905.11.1 through 905.11.6. In addition to painting, standpipe piping may also be identified by lettered legend in accordance with ANSI A13.1. Where the piping is required to be listed and labeled such painting shall not obscure such labeling.

Exceptions:

1. Attachments, gauges, valves and operable parts of standpipes other than valve handles.
2. Where different color coding may be required by Section 3406 of the *New York City Fire Code* for facilities storing, handling, and using flammable and combustible liquids in connection with special operations.

905.11.1 New buildings. All portions of a standpipe system and the handles of valves serving the standpipe system in new buildings, including buildings constructed pursuant to Section 28-101.4.2 of the *Administrative Code*, shall be painted red prior to the hydrostatic pressure test whether or not they are intended to be enclosed at the end of construction.

905.11.2 Alterations. Existing handles of valves serving existing standpipe systems and existing unpainted standpipe risers that are exposed during alterations, including alterations pursuant to Section 28-101.4.2 of the *Administrative Code* shall be painted red. Where the alteration requires a hydrostatic pressure test such painting shall be completed prior to such test.

905.11.3 Retroactive requirement for completed buildings. Notwithstanding any other provision of law, all portions of exposed standpipe systems and handles of valves serving the standpipe system of completed buildings in existence March 2, 2010 shall be painted red.

905.11.4 Buildings under construction on March 2, 2010. Notwithstanding any other provision of law, where construction documents were approved and permits issued for the construction of a new building or alteration of an existing building prior to March 2, 2010, and the work is not signed off by the department prior to such date, all exposed portions of the standpipe system and handles of valves serving the standpipe system shall be painted red prior to the hydrostatic pressure test, including portions that will be enclosed at a later point in time.

Exception: Portions of the standpipe system enclosed prior to March 2, 2010, need not be painted.

905.11.5 Combination standpipe and sprinkler systems. Where a standpipe system that is used as a combination standpipe and sprinkler system is required to be painted pursuant to Section 905.11.1, 905.11.2, 905.11.3 or 905.11.4, the sprinkler risers and cross connections that are also used for the standpipe system shall be painted red, and the handles of valves serving such combination standpipe and sprinkler system shall be painted yellow.

905.11.6 Certification of completion of system painting. For all buildings where standpipe and combination sprinkler and standpipe systems are not subject to a special inspection pursuant to Section 1705.30 of this code, a licensed master plumber, licensed master fire suppression piping contractor, registered design professional or an individual holding an appropriate certificate of fitness from the Fire Department for the operation and/or maintenance of such system shall certify on forms provided by the department that all required painting has been completed in accordance with Section 905.11. Such certification shall be maintained on the premises and made available for inspection by the department and the Fire Department.

SECTION BC 906
PORTABLE FIRE EXTINGUISHERS

906.1 General. Portable fire extinguishers shall be provided in occupancies and locations as required by the *New York City Fire Code*.

SECTION BC 907
FIRE ALARM AND DETECTION SYSTEMS

907.1 General. This section covers the application, installation, performance and maintenance of fire alarm systems and their components. Systems shall be designed and installed in accordance with NFPA 72 and the *New York City Electrical Code*. Systems shall be tested and maintained in accordance with this code and the *New York City Fire Code*.

907.1.1 Construction documents. Construction documents for fire alarm systems shall be submitted for review and approval to the Fire Department prior to system installation. Construction documents shall include, but not be limited to, all of the following:

1. A floor plan that indicates the use of all rooms.
2. Locations of alarm-initiating devices.
3. Locations of alarm notification appliances, including candela ratings for visible alarm notification appliances.
4. Location of fire command center, fire alarm control units, transponders and notification power supplies.
5. Location of remote annunciators.
6. Location of all primary, secondary and local sources of power.
7. Fire alarm riser diagram showing all fire alarm devices indicated on the floor plans. Quantities of devices on the floor plans shall match the quantities indicated on the riser diagram. Riser diagram shall include class and style of circuits and levels of survivability. The riser diagram shall show the interface of fire safety control functions.
8. Copies of any variances granted by the department or the Fire Department.
9. Legend of all fire alarm symbols and abbreviations used.
10. Design criteria for fire alarm audibility in various occupancies indicated on plans.
11. Fire alarm sequence of operation for the fire alarm system in a matrix format.
12. Classification of the central supervising station.

907.1.1.1 Amended construction documents. Amendments to approved construction documents shall be submitted and approved by the Fire Department before the final inspection of the work or equipment is completed, and such amendments when approved shall be deemed part of the original construction documents. The Fire Department may allow minor revisions of construction documents to be made and submitted to the Fire Department after the completion of work but prior to sign-off of the work in accordance with rules promulgated by the Fire Department regarding such amendments.

907.1.2 Equipment. Systems and their components shall be listed for the purpose for which they are installed. The fire alarm control unit shall meet the requirements of the Fire Department.

907.2 Where required. An approved fire alarm system installed in accordance with the provisions of this code and NFPA 72 shall be provided in accordance with Sections 907.2.1 through 907.2.23 and provide occupant notification in accordance with Section 907.5, unless other requirements are provided by another section of this code.

A minimum of one manual fire alarm box shall be provided in an approved location to initiate a fire alarm signal for fire alarm systems employing automatic fire detectors or waterflow detection devices. Where other sections of this code allow elimination of fire alarm boxes due to sprinklers, a single fire alarm box shall be installed.

Exceptions:

1. The manual fire alarm box is not required for fire alarm systems dedicated to elevator recall control and supervisory service.
2. The manual fire alarm box is not required for Group R-2 occupancies unless required by the Fire Department to provide a means for fire watch personnel to initiate an alarm during a sprinkler system impairment event. Where provided, the manual fire alarm box shall not be located in an area that is accessible to the public.

In all occupancies where an automatic fire alarm system is required by this section, selective coverage smoke detectors shall be located as follows, unless partial or total coverage automatic detection is specified.

1. In each mechanical equipment, electrical, transformer, telephone equipment or similar room, in elevator machine rooms, and in elevator lobbies.
2. In air distribution systems in accordance with Section 606 of the *New York City Mechanical Code*.

907.2.1 Group A. A manual and automatic fire alarm system that activates the occupant notification system in accordance with Section 907.5 shall be installed in Group A occupancies where the occupant load due to the assembly occupancy is 300 or more. Group A occupancies not separated from one another in accordance with Section 707.3.10 shall be considered as a single occupancy for the purposes of applying this section. Portions of Group E occupancies occupied for assembly purposes shall be provided with a fire alarm system as required for the Group E occupancy.

Exceptions:

1. Manual fire alarm boxes are not required where the building is equipped throughout with an automatic sprinkler system and the notification appliances will activate upon sprinkler water flow. This exception shall not apply to Group A-2 occupancies used as a cabaret.
2. A Group A-2 occupancy used as a cabaret with an occupant load of 75 or more, including associated stages, dressing rooms, and property rooms, shall be equipped with a manual fire alarm system. Such a Group A-2 occupancy with an occupant load of 300 or more shall also be equipped with an automatic fire alarm system.
3. Group A occupancies with a stage in accordance with Section 410, and having an occupant load of 75 or more, shall be provided with a voice/alarm communication system as required by Sections 410.9 and 907.2.1.1.

907.2.1.1 System initiation in Group A occupancies. Activation of the fire alarm in Group A-1 occupancies with an occupant load of 300 or more, and in all other Group A occupancies with an occupant load of 1,000 or more, shall initiate a presignal system in accordance with NFPA 72 at a constantly attended location from which the Fire Department shall be notified and live voice evacuation instructions shall be initiated using an emergency voice/alarm communications system in accordance with Section 907.5.2.2.

907.2.1.2 Emergency voice/alarm communication captions. Stadiums, arenas and grandstands required to caption audible public announcements shall be in accordance with Section 907.5.2.2.4.

907.2.2 Group B. A manual and automatic fire alarm system shall be installed in Group B occupancies where one of the following conditions exists:

1. The combined Group B occupant load of all floors is 500 or more.
2. The Group B occupant load is more than 100 persons above or below the lowest level of exit discharge.
3. The fire area contains an ambulatory care facility, which shall comply with Section 907.2.2.1.

Where such Group B occupancies meeting any one of the above conditions are not protected by an automatic sprinkler system, a partial coverage automatic smoke detection system shall be installed in accordance with NFPA 72.

907.2.2.1 Ambulatory care facilities. Fire areas containing ambulatory care facilities shall be provided with an electronically supervised automatic partial-coverage smoke detection system installed within the ambulatory care facility and in public use areas outside of tenant spaces and along the path of egress, including public corridors and elevator lobbies.

Exception: Buildings equipped throughout with an automatic sprinkler system in accordance with Section 903.3.1.1, provided the occupant notification appliances will activate throughout the notification zones upon sprinkler waterflow.

907.2.2.2 Large-area buildings. Group B occupancies having a total gross area exceeding 100,000 square feet (9290.3 m^2) located in buildings where the highest occupied floor is 75 feet (22 860 mm) or less above the lowest level of Fire Department vehicle access shall be provided with automatic smoke detection connected to an automatic fire alarm system in accordance with Section 907.2.13.1 and emergency voice/alarm communication system in accordance with Section 907.5.2.2 that initiates a total evacuation signal.

907.2.3 Group E. A manual and automatic fire alarm system shall be installed in Group E occupancies. An emergency voice/alarm communication system meeting the requirements of Section 907.5.2.2 and installed in accordance with Section 907.6 shall be installed in Group E occupancies. Automatic sprinkler systems, smoke detectors and/or other initiating devices, shall be installed in accordance with other sections of this code. When such systems, initiating devices, and/or detectors are installed, they shall be connected to the building fire alarm system.

Exception: Emergency voice/alarm communication systems meeting the requirements of Section 907.5.2.2 and installed in accordance with Section 907.6 shall not be required in Group E occupancies with occupant loads of 100 or less, provided that activation of the manual and automatic fire alarm system initiates an approved occupant notification signal in accordance with Section 907.5.

907.2.4 Group F. A manual and automatic fire alarm system shall be installed in Group F occupancies that are two or more stories in height and have an occupant load of 100 or more, or when 25 persons or more are above or below the lowest level of exit discharge.

907.2.5 Group H. A manual and automatic fire alarm system shall be installed in Group H-5 occupancies and in occupancies used for the manufacture of organic coatings. In addition to the automatic fire alarm system requirements of Section 907.2, an automatic smoke detection system shall be installed for highly toxic gases, organic peroxides and oxidizers in accordance with the *New York City Fire Code*.

Exception: Where exempt under the *New York City Fire Code*.

907.2.6 Group I. A manual and automatic fire alarm system that activates the occupant notification system in accordance with Section 907.5 shall be installed in Group I occupancies. An automatic smoke detection system that activates the occupant notification system shall be provided in accordance with Sections 907.2.6.1. 907.2.6.2 and 907.2.6.3.3.

907.2.6.1 Group I-1. An automatic smoke detection system shall be installed in corridors, waiting areas open to corridors and habitable spaces other than sleeping units and kitchens. The system shall be activated in accordance with Section 907.5.

Exception: Smoke detection is not required for exterior balconies.

907.2.6.1.1 Smoke detectors within dwelling and sleeping units. Smoke detectors and notification appliances shall be installed in dwelling units and such notification appliances shall activate only in the unit in which the alarm originates. Such detectors and appliances shall be annunciated by dwelling unit at a constantly attended location from which the fire alarm system is capable of being manually activated. Smoke detectors are required in the following areas:

1. Sleeping areas;
2. Every room in the path of the means of egress from the sleeping area to the door leading from the dwelling unit; and
3. Each story within the unit, including below-grade stories. For dwelling units with split levels and without an intervening door between the adjacent levels, a smoke detector installed on the upper level shall suffice for the adjacent lower level.

907.2.6.2 Group I-2. An automatic smoke detection system shall be installed in corridors in nursing homes (both intermediate-care and skilled nursing facilities), corridors in detoxification facilities and spaces permitted to be open to the corridors by Section 407.2. The system shall be activated in accordance with Section 907.5. Hospitals shall be equipped with smoke detection as required in Section 407. A one-way voice communication system activated in accordance with Section 907.5.2.2 shall be provided at the fire command center for use by Fire Department personnel. A two-way voice communication system in accordance with Section 907.2.13.3 shall be provided for Group I-2 buildings.

907.2.6.3 Group I-3. Group I-3 occupancies shall be equipped with a manual and automatic fire alarm system and automatic smoke detection system installed for alerting staff.

907.2.6.3.1 System initiation. Actuation of an automatic fire-extinguishing system, automatic sprinkler system, a manual fire alarm box, a fire detector, or a smoke detector shall initiate an approved alarm signal that automatically notifies staff.

907.2.6.3.2 Manual fire alarm boxes. Manual fire alarm boxes are not required to be located in accordance with Section 907.4.2 where the fire alarm boxes are provided at staff-attended locations having direct supervision over areas where manual fire alarm boxes have been omitted.

907.2.6.3.2.1 Manual fire alarm boxes in detainee areas. Manual fire alarm boxes are allowed to be locked in areas occupied by detainees, provided that staff members are present within the subject area and have keys readily available to operate the manual fire alarm boxes.

907.2.6.3.3 Automatic smoke detection system. An automatic smoke detection system shall be installed throughout resident housing areas, including sleeping units and contiguous day rooms, group activity spaces and other common spaces normally accessible to residents.

Exceptions:

1. Other approved smoke detection arrangements providing equivalent protection, including, but not limited to, placing detectors in exhaust ducts from cells or behind protective guards listed for the purpose, are allowed when necessary to prevent damage or tampering.
2. Sleeping units in Use Conditions 2 and 3 as described in Section 308.
3. Smoke detectors are not required in sleeping units with four or fewer occupants in smoke compartments that are equipped throughout with an approved automatic sprinkler system installed in accordance with Section 903.3.1.1.

907.2.7 Group M. A manual and automatic fire alarm system shall be installed in Group M occupancies where any one of the following conditions exists:

1. Where a Group M fire area exceeds 12,000 square feet (1114.8 m^2);
2. Where a Group M fire area is located more than three stories above grade;
3. Where the combined area of all Group M fire areas on all floors, including mezzanines, exceeds 24,000 square feet (2229.7 m^2); or
4. Where a Group M fire area in a below-grade story exceeds 1,500 square feet (139.4 m^2).

Where such occupancies are not protected by an automatic sprinkler system, a manual fire alarm and partial coverage automatic smoke detection or automatic heat detection system shall be installed in accordance with NFPA 72.

907.2.7.1 Large-area buildings. Group M occupancies having a total gross area exceeding 100,000 square feet (9290.3 m^2) located in buildings where the highest occupied floor is 75 feet (22 860 mm) or less above the lowest level of Fire Department vehicle access and covered mall buildings having a total gross area exceeding 50,000 square feet (4645.2 m^2) shall be provided with automatic smoke detection connected to an automatic fire alarm system in accordance with Section 907.2.13.1 and an emergency voice/alarm communication system in accordance with Section 907.5.2.2 initiating a total evacuation signal.

907.2.8 Group R-1. Fire alarm systems shall be installed in Group R-1 occupancies as required in Sections 907.2.8.1 through 907.2.8.5.

907.2.8.1 Manual fire alarm system. A manual fire alarm system that activates the occupant notification system in accordance with Section 907.5 shall be installed in Group R-1 occupancies.

Exception: A manual fire alarm system is not required in buildings not over two stories in height where all individual sleeping units and contiguous attic and crawl spaces are separated from each other and public or common areas by at least 1-hour fire partitions and each individual dwelling unit has an exit directly to a public way, exit court or yard.

907.2.8.2 Automatic smoke detection system. An automatic smoke detection system that activates the occupant notification system in accordance with Section 907.5 shall be installed in all public corridors serving dwelling units or sleeping units in accordance with Section 907.2.8.3.

Exception: An automatic fire detection system is not required in buildings that do not have public corridors serving sleeping units and each sleeping unit has a means of egress door opening directly to an exterior exit access that leads directly to an exit.

907.2.8.3 Smoke detectors within dwelling units and sleeping units. Smoke detectors and audible notification appliances shall be installed in dwelling units and sleeping units and shall be annunciated by dwelling unit and sleeping unit at a constantly attended location from which the fire alarm system is capable of being manually activated. Smoke detectors are required in the following areas:

1. In sleeping areas.
2. In every room in the path of the means of egress from the sleeping area to the door leading from the dwelling unit and sleeping unit.
3. In each story within the unit, including below-grade stories. For dwelling units and sleeping units with split levels and without an intervening door between the adjacent levels, a smoke alarm installed on the upper level shall suffice for the adjacent lower level.

907.2.8.4 Group R-1 student dormitories. A manual and automatic smoke detection system shall be installed in occupancies for student or school staff dormitory housing in all of the following locations:

1. Common spaces outside of sleeping units.
2. Laundry rooms, mechanical equipment rooms and storage rooms.
3. Interior corridors serving sleeping units.

Exception: An automatic smoke detection system is not required in buildings that do not have interior corridors serving sleeping units or dwelling units and where each sleeping unit or dwelling unit either has a means of egress door opening directly to an exterior exit access that leads directly to an exit or a means of egress door opening directly to an exit.

Required smoke alarms for student or staff dormitory housing shall be interconnected with the fire alarm system in accordance with NFPA 72.

907.2.8.5 Large Group R-1 occupancies. Group R-1 occupancies with a total of more than 50 sleeping rooms above street level located in buildings where the highest occupied floor is 75 feet (22 860 mm) or less above the lowest level of Fire Department vehicle access, or communal sleeping facilities above street level occupied or designed to be occupied by more than 50 lodgers, shall be provided with automatic smoke detection connected to an automatic fire alarm system in accordance with Section 907.2.13.1 and an emergency voice/alarm communication system in accordance with Section 907.5.2.2 that initiates a total evacuation signal.

907.2.9 Group R-2. A fire alarm system without alarm notification appliances and smoke alarms shall be installed in accordance with this section in Group R-2 occupancies, other than student apartments, where such occupancy satisfies any one of the following conditions:

1. Any dwelling unit is located three or more stories above the lowest level of exit discharge, including dwelling units in penthouses of any area;
2. Any dwelling unit is located more than one story below the highest level of exit discharge of exits serving the dwelling unit; or
3. The building contains more than 16 dwelling units.

Actuation of smoke detectors shall not initiate a signal to alarm notification appliances. The activation of any detector required by this section shall initiate a signal at a central station or a constantly attended location. Smoke detectors shall be located as follows:

1. In each mechanical equipment, electrical, transformer, telephone equipment or similar room.
2. In air distribution systems in accordance with Section 606 of the *New York City Mechanical Code*.
3. In elevator machine rooms and in elevator lobbies.

907.2.9.1 Group R-2 student apartments. Where the main use or dominant occupancy of a building is classified as R-2 student apartments, as defined in Section 310.2, fire alarm systems shall be installed in accordance with Section 907.2.8. Where the main use or dominant occupancy of a building is not classified as R-2 student apartments and the building is occupied partially by Group R-2 student apartments, fire alarm systems shall be installed in accordance with Sections 907.2.9.1.1 through 907.2.9.2.

907.2.9.1.1 Manual fire alarm system. A manual fire alarm system shall be installed throughout all public corridors serving student apartments and student-related uses. Student-related uses shall include common spaces such as recreation rooms, lounges, dining rooms, laundry rooms and storage rooms.

Exceptions:

1. A manual fire alarm system is not required in buildings not over two stories in height where all individual dwelling units and contiguous attic and crawl spaces are separated from each other and public or common areas by at least 1-hour fire partitions and each individual dwelling unit has an exit directly to a public way, exit court or yard.
2. A manual fire alarm system is not required in buildings containing fewer than 15 student apartments.

907.2.9.1.2 Automatic fire alarm system. An automatic fire alarm system without alarm notification appliances shall be installed in accordance with this section in Group R-2 student apartments and student-related uses. The activation of any smoke detector required by this section shall initiate a signal at a central station or a constantly attended location. Smoke detectors shall be located as follows:

1. In each mechanical equipment, electrical, transformer, telephone equipment or similar room, in elevator machine rooms, and in elevator lobbies.
2. In air distribution systems in accordance with Section 606 of the *New York City Mechanical Code*.
3. Throughout all public corridors serving student apartments and student-related uses. Student-related uses shall include common spaces such as recreation rooms, lounges, dining rooms, laundry rooms and storage rooms. However, smoke detectors shall not be required in such public corridors in buildings containing fewer than 15 student apartments.

   Exception: An automatic fire alarm system is not required in buildings not over two stories in height where all individual dwelling units and contiguous attic and crawl spaces are separated from each other and public or common areas by at least 1-hour fire barriers and each individual dwelling unit has an exit directly to a public way, exit court or yard.

907.2.9.2 Smoke alarms. Single- and multiple-station smoke alarms shall be installed in accordance with Section 907.2.11.

907.2.10 Group S. A manual and automatic fire alarm system shall be installed in Group S occupancies where any one of the following conditions exists:

1. Group S fire area has an occupant load of 300 occupants or more;
2. The combined occupant load of all Group S fire areas on all floors, including mezzanines, is 300 or more.

907.2.10.1 Large-area buildings. Group S occupancies having a total gross area exceeding 500,000 square feet (46 451.5 m^2) located in buildings, where the highest occupied floor is 75 feet (22 860) or less above the lowest level of Fire Department vehicle access, shall be provided with automatic smoke detection connected to an automatic fire alarm system in accordance with Section 907.2.13.1 and an emergency voice/alarm communication system in accordance with Section 907.5.2.2 that initiates a total evacuation signal.

907.2.11 Single- and multiple-station smoke alarms. Listed single- and multiple-station smoke alarms complying with UL 217 shall be installed in accordance with Sections 907.2.11.1 through 907.2.11.7 and NFPA 72.

907.2.11.1 Smoke alarms in Groups R-2 and R-3. Single- or multiple-station smoke alarms shall be installed and maintained in Groups R-2 and R-3, regardless of occupant load at all of the following locations within all dwelling units:

1. On the ceiling or wall outside of each room used for sleeping purposes within 15 feet (4572 mm) from the door to such room.
2. In each room used for sleeping purposes.
3. In each story within a dwelling unit, including below-grade stories and penthouses of any area, but not including crawl spaces and uninhabitable attics. In dwellings or dwelling units with split levels and without an intervening door between the adjacent levels, a smoke alarm installed on the upper level shall suffice for the adjacent lower level provided that the lower level is less than one full story below the upper level.

907.2.11.1.1 Group R-2 occupancy. Smoke alarms shall be provided with the capability to support visible alarm notification appliances in accordance with ICC/ANSI A117.1.

907.2.11.2 Power source. Required smoke alarms shall receive their primary power from a dedicated branch circuit or the unswitched portion of a branch circuit also used for power or lighting, or both, and shall be equipped with a battery backup. Smoke alarms shall emit a signal when the batteries are low. Wiring shall be permanent and without a disconnecting switch other than as required for over-current protection.

Exception: Smoke alarms with integral strobes shall not require battery backup provided that the smoke alarms are connected to an emergency or standby power electrical source.

907.2.11.3 Installation near cooking appliances. Smoke alarms shall not be installed in the following locations unless this would prevent placement of a smoke alarm in a location required by Section 907.2.11.1:

1. Ionization smoke alarms shall not be installed less than 20 feet (6096 mm) horizontally from a permanently installed cooking appliance.
2. Ionization smoke alarms with an alarm-silencing switch shall not be installed less than 10 feet (3048 mm) horizontally from a permanently installed cooking appliance.
3. Photoelectric smoke alarms shall not be installed less than 6 feet (1828.8 mm) horizontally from a permanently installed cooking appliance.

907.2.11.4 Installation near bathrooms. Smoke alarms shall be installed not less than 3 feet (914.4 mm) horizontally from the door or opening of a bathroom that contains a bathtub or shower unless this would prevent placement of a smoke alarm required by Section 907.2.11.1.

907.2.11.5 Interconnection. Where more than one smoke alarm is required to be installed within an individual dwelling unit or sleeping unit in Group R-2 and R-3 occupancies, the smoke alarms shall be interconnected in such a manner that the activation of one alarm will activate all of the alarms in the individual unit. Physical interconnection of smoke alarms shall not be required where listed wireless alarms are installed and all alarms sound upon activation of one alarm. The alarm shall be clearly audible in all bedrooms over background noise levels with all intervening doors closed.

907.2.11.6 Smoke alarms and smoke detectors in group R occupancies. On and after January 1, 2021, smoke alarms and smoke detectors shall not be installed or replaced in an individual dwelling or sleeping unit, including dwellings or units in prior code buildings, within an area of exclusion determined by a 20 foot (6096 mm) radial distance along a horizontal flow path from a stationary or fixed cooking appliance, unless listed in accordance with the 8th edition of UL 217 for smoke alarms or the 7th edition of UL 268 for smoke detectors.

907.2.11.7 Smoke detection system. Smoke detectors listed in accordance with UL 268 and provided as part of the building fire alarm system shall be an acceptable alternative to single and multiple station smoke alarms and shall comply with the following:

1. The fire alarm system shall comply with all applicable requirements in Section 907 of this code.
2. Activation of a smoke detector in a dwelling unit or sleeping unit shall initiate alarm notification in the dwelling unit or sleeping unit in accordance with Section 907.5.2 of this code.
3. Activation of a smoke detector in a dwelling unit or sleeping unit shall not activate alarm notification appliances outside of the dwelling unit or sleeping unit, provided that a supervisory signal is generated and monitored in accordance with Section 907.6.5 of this code.

907.2.12 Special amusement buildings. An automatic smoke detection system shall be provided in special amusement buildings in accordance with Sections 907.2.12.1 through 907.2.12.3.

Exception: In areas where ambient conditions will cause a smoke detection system to alarm, an approved alternative type of automatic detector shall be installed.

907.2.12.1 Alarm. Actuation of a single smoke detector, automatic sprinkler system or other automatic fire detection system shall initiate a pre-signal system in accordance with NFPA 72 at a constantly attended location from which the Fire Department shall be notified and live voice evacuation instructions shall be initiated using an emergency voice/alarm communications system in accordance with Section 907.5.2.2.

907.2.12.2 System response. The following minimum system actuations and responses shall be required upon approval by the department and the Fire Department. The activation of two or more smoke detectors, a single smoke detector equipped with an alarm verification feature, the automatic sprinkler system or other approved fire detection device shall automatically do all of the following:

1. Cause illumination of the means of egress with light of not less than 1 footcandle (11 lux) at the walking surface level.
2. Stop any conflicting or confusing sounds and visual distractions.
3. Activate an approved directional exit marking that will become apparent in an emergency.

907.2.12.3 Emergency voice/alarm communication system. An emergency voice/alarm communication system, which is also allowed to serve as a public address system, shall be installed in accordance with Section 907.5.2.2 and be audible throughout the entire special amusement building.

907.2.13 High-rise buildings. In addition to the requirements of Sections 907.2.1 through 907.2.12, buildings constructed in accordance with Section 403 and having floors used for human occupancy located more than 75 feet (22 860 mm) above the lowest level of Fire Department vehicle access shall be provided with an automatic smoke detection connected to an automatic fire alarm system in accordance with Section 907.2.13.1, a Fire Department communication system in accordance with Section 907.2.13.2, a two-way communication system in accordance with Section 907.2.13.3, and an emergency voice/alarm communication system in accordance with Section 907.5.2.2.

Exceptions:

1. Open parking garages in accordance with Section 406.5.
2. Buildings with an occupancy in Group A-5 in accordance with Section 303.6.
3. Low-hazard special occupancies in accordance with Section 503.1.1.
4. Buildings with an occupancy in Group H-1, H-2 or H-3 in accordance with Section 415.

907.2.13.1 Automatic smoke detection. In addition to smoke detection otherwise required by this code, automatic smoke detection in high-rise buildings shall be in accordance with Sections 907.2.13.1.1 and 907.2.13.1.2.

Exception for Group R-2 occupancies: In R-2 occupancies, the activation of smoke detectors shall initiate a signal at a central supervising station or a constantly attended location and shall not initiate a signal to alarm notification appliances.

907.2.13.1.1 Automatic smoke detection. Automatic smoke detectors shall be provided in accordance with this section. Smoke detectors shall be connected to an automatic fire alarm system. The activation of any detector required by this section shall activate the emergency voice/alarm communication system. Smoke detectors shall be located as follows:

1. In each mechanical equipment, electrical, transformer, telephone equipment or similar room.
2. In each elevator machine room and in elevator lobbies.

907.2.13.1.2 Duct smoke detection. Duct smoke detectors complying with Section 907.3.1 shall be located in air distribution systems in accordance with Section 606 of the *New York City Mechanical Code*.

907.2.13.2 Fire Department communication system. The Fire Department Auxiliary Radio Communication System (ARCS) shall be in accordance with Section 916.

‡‡‡ **907.2.13.3 Two-way voice communication system.** A two-way voice communication system (warden) phone that complies with the requirements of NFPA 72 shall be provided in the following locations and shall comply with the following requirements. Such phones shall communicate with the fire command center.

1. In Group B high-rise office buildings and large area office buildings, there shall be a minimum of two phones located on every floor accessible to all occupants, with each phone located within 5 feet (1524 mm) of a different exit stair.
2. Where elevator lobbies are permitted to be locked, the phones provided are permitted to be connected to the fire alarm system.
3. If phones are provided in areas of rescue assistance and refuge areas, the phones are permitted to be connected to the fire alarm system.
4. Where phones are provided to meet the requirements for stairway communication systems in Section 403.5.3.1, the phones are permitted to be connected to the fire alarm system.

5. In all Group I-2 buildings, there shall be a phone located at staff attended locations, such as nurses' stations or similar locations accessible to all staff members, on every patient floor per fire/smoke zone. Phones shall also be located in areas of the building where the fire alarm does not sound.

   Exception: Group R-2 occupancies.

907.2.14 Atriums connecting more than two stories. A fire alarm system shall be installed in occupancies with an atrium that connects more than two stories, with smoke detection installed in locations as required by a rational analysis in Section 909.4 and in accordance with the system operation requirements in Section 909.17. The system shall be activated in accordance with Section 907.5. Such occupancies in Group A, E or M shall be provided with an emergency voice/alarm communication system complying with the requirements of Section 907.5.2.2.

907.2.15 High-piled combustible storage areas. An automatic fire detection system shall be installed throughout high-piled combustible storage areas where required by the *New York City Fire Code*.

907.2.16 Aerosol storage uses. Aerosol storage rooms and general-purpose warehouses containing aerosols shall be provided with an approved manual fire alarm system where required by the *New York City Fire Code*.

907.2.17 Lumber, wood structural panel and veneer mills. Lumber, wood structural panel and veneer mills shall be provided with a manual fire alarm system.

907.2.18 Underground buildings with compartment smoke control system. Where a compartment smoke control system is installed in an underground building as required by Section 405, automatic smoke detectors shall be provided in accordance with Section 907.2.18.1.

907.2.18.1 Smoke detectors. Not fewer than one smoke detector listed for the intended purpose shall be installed in all of the following areas:

1. Mechanical equipment, electrical, transformer, telephone equipment, elevator machine or similar rooms.
2. Elevator lobbies.
3. The main supply, main return, and exhaust air plenum of each air-conditioning system serving more than one story and located in a serviceable area downstream from filters on supply ducts and in return/exhaust ducts downstream of the last duct inlet.
4. Each connection to a vertical duct or riser serving two or more floors from return air ducts or plenums of heating, ventilating and air-conditioning systems, except that in Group R occupancies, a listed smoke detector is allowed to be used in each return air riser carrying not more than 5,000 cfm (2.4 m^3/s) and serving not more than 10 air inlet openings.

907.2.18.2 Alarm required. Activation of the smoke exhaust system shall activate an audible alarm at a constantly attended location.

907.2.19 Underground buildings. In underground buildings complying with Section 405 where the lowest level of a structure is more than 30 feet (9144 mm) below the lowest level of exit discharge, the structure shall be equipped throughout with a manual and automatic fire alarm system, including an emergency voice/alarm communication system installed in accordance with Section 907.5.2.2.

907.2.20 Covered and open mall buildings. Where the total floor area exceeds 50,000 square feet (4645.2 m^2) within either a covered mall building or within the perimeter line of an open mall building, an emergency voice/alarm communication system shall be provided. Emergency voice/alarm communication systems serving a mall, required or otherwise, shall be accessible to the Fire Department. The system shall be provided in accordance with Section 907.5.2.2.

907.2.21 Reserved.

907.2.22 Airport traffic control towers. An automatic smoke detection system that activates the occupant notification system in accordance with Section 907.5 shall be provided in airport control towers in accordance with Sections 907.2.22.1 and 907.2.22.2.

Exception: Audible appliances shall not be installed within the control tower cab.

907.2.22.1 Airport traffic control towers with multiple exits and automatic sprinklers. Airport traffic control towers with multiple exits and equipped throughout with an automatic sprinkler system in accordance with Section 903.3.1.1 shall be provided with smoke detectors in all of the following locations:

1. Airport traffic control cab.
2. Electrical and mechanical equipment rooms.
3. Airport terminal radar and electronics rooms.
4. Outside each opening into interior exit stairways.
5. Along the single means of egress permitted from observation levels.
6. Outside each opening into the single means of egress permitted from observation levels.

907.2.22.2 Other airport traffic control towers. Airport traffic control towers with a single exit or where sprinklers are not installed throughout shall be provided with smoke detectors in all of the following locations:

1. Airport traffic control cab.
2. Electrical and mechanical equipment rooms.
3. Airport terminal radar and electronics rooms.
4. Office spaces incidental to the tower operation.
5. Lounges for employees, including sanitary facilities.
6. Means of egress.
7. Accessible utility shafts.

907.2.23 Battery rooms. An approved automatic smoke detection system shall be installed in areas containing any battery systems. Where the battery room is located in a building or space that is provided with a fire alarm system or subsystem, the smoke detectors shall be connected to such building fire alarm system or subsystem. The detection system shall be supervised by a central supervising station, or a local alarm that will sound an audible signal at a constantly attended location. Battery rooms shall also comply with the requirements of the *New York City Fire Code*.

907.3 Fire safety functions. Automatic fire detectors utilized for the purpose of performing fire safety functions shall be connected to the building's fire alarm control unit where a fire alarm system is required by Section 907.2. Detectors shall, upon actuation, perform the intended function and activate the alarm notification appliances or activate a visible and audible supervisory signal at a constantly attended location.

Exception: In buildings not equipped with a fire alarm system, the automatic fire detector shall be powered by normal electrical service and, upon actuation, perform the intended function. The detectors shall be located in accordance with NFPA 72.

907.3.1 Duct smoke detectors. Smoke detectors installed in ducts shall be listed for the air velocity, temperature and humidity present in the duct. Duct smoke detectors shall be connected to the building's fire alarm control unit when a fire alarm system is required by Section 907.2. Activation of a duct smoke detector shall initiate a visible and audible alarm signal at a constantly attended location and shall perform the intended fire safety function in accordance with this code and the *New York City Mechanical Code*. Duct smoke detectors shall not be used as a substitute for required open area detection.

Exceptions:

1. The alarm signal at a constantly attended location is not required where duct smoke detectors activate the building's alarm notification appliances.
2. In occupancies not required to be equipped with a fire alarm system, actuation of a smoke detector shall activate a visible and an audible signal in an approved location. Smoke detector trouble conditions shall activate a visible or audible signal in an approved location and shall be identified as air duct detector trouble.

907.3.2 Delayed egress locks. Where delayed egress locks are installed on means of egress doors in accordance with Section 1010.1.9.7, an automatic smoke or heat detection system shall be installed as required by that section.

907.3.3 Elevator emergency operation. Automatic fire detectors installed for elevator emergency operation shall be installed in accordance with the provisions of Chapter 30 of this code, ASME A17.1 and NFPA 72.

907.3.4 Wiring. The wiring to the auxiliary devices and equipment used to accomplish fire safety functions shall be monitored for integrity in accordance with NFPA 72 and the *New York City Electrical Code*.

907.3.5 Monitoring of hold-open devices and closers. All hold-open devices used in automatic-closing doors pursuant to the exception to Section 713.7 shall be electrically supervised to monitor the integrity of the wiring connections among the fire alarm system, the smoke detection system, and the hold-open devices.

907.4 Initiating devices. Where manual or automatic alarm initiation is required as part of a fire alarm system, the initiating devices shall be installed in accordance with Sections 907.4.1 through 907.4.4.

907.4.1 Protection of fire alarm control unit. In areas that are not continuously occupied, a single smoke detector shall be provided at the location of each fire alarm control unit, notification appliance circuit power extenders, and supervising station transmitting equipment.

Exceptions:

1. Where ambient conditions prohibit installation of a smoke detector, a heat detector shall be permitted.
2. In prior code buildings, where a fire alarm control unit is installed in an interior exit stairway or ramp or exit passageway, a smoke detector is not required at the location of such fire alarm control unit.

907.4.2 Manual fire alarm boxes. Where a manual fire alarm system is required by another section of this code, it shall be activated by fire alarm boxes installed in accordance with Sections 907.4.2.1 through 907.4.2.6.

907.4.2.1 Location. Manual fire alarm boxes shall be located not more than 5 feet (1524 mm) from the entrance to each exit. Additional manual fire alarm boxes shall be located so that travel distance to the nearest box does not exceed 200 feet (60 960 mm).

907.4.2.2 Height. The height of the manual fire alarm boxes shall be not less than 42 inches (1066.8 mm) and not more than 48 inches (1220 mm) measured vertically, from the floor level to the activating handle or lever of the box.

907.4.2.3 Color. Manual fire alarm boxes shall be red in color.

907.4.2.4 Signs. Where fire alarm systems are not required to be monitored by a supervising station, an approved permanent sign that reads: WHEN ALARM SOUNDS—CALL 911 shall be installed adjacent to each manual fire alarm box.

Exception: Where the manufacturer has permanently provided this information on the manual fire alarm box.

907.4.2.5 Protective covers. The Fire Department is authorized to require the installation of listed manual fire alarm box protective covers to prevent malicious false alarms or provide the manual fire alarm box with protection from physical damage. The protective cover shall be transparent or red in color with a transparent face to permit visibility of the manual fire alarm box. Each cover shall include proper operating instructions. A protective cover that emits a local alarm signal shall not be installed unless approved. Protective covers shall not project more than that permitted by Section 1003.3.3.

907.4.2.6 Unobstructed and unobscured. Manual fire alarm boxes shall be readily accessible, unobstructed, unobscured and visible at all times.

907.4.3 Automatic smoke detection. Where an automatic smoke detection system is required, it shall utilize smoke detectors unless ambient conditions prohibit such an installation. In spaces where smoke detectors cannot be utilized due to ambient conditions, approved automatic heat detectors shall be permitted.

907.4.3.1 Automatic sprinkler system. For conditions other than specific fire safety functions noted in Section 907.3, in areas where ambient conditions prohibit the installation of smoke detectors, an automatic sprinkler system installed in such areas in accordance with Section 903.3.1.1 or 903.3.1.2 and that is connected to the fire alarm system shall be approved as automatic heat detection.

907.4.4 Fire-extinguishing systems. Where a fire alarm system is required by another section of this code or is otherwise installed, automatic fire-extinguishing systems installed in accordance with Section 904 shall be monitored by the fire alarm system.

907.5 Occupant notification systems. A fire alarm system shall annunciate at the fire alarm control unit and shall initiate occupant notification upon activation, in accordance with Sections 907.5.1 through 907.5.2.3.2. Where a fire alarm system is required by another section of this code, it shall be activated by:

1. Automatic fire detectors.
2. Automatic sprinkler system waterflow devices.
3. Manual fire alarm boxes.
4. Automatic fire-extinguishing systems.

Exception: Where notification systems are allowed elsewhere in Section 907 to annunciate at a constantly attended location or to a central supervising station.

907.5.1 Presignal feature. A presignal feature shall not be installed unless approved by the Fire Department. Where a presignal feature is provided, a signal shall be annunciated at a constantly attended location approved by the Fire Department, in order that occupant notification can be activated in the event of fire or other emergency.

907.5.2 Alarm notification appliances. Alarm notification appliances shall be provided and shall be listed for their purpose.

907.5.2.1 Audible alarms. Audible alarm notification appliances shall be provided and emit a distinctive sound that is not to be used for any purpose other than that of a fire alarm.

Exceptions:

1. Visible alarm notification appliances shall be allowed in lieu of audible alarm notification appliances in critical care areas of Group I-2 occupancies.
2. Where provided, audible notification appliances located in each occupant evacuation elevator lobby in accordance with Section 3008.10.1 shall be connected to a separate notification zone for manual paging only.

907.5.2.1.1 Average sound pressure. The audible alarm notification appliances shall provide a sound pressure level of 15 decibels (dBA) above the average ambient sound level or 5 dBA above the maximum sound level having a duration of not less than 60 seconds, whichever is greater, in every occupiable space within the building. The minimum sound pressure levels shall be: 75 dBA in occupancies in Groups R and I-1; 90 dBA in mechanical equipment rooms and 60 dBA in other occupancies. For one-way voice communication system in Group R-2 occupancies, the minimum sound pressure level of the alert tone shall be 75 dBA throughout the dwelling unit.

907.5.2.1.2 Maximum sound pressure. The maximum sound pressure level for audible alarm notification appliances shall be 110 dBA at the minimum hearing distance from the audible appliance. Where the average ambient noise is greater than 95 dBA, visible alarm notification appliances shall be provided in accordance with NFPA 72 and audible alarm notification appliances shall not be required.

907.5.2.2 Emergency voice/alarm communication systems. Emergency voice/alarm communication systems required by this code shall be designed and installed in accordance with NFPA 72. The operation of any automatic fire detector, sprinkler waterflow device or manual fire alarm box shall automatically sound an alert tone followed by voice instructions giving approved information and directions for a general or staged evacuation in accordance with the building's fire safety and evacuation plans required by the *New York City Fire Code*. In high-rise buildings, the system shall operate on at least the alarming floor, the floor above and the floor below. Speakers shall be provided throughout the building by paging zones. At a minimum, paging zones shall be provided as follows:

1. Each exit stairway.
2. Each floor.
3. Refuge areas as defined in Chapter 2 of this code.

Exceptions:

1. **Group I-1 and I-2 occupancies.** In Group I-1 and I-2 occupancies, the alarm shall sound in a constantly attended area and a general occupant notification shall be broadcast over the overhead page.

2. **Group R-2 occupancies greater than 75 feet (22 860 mm) in height.** In Group R-2 occupied buildings greater than 75 feet (22 860 mm) in height above the lowest level of Fire Department vehicle access, activation of any smoke detector or sprinkler water flow device shall initiate a signal at a central supervising station or constantly attended location and shall not initiate a signal to an alarm notification appliance. An emergency voice/alarm communication system shall not be required. However, a one-way voice communication shall be provided between the fire command center for use by Fire Department personnel and the following terminal areas:
   - 2.1. Within dwelling units. An intercom system may be utilized when provided with an override feature for use by Fire Department personnel. Such intercom system shall comply with rules promulgated by the commissioner establishing instal-lation requirements.
   - 2.2. Within required exit stairs. Annunciation devices shall be located at least on every other story units. Such annunciation devices shall comply with rules promulgated by the commissioner establishing instal-lation requirements.

907.5.2.2.1 Manual override. A manual override for emergency voice communication shall be provided on a selective and all-call basis for all paging zones.

907.5.2.2.2 Live voice messages. The emergency voice/alarm communication system shall have multi-channel capability to broadcast live voice messages by paging zones on a selective and all-call basis without automatic interruption of the alarm tones on the floor of incidence, floor above or floor below.

907.5.2.2.3 Alternate uses. When approved by the Fire Commissioner, the emergency voice/alarm communication system may be allowed to be used for other announcements.

907.5.2.2.4 Emergency voice/alarm communication captions. Where stadiums, arenas and grandstands provide audible public announcements, the emergency/voice alarm communication system shall be captioned. Emergency captions shall be approved by the Fire Department.

907.5.2.2.5 Emergency power. Emergency voice/alarm communications systems shall be provided with an approved emergency power source in accordance with Section 2702 of this code and the *New York City Electrical Code.*

907.5.2.3 Visible alarms. Visible alarm notification appliances shall be provided in accordance with Sections 907.5.2.3.1 through 907.5.2.3.2.

Exceptions:

1. Visible alarm notification appliances shall not be required in exits as defined in Chapter 2.
2. Visible alarm notification appliances shall not be required in elevator cars.

907.5.2.3.1 Public and common areas. Visible alarm notification appliances shall be provided in public use areas and common use areas, as defined in Chapter 2.

Exception: Where employee work areas have audible alarm coverage, the notification appliance circuits serving the employee work areas shall be initially designed with not less than 20-percent spare capacity to account for the potential of adding visible notification appliances in the future to accommodate hearing impaired employee(s).

907.5.2.3.2 Groups I-1 and R-1. Group I-1 and R-1 dwelling units or sleeping units in accordance with Table 907.5.2.3.2 shall be provided with a visible alarm notification appliance, activated by both the in-room smoke detector and the building fire alarm system.

TABLE 907.5.2.3.2
VISIBLE ALARMS

NUMBER OF UNITS	UNITS WITH VISIBLE ALARMS
6 to 25	2
26 to 50	4
51 to 75	7
76 to 100	9

101 to 150	12
151 to 200	14
201 to 300	17
301 to 400	20
401 to 500	22
501 to 1,000	5% of total
1,001 and over	50 plus 3 for each 100 over 1,000

907.6 Installation and monitoring. A fire alarm system shall be installed and monitored in accordance with Sections 907.6.1 through 907.6.5.2 of this code and NFPA 72.

907.6.1 Wiring. Wiring shall comply with the requirements of the *New York City Electrical Code* and NFPA 72. Wireless protection systems utilizing radio-frequency transmitting devices shall comply with the special requirements for supervision of low-power wireless systems in NFPA 72.

907.6.2 Power supply. The primary and secondary power supply for the fire alarm system shall be provided in accordance with the *New York City Electrical Code*.

Exception: Secondary power for single-station and multiple-station smoke alarms as required in Section 907.2.11.2.

907.6.3 Zones. Each floor shall be zoned separately. For non-addressable systems, a zone shall not exceed 22,500 square feet (2090.3 m^2). The length of any zone shall not exceed 300 feet (91 440 mm) in any direction.

Exception: Automatic sprinkler system zones shall not exceed the area permitted by NFPA 13.

907.6.3.1 Zoning indicator panel. A zoning indicator panel and the associated controls shall be provided at the main building entrance accessible to responding Fire Department personnel and in other locations approved by the commissioner and the Fire Department. The visual zone indication shall lock in until the system is reset and shall not be canceled by the operation of an audible-alarm silencing switch.

907.6.3.2 High-rise buildings. In high-rise buildings constructed in accordance with Section 403, a separate zone by floor shall be provided for each of the following types of alarm-initiating devices where provided:

1. Smoke detectors.
2. Sprinkler waterflow devices.
3. Manual fire alarm boxes.
4. Other approved types of automatic fire detection devices or suppression systems.

907.6.4 Access. Access shall be provided to each fire alarm device and notification appliance for periodic inspection, maintenance and testing.

907.6.5 Monitoring. Fire alarm systems required by this chapter or by the *New York City Fire Code* shall be monitored by a central supervising station in accordance with NFPA 72 and approved by the Fire Commissioner.

Exception: Monitoring by a central supervising station is not required for:

1. Single- and multiple-station smoke alarms required by Section 907.2.11.
2. Smoke detectors in Group I-3 occupancies.
3. Automatic sprinkler systems in one- and two-family dwellings.

907.6.5.1 Automatic telephone-dialing devices. Automatic telephone-dialing devices used to transmit an emergency alarm shall not be connected to any Fire Department telephone number unless approved by the Fire Commissioner.

907.6.5.2 Termination of monitoring service. Termination of fire alarm monitoring services shall be in accordance with the *New York City Fire Code*.

907.7 Acceptance tests and completion. Upon completion of the installation, the fire alarm system and all fire alarm components shall be tested in accordance with NFPA 72.

907.7.1 Single- and multiple-station alarm devices. When the installation of the alarm devices is complete, each device and interconnecting wiring for multiple-station alarm devices shall be tested in accordance with the smoke alarm provisions of NFPA 72.

907.7.2 Record of completion. A record of completion in accordance with NFPA 72 verifying that the system has been installed and tested in accordance with the approved construction documents and specifications shall be provided.

907.7.3 Instructions. Operating, testing and maintenance instructions, and record drawings ("as builts") and equipment specifications shall be provided at an approved location.

907.8 Inspection, testing and maintenance. The maintenance and testing schedules and procedures for fire alarm and fire detection systems shall be in accordance with the *New York City Fire Code*.

SECTION BC 908
EMERGENCY ALARM SYSTEMS

908.1 Group H occupancies. Emergency alarms for the detection and notification of an emergency condition in Group H occupancies shall be provided in accordance with Section 415.5 of this code and the *New York City Fire Code*.

908.2 Group H-5 occupancy. Emergency alarms for notification of an emergency condition in a Hazardous Production Material (HPM) facility shall be provided as required in Section 415.11.3.5. A continuous gas detection system shall be provided for HPM gases in accordance with Section 415.11.7 of this code and the *New York City Fire Code*.

908.3 Highly toxic and toxic materials. A gas detection system shall be provided to detect the presence of highly toxic or toxic gas at or below the permissible exposure limit (PEL) or ceiling limit of the gas for which detection is provided. The system shall be capable of monitoring the discharge from the treatment system at or below one-half the immediately dangerous to life and health (IDLH) limit and shall comply with the *New York City Fire Code*.

Exception: A gas detection system is not required for toxic gases when the physiological warning threshold level for the gas is at a level below the accepted PEL for the gas.

908.3.1 Alarms. The gas detection system shall initiate a local alarm and transmit a signal to a constantly attended control station when a short-term hazard condition is detected. The alarm shall be both visible and audible and shall provide warning both inside and outside the area where gas is detected. The audible alarm shall be distinct from all other alarms.

Exception: Signal transmission to a constantly attended control station is not required when not more than one cylinder of highly toxic or toxic gas is stored.

908.3.2 Shutoff of gas supply. The gas detection system shall automatically close the shutoff valve at the source on gas supply piping and tubing related to the system being monitored for whichever gas is detected.

Exception: Automatic shutdown is not required for reactors utilized for the production of highly toxic or toxic compressed gases where such reactors are:

1. Operated at pressures less than 15 pounds per square inch gauge (psig) (103.4 kPa);
2. Constantly attended; and
3. Provided with readily accessible emergency shutoff valves.

908.3.3 Valve closure. The automatic closure of shutoff valves shall be in accordance with the following:

1. When the gas-detection sampling point initiating the gas detection system alarm is within a gas cabinet or exhausted enclosure, the shutoff valve in the gas cabinet or exhausted enclosure for the specific gas detected shall automatically close.

2. Where the gas-detection sampling point initiating the gas detection system alarm is within a gas room and compressed gas containers are not in gas cabinets or exhausted enclosures, the shutoff valves on all gas lines for the specific gas detected shall automatically close.

3. Where the gas-detection sampling point initiating the gas detection system alarm is within a piping distribution manifold enclosure, the shutoff valve for the compressed container of specific gas detected supplying the manifold shall automatically close.

   Exception: When the gas-detection sampling point initiating the gas detection system alarm is at a use location or within a gas valve enclosure of a branch line downstream of a piping distribution manifold, the shutoff valve in the gas valve enclosure for the branch line located in the piping distribution manifold enclosure shall automatically close.

908.4 Ozone gas-generator rooms. Ozone gas-generator rooms shall be equipped with a continuous gas detection system that will shut off the generator and sound a local alarm when concentrations above the PEL occur and shall comply with the *New York City Fire Code*.

908.5 Repair garages. A flammable-gas detection system shall be provided in repair garages for vehicles fueled by enumerated gases in accordance with Section 406.8.5 of this code and the *New York City Fire Code*.

908.6 Refrigerant detectors. Machinery rooms shall contain refrigerant detectors with audible and visual alarms. The detectors, or sampling tubes that draw air to each detector, shall be located in all areas where refrigerant from a leak will concentrate. The alarm shall be actuated at a value not greater than the corresponding TLV-TWA values for the refrigerant classification indicated in the *New York City Mechanical Code*. Detectors and alarms shall be placed in approved locations. Refrigerant detectors shall initiate all functions as required by the *New York City Mechanical Code* and *New York City Fire Code*.

908.7 Carbon dioxide (CO_2) systems. The emergency alarm system for a carbon dioxide system, including detection, pre-discharge and discharge alarms, shall be provided in accordance with the *New York City Fire Code*.

908.8 Gas detection systems. Gas detection systems shall be provided in accordance with Section 918.

908.9 Medical gas. Medical gas pressure monitoring and alarm systems shall be provided in accordance with NFPA 99.

908.10 Flammable gas. Rooms and spaces containing flammable gas distribution piping operating at levels above 15 pounds per square inch gauge (psig) (103.4 kPa) shall be provided with an approved flammable gas detection-alarm system and shall comply with the *New York City Fire Code*.

908.11 Construction documents. Construction documents for emergency alarm systems shall be submitted for review and approval to the Fire Department prior to system installation.

908.12 Acceptance testing and maintenance. Acceptance testing and maintenance of emergency alarm systems shall be performed in accordance with the *New York City Fire Code*.

***908.13 Natural gas alarms.** Natural gas alarms listed in accordance with a standard established or adopted by department rule shall be provided and installed in accordance with department rules.

**Section 908.13 was added by [Local Law 102 of 2025.](#) This law has an effective date of July 30, 2025.*

SECTION BC 909
SMOKE CONTROL SYSTEMS

909.1 Scope and purpose. This section applies to mechanical or passive smoke control systems where they are required by other provisions of this code. A smoke control system, where required, facilitates the evacuation of the occupants. The purpose of this section is to establish minimum requirements for the design, installation and acceptance testing of smoke control systems that are intended to provide a tenable environment for the evacuation or relocation of occupants. These provisions are not intended for the preservation of contents, the timely restoration of operations or for assistance in fire suppression or post-fire smoke purge. Smoke control systems regulated by this section serve a different purpose than the smoke- and heat-venting provisions found in Section 910. Mechanical smoke control systems shall not be considered exhaust systems under Chapter 5 of the *New York City Mechanical Code*.

909.1.1 Definitions. The following terms are defined in Chapter 2 of this code:

ELEVATOR LANDING.

PRESSURIZATION.

SMOKE.

SMOKE BARRIER.

SMOKE CONTROL MODE.

SMOKE CONTROL SYSTEM, MECHANICAL.

SMOKE CONTROL SYSTEM, PASSIVE.

SMOKE CONTROL ZONE.

SMOKE DAMPER.

STACK EFFECT.

TENABLE ENVIRONMENT.

909.2 General design requirements. Buildings, structures or parts thereof required by this code to have a smoke control system or systems shall have such systems designed in accordance with the applicable requirements of Section 909 and the generally accepted and well-established principles of engineering relevant to the design. The construction documents shall include sufficient information and detail to adequately describe the elements of the design necessary for the proper implementation of the smoke control systems. These documents shall be accompanied by sufficient information and analysis to demonstrate compliance with these provisions.

909.3 Special inspection and test requirements. In addition to the ordinary inspection and test requirements that buildings, structures and parts thereof are required to undergo, smoke control systems subject to the provisions of Section 909 shall undergo special inspections and tests sufficient to verify the proper commissioning of the smoke control design in its final installed condition. The design submission accompanying the construction documents shall clearly detail procedures and methods to be used and the devices, flow measurement, and other items subject to such inspections and tests. Such commissioning shall be in accordance with generally accepted engineering practice and, where possible, based on published standards for the particular testing involved. The special inspections and tests required by this section shall be conducted under the same terms in Section 1705. Records of the special inspection, including device locations, duct air leakage, pressure differentials, air/smoke flow measurements, smoke detection and control verification shall be maintained on the premises as a baseline against which future tests can be compared.

909.3.1 Periodic testing. Smoke control systems shall be verified weekly through the automatic control system in accordance with Section 909.12 and shall be tested annually to ensure proper operation of detection devices, dampers, fans and controls in accordance with the requirements of Sections 909.18.1, 909.18.3, 909.18.5 and 909.18.7. Full testing of smoke control systems in accordance with Sections 909.18 through 909.18.7 shall be conducted at 5-year intervals by an inspector qualified in accordance with Section 909.18.8.2. Test reports shall include all information required by Section 909.18.8.3 and shall be compared against the baseline special inspection report. Causes for any significant deviations from the baseline report shall be identified and corrected. A record of each inspection and test shall be maintained on the premises by the owner or lessee, and the records for at least the last 5 years of operation shall be made available for inspection by the department and the Fire Commissioner.

909.4 Analysis. A rational analysis supporting the types of smoke control systems to be employed, their methods of operation, the systems supporting them and the methods of construction to be utilized shall accompany the submitted construction documents and shall include, but not be limited to, the items indicated in Sections 909.4.1 through 909.4.7. The basis of design and design analysis of the smoke control system shall be submitted to the department and the Fire Department.

909.4.1 Stack effect. The system shall be designed such that the maximum probable normal or reverse stack effect will not adversely interfere with the system's capabilities. In determining the maximum probable stack effect, altitude, elevation, weather history and interior temperatures shall be used.

909.4.2 Temperature effect of fire. Buoyancy and expansion caused by the design fire in accordance with Section 909.9 shall be analyzed. The system shall be designed such that these effects do not adversely interfere with the system's capabilities.

909.4.3 Wind effect. The design shall consider the adverse effects of wind. Such consideration shall be consistent with the wind-loading provisions of Chapter 16.

909.4.4 HVAC systems. The design shall consider the effects of the heating, ventilating and air-conditioning (HVAC) systems on both smoke and fire transport. The analysis shall include all permutations of systems' status. The design shall consider the effects of the fire on the HVAC systems.

909.4.5 Climate. The design shall consider the effects of low temperatures on systems, property and occupants. Air inlets and exhausts shall be located so as to prevent snow or ice blockage.

909.4.6 Duration of operation. All portions of active or passive smoke control systems shall be capable of continued operation after detection of the fire event for a period of not less than either 20 minutes or 1.5 times the required safe egress time, whichever is greater.

909.4.7 Smoke control system interaction. The design shall consider the interaction effects of the operation of multiple smoke control systems for all design scenarios.

‡‡‡ **909.5 Smoke barrier construction.** Smoke barriers required for passive smoke control and a smoke control system using the pressurization method shall comply with Section 709. Smoke barriers shall be constructed and sealed to limit leakage areas exclusive of protected openings. The maximum allowable leakage area shall be the aggregate area calculated using the following leakage area ratios:

1. Walls: $A/A_W = 0.00100$
2. Interior exit stairways and ramps and exit passageways: $A/A_w = 0.00035$
3. Enclosed exit access stairways and ramps and all other shafts: $A/A_w = 0.00150$
4. Floors and roofs: $A/A_F = 0.00050$

where:

A = Maximum allowable leakage area, square feet (m^2).

A_F = Unit floor or roof area of barrier, square feet (m^2).

A_w = Unit wall area of barrier, square feet (m^2).

The leakage area ratios shown do not include openings due to gaps around doors and operable windows. The total leakage area of the smoke barrier shall be determined in accordance with Section 909.5.1 and tested in accordance with Section 909.5.2.

909.5.1 Total leakage area. Total leakage area of the barrier is the product of the smoke barrier gross area multiplied by the allowable leakage area ratio, plus the area of other openings such as gaps around doors and operable windows.

909.5.2 Testing of leakage area. Compliance with the maximum allowable leakage area shall be determined by achieving the minimum air pressure difference across the barrier with the system in the smoke control mode for mechanical smoke control systems utilizing the pressurization method. Compliance with the maximum allowable leakage area of passive smoke control systems shall be verified through methods such as door fan testing or other methods, as approved by the commissioner and the Fire Commissioner.

909.5.3 Opening protection. Openings in smoke barriers shall be protected by automatic-closing devices actuated by the required controls for the mechanical smoke control system. Door openings shall be protected by fire door assemblies complying with Section 716.5.3.

Exceptions:

1. Passive smoke control systems with automatic-closing devices actuated by spot-type smoke detectors listed for releasing service installed in accordance with Section 907.3.
2. Fixed openings between smoke zones that are protected utilizing the airflow method.

3. In Group I-1, Group I-2, and Group B ambulatory care facilities, where a pair of opposite-swinging doors are installed across a corridor in accordance with Section 909.5.3.1, the doors shall not be required to be protected in accordance with Section 716. The doors shall be close-fitting within operational tolerances and shall not have a center mullion, louvers, grilles, or door undercuts in excess of $^3/_4$ inch (19.1 mm). The doors shall have head and jamb stops and astragals or rabbets at meeting edges. If allowed by the door manufacturer's listing, positive-latching devices are not required.
4. In Group I-1, Group I-2 and Group B ambulatory care facilities, where such doors are special-purpose horizontal sliding, accordion or folding door assemblies installed in accordance with Section 1010.1.4.3 and are automatic closing by smoke detection in accordance with Section 716.5.9.3.
5. Group I-3.
6. Openings between smoke zones with clear ceiling heights of 14 feet (4267 mm) or greater and bank-down capacity of greater than 20 minutes as determined by the design fire size.

909.5.3.1 Group I-2 and Group B ambulatory care facilities. In Group I-2 and Group B ambulatory care facilities, where doors are installed across a corridor, the doors shall be automatic closing by smoke detection in accordance with Section 716.5.9.3 and shall have a vision panel with fire protection-rated glazing materials in fire protection-rated frames, the area of which shall not exceed that tested.

909.5.3.2 Ducts and air transfer openings. Ducts and air transfer openings are required to be protected with a minimum Class II, 250°F (121.1°C) smoke damper complying with Section 717.

909.6 Pressurization method. The primary mechanical means of controlling smoke shall be by pressure differences across smoke barriers. Maintenance of a tenable environment is not required in the smoke control zone of fire origin.

909.6.1 Minimum pressure difference. The minimum pressure difference across a smoke barrier shall be 0.05-inch water gauge (0.0124 kPa) in fully sprinklered buildings. In buildings permitted to be other than fully sprinklered, the smoke control system shall be designed to achieve pressure differences not less than two times the maximum calculated pressure difference produced by the design fire.

909.6.2 Maximum pressure difference. The maximum air pressure difference across a smoke barrier shall be determined by required door-opening or closing forces. The actual force required to open exit doors when the system is in the smoke control mode shall be in accordance with Section 1010.1.3. Opening and closing forces for other doors shall be determined by standard engineering methods for the resolution of forces and reactions. The calculated force to set a side-hinged, swinging door in motion shall be determined by:

$$F = F_{dc} + K(WA\Delta P)/2(W - d)$$ **(Equation 9-1)**

where:

A = Door area, square feet (m^2).

d = Distance from door handle to latch edge of door, feet (m).

F = Total door opening force, pounds (N).

F_{dc} =Force required to overcome closing device, pounds (N).

K = Coefficient 5.2 (1.0).

W = Door width, feet (m).

ΔP= Design pressure difference, inches of water (Pa).

909.6.3 Pressurized stairways and elevator hoistways. Where stairways or elevator hoistways are pressurized, such pressurization systems shall comply with Section 909 of this code as smoke control systems, in addition to the requirements of Sections 909.20 and 909.21 of this code and the *New York City Fire Code.*

909.7 Air flow design method. Where approved by the commissioner, smoke migration through openings fixed in a permanently open position, which are located between smoke control zones by the use of the airflow method, shall be permitted. The design air flow shall be in accordance with this section. Air flow shall be directed to limit smoke migration from the fire zone. The geometry of openings shall be considered to prevent flow reversal from turbulent effects. Smoke control systems using the airflow method shall be designed in accordance with NFPA 92.

909.7.1 Prohibited conditions. This method shall not be employed where either the quantity of air or the velocity of the airflow will adversely affect other portions of the smoke control system, unduly intensify the fire, disrupt plume dynamics or interfere with exiting. In no case shall airflow toward the fire exceed 200 feet per minute (1.02 m/s). Where the calculated airflow exceeds this limit, the airflow method shall not be used.

909.8 Exhaust method. Where approved by the commissioner, mechanical smoke control for large enclosed volumes, such as in atriums or malls, shall be permitted to utilize the exhaust method. Smoke control systems using the exhaust method shall be designed in accordance with NFPA 92.

909.8.1 Smoke layer. The height of the lowest horizontal surface of the accumulating smoke layer shall be maintained not less than 6 feet (1828.8 mm) above any walking surface that forms a portion of a required egress system within the smoke zone.

909.9 Design fire. The design fire shall be based on a rational analysis performed by the registered design professional and approved by the commissioner. The design fire shall be based on the analysis in accordance with Section 909.4 and this section.

909.9.1 Factors considered. The engineering analysis shall include the characteristics of the fuel, fuel load, effects included by the fire and whether the fire is likely to be steady or unsteady.

909.9.2 Design fire fuel. Determination of the design fire shall include consideration of the type of fuel, fuel spacing and configuration.

909.9.3 Heat-release assumptions. The analysis shall make use of best available data from approved sources and shall not be based on excessively stringent limitations of combustible material.

909.9.4 Sprinkler effectiveness assumptions. A documented engineering analysis shall be provided for conditions that assume fire growth is halted at the time of sprinkler activation.

909.10 Equipment. Equipment including, but not limited to, fans, ducts, automatic dampers and balance dampers, shall be suitable for its intended use, suitable for the probable exposure temperatures that the rational analysis indicates and as approved by the commissioner.

909.10.1 Exhaust fans. Components of exhaust fans shall be rated and certified by the manufacturer for the probable temperature rise to which the components will be exposed. This temperature rise shall be computed in accordance with NFPA 92.

909.10.2 Ducts. Duct materials and joints shall be capable of withstanding the probable temperatures and pressures to which they are exposed as determined in accordance with Section 909.10.1. Ducts shall be constructed and supported in accordance with the *New York City Mechanical Code*. Ducts shall be leak tested to 1.5 times the maximum design pressure in accordance with nationally accepted practices. Measured leakage shall not exceed 5 percent of design flow. Results of such testing shall be a part of the documentation procedure. Ducts shall be supported directly from fire-resistance-rated structural elements of the building by substantial, noncombustible supports.

Exception: Flexible connections, for the purpose of vibration isolation, complying with the *New York City Mechanical Code* and that are constructed of approved fire-resistance-rated materials.

909.10.3 Equipment, inlets and outlets. Equipment shall be located so as not to expose uninvolved portions of the building to an additional fire hazard. Outside air inlets shall be located so as to minimize the potential for introducing smoke or flame into the building. Exhaust outlets shall be located so as to minimize reintroduction of smoke into the building and to limit exposure of the building or adjacent buildings to an additional fire hazard.

909.10.4 Automatic dampers. Automatic dampers, regardless of the purpose for which they are installed within the smoke control system, shall comply with Section 717.3.1.

909.10.5 Fans. In addition to other requirements, belt-driven fans shall have 1.5 times the number of belts required for the design duty with the minimum number of belts being two. Fans shall be selected for stable performance based on normal temperature and, where applicable, elevated temperature. Calculations and manufacturer's fan curves shall be part of the documentation procedures. Fans shall be supported and restrained by noncombustible devices in accordance with the requirements of Chapter 16. Motors driving fans shall not be operated beyond their nameplate horsepower (kilowatts), as determined from measurement of actual current draw, and shall have a minimum service factor of 1.15.

‡‡‡ **909.10.6 Seismic requirements.** Smoke control systems covered by Section 909 are required to function after an earthquake. Such smoke control systems shall be seismically designed in accordance with Section 1613 of this code and ASCE 7. The component importance factor, I_p, shall be taken as 1.5 in accordance with ASCE 7, Section 13.1.3. The smoke control system includes all components required for its operation, including but not limited to fans, ducts, electrical power, switchboards, motor control centers, starters, and controls.

Exception: Smoke control systems in structures classified in Seismic Design Categories A or B shall have a component importance factor, I_p, of 1.0.

909.11 Standby power. The smoke control system shall be supplied with two sources of power. The primary power supply shall be from the normal building power systems, while the secondary power source shall be from a standby power system complying with Section 2702.

909.11.1 Equipment room. The standby power source shall be located in a room separate from the normal power transformers and switch gears, and ventilated directly to and from the exterior. The room shall be enclosed with not less than 2-hour fire barriers constructed in accordance with Section 707, or with not less than 2-hour fire-resistance-rated horizontal assemblies constructed in accordance with Section 711, or both.

909.11.2 Power sources and power surges. Elements of the smoke control system relying on electronic volatile memories or similar systems shall be supplied with uninterruptable power sources of sufficient duration to span a 15-minute primary power interruption. Elements of the smoke control system susceptible to power surges shall be suitably protected by conditioners, suppressors or other approved means.

909.12 Detection and control systems. Fire detection systems providing control input or output signals to mechanical smoke control systems or elements thereof shall comply with the requirements of Section 907. Such systems shall be equipped with a control unit complying with UL 864 and listed as smoke control equipment.

909.12.1 Verification. Control systems for mechanical smoke control systems shall include provisions for verification. Verification shall include positive confirmation of actuation, testing, manual override and the presence of power downstream of all disconnects. A preprogrammed weekly test sequence shall report, abnormal conditions audibly, visually and by printed report. The preprogrammed weekly test shall operate all devices, equipment and components used for smoke control.

Exception: Where verification of individual components tested through the preprogrammed weekly testing sequence will interfere with, and produce unwanted effects to, normal building operation, such individual components are permitted to be bypassed from the preprogrammed weekly testing, where approved by the Fire Department and in accordance with both of the following:

1. Where the operation of components is bypassed from the preprogrammed weekly test, presence of power downstream of all disconnects shall be verified weekly by a listed control unit.
2. Testing of all components bypassed from the preprogrammed weekly test shall be tested semi-annually, and be tested under standby power conditions in accordance with Section 909 of the *New York City Fire Code*.

909.12.2 Wiring. In addition to meeting requirements of the *New York City Electrical Code*, all wiring, regardless of voltage, shall be fully enclosed within continuous raceways.

909.12.3 Activation. Smoke control systems shall be activated in accordance with this section.

909.12.3.1 Pressurization, airflow or exhaust method. Mechanical smoke control systems using the pressurization, airflow or exhaust method shall have completely automatic control.

909.12.3.2 Passive method. Passive smoke control systems actuated by spot-type detectors listed for releasing service shall be permitted.

909.12.4 Automatic control. Where completely automatic control is required or used, the automatic-control sequences shall be initiated from an appropriately zoned automatic sprinkler system complying with Section 903.3.1.1, manual controls that are readily accessible to the Fire Department and any smoke detectors required by engineering analysis. See Section 909.16 for manual control requirements.

909.12.4.1 Building Management System. Automatic and manual operation of the smoke control system may alternately be done through a Building Management System (BMS) that is approved by the Fire Department and meets the following requirements:

1. The BMS system shall be listed for UL 864 UUKL Smoke Control.
2. The BMS Control Center shall be staffed 24 hours a day by operators trained in the building's smoke control systems and their operation. The smoke control system shall be operated by a certificate of fitness holder where required by the *New York City Fire Code*.
3. The control room shall be 2-hour fire-resistance-rated construction.
4. BMS annunciation and additional control station locations shall be located in the fire command center in accordance with Section 909.16.

909.13 Control air tubing. Control air tubing shall be of sufficient size to meet the required response times. Tubing shall be flushed clean and dry prior to final connections and shall be adequately supported and protected from damage. Tubing passing through concrete or masonry shall be sleeved and protected from abrasion and electrolytic action.

909.13.1 Materials. Control air tubing shall be hard drawn copper, Type L, ACR in accordance with ASTM B 42, ASTM B 43, ASTM B 68, ASTM B 88, ASTM B 251 and ASTM B 280. Fittings shall be wrought copper or brass, solder type, in accordance with ASME B16.18‡ or ASME B16.22. Changes in direction shall be made with appropriate tool bends. Brass compression-type fittings shall be used at final connection to devices; other joints shall be brazed using a BCuP-5 brazing alloy with solidus above 1,100°F (593.3°C) and liquids below 1,500°F (815.6°C). Brazing flux shall be used on copper-to-brass joints only.

Exception: Nonmetallic tubing used within control panels and at the final connection to devices provided all of the following conditions are met:

1. Tubing shall comply with the optical density, flame spread, and the listing & labeling requirements of Section 602.2.1.3 of the *New York City Mechanical Code*.
2. Tubing and connected devices shall be completely enclosed within a galvanized or paint-grade steel enclosure having a minimum thickness of 0.0296 inch (0.7534 mm) (No. 22 gage). Entry to the enclosure shall be by copper tubing with a protective grommet of neoprene or Teflon or by suitable brass compression to male-barbed adapter.
3. Tubing shall be identified by appropriately documented coding.
4. Tubing shall be neatly tied and supported within the enclosure. Tubing bridging cabinets and doors or moveable devices shall be of sufficient length to avoid tension and excessive stress. Tubing shall be protected against abrasion. Tubing serving devices on doors shall be fastened along hinges.

909.13.2 Isolation from other functions. Control tubing serving other than smoke control functions shall be isolated by automatic isolation valves or shall be an independent system.

909.13.3 Testing. Control air tubing shall be tested at three times the operating pressure for not less than 30 minutes without any noticeable loss in gauge pressure prior to final connection to devices.

909.14 Marking and identification. The detection and control systems shall be clearly marked at all junctions, accesses and terminations.

909.15 Control diagrams. Identical control diagrams showing all devices in the system and identifying their location and function shall be maintained current and kept on file with the department, the Fire Department and in the fire command center in a format and manner approved by the Fire Commissioner.

909.16 Firefighter's smoke control panel. A firefighter's smoke control panel for Fire Department emergency response purposes only shall be provided and shall include manual control or override of automatic control for mechanical smoke control systems. The panel shall be located in a fire command center complying with Section 911 in high-rise buildings or buildings with smoke-protected assembly seating. In all other buildings, the firefighter's smoke control panel shall be installed in the ground floor lobby of the building, adjacent to the fire alarm control panel or remote annunciator, or in another approved location. The firefighter's smoke control panel shall either be a separate panel or can be integrated with a UUKL listed fire alarm control panel. The firefighter's smoke control panel shall comply with Sections 909.16.1 through 909.16.3. Where required in Section 917, the post-fire smoke purge system shall be manually activated from the firefighter's control panel or an adjacent panel.

909.16.1 Panel indicators. Fans within the building shall be shown on the firefighter's control panel. A clear indication of the direction of airflow and the relationship of components shall be displayed. Status indicators shall be provided for all smoke control equipment, annunciated by fan and zone, and by pilot-lamp-type indicators as follows:

1. Fans, dampers and other operating equipment in their normal status—WHITE.
2. Fans, dampers and other operating equipment in their on or open status—GREEN.
3. Fans, dampers and other operating equipment in a fault status—YELLOW/AMBER.
4. Fans, dampers and other operating equipment in their off or closed status—RED.

The indicators shall be provided in the following order: WHITE, GREEN, YELLOW/AMBER then RED.

909.16.2 Panel controls. The firefighter's control panel shall provide control capability over the complete smoke control system equipment within the building as follows:

1. ON-AUTO-OFF control over each individual piece of operating smoke control equipment that can also be controlled from other sources within the building. This includes stairway pressurization fans; smoke exhaust fans; supply, return and exhaust fans; elevator shaft fans and other operating equipment used or intended for smoke control purposes.
2. OPEN-AUTO-CLOSE control over dampers relating to smoke control and that are also controlled from other sources within the building. Dampers are permitted to be controlled by individual damper or grouped by smoke zone.
3. ON-OFF or OPEN-CLOSE control over smoke control and other critical equipment associated with a fire or smoke emergency and that can only be controlled from the firefighter's control panel. The firefighter's control panel shall be configured as described in Section 911.

   Exceptions:

   1. Systems, where approved by the commissioner and the Fire Department, where the controls and indicators are combined to control and indicate all elements of a single smoke zone as a unit.
   2. Systems, where approved by the commissioner and the Fire Department, where the control is accomplished by computer interface using approved, plain English commands.

909.16.3 Control action and priorities. The firefighter's control panel actions shall be as follows:

1. ON-OFF and OPEN-CLOSE control actions shall have the highest priority of any control point within the building. Once issued from the firefighter's control panel, automatic or manual control from any other control point within the building shall not contradict the control action. Where automatic means are provided to interrupt normal, nonemergency equipment operation or produce a specific result to safeguard the building or equipment including, but not limited to, duct freezestats, duct smoke detectors, high-temperature cutouts, temperature-actuated linkage and similar devices, such means shall be capable of being overridden by the firefighter's control panel. The last control action as indicated by each firefighter's control panel switch position shall prevail. Control actions shall not require the smoke control system to assume more than one configuration at any one time.

   Exception: Power disconnects required by the *New York City Electrical Code*.
2. Only the AUTO position of each three-position firefighter's control panel switch shall allow automatic or manual control action from other control points within the building. The AUTO position shall be the NORMAL, nonemergency, building control position. Where a firefighter's control panel is in the AUTO position, the actual status of the device (on, off, open, closed) shall continue to be indicated by the status indicator described in Section 909.16.1. Where directed by an automatic signal to assume an emergency condition, the NORMAL position shall become the emergency condition for that device or group of devices within the zone. Control actions shall not require the smoke control system to assume more than one configuration at any one time.

909.17 System response time. Smoke-control system activation shall be initiated immediately after receipt of an appropriate automatic or manual activation command. Smoke control systems shall activate individual components (such as dampers and fans) in the sequence necessary to prevent physical damage to the fans, dampers, ducts and other equipment. For purposes of smoke control, the firefighter's control panel response time shall be the same for automatic or manual smoke control action initiated from any other building control point. The total response time, including that necessary for detection, shutdown of operating equipment and smoke control system startup, shall allow for full operational mode to be achieved before the conditions in the space exceed the design smoke condition. The system response time for each component and their sequential relationships shall be detailed in the required rational analysis and verification of their installed condition reported in the required final report.

909.18 Acceptance testing. Devices, equipment, components and sequences shall be individually tested. These tests, in addition to those required by other provisions of this code, shall consist of determination of function, sequence and, where applicable, capacity of their installed condition.

909.18.1 Detection devices. Smoke or fire detectors that are a part of a smoke control system shall be tested in accordance with Chapter 9 in their installed condition. Where applicable, this testing shall include verification of airflow in both minimum and maximum conditions.

909.18.2 Ducts. Ducts that are part of a smoke control system shall be traversed using generally accepted practices to determine actual air quantities.

909.18.3 Dampers. Dampers shall be tested for function in their installed condition.

909.18.4 Inlets and outlets. Inlets and outlets shall be read using generally accepted practices to determine air quantities.

909.18.5 Fans. Fans shall be examined for correct rotation. Measurements of voltage, amperage, revolutions per minute (rpm) and belt tension shall be made.

909.18.6 Smoke barriers. Measurements using inclined manometers or other approved calibrated measuring devices shall be made of the pressure differences across smoke barriers. Such measurements shall be conducted for each possible smoke control condition.

909.18.7 Controls. Each smoke zone equipped with an automatic-initiation device shall be put into operation by the actuation of one such device. Each additional device within the zone shall be verified to cause the same sequence without requiring the operation of fan motors in order to prevent damage. Control sequences shall be verified throughout the system, including verification of override from the firefighter's control panel and simulation of standby power conditions.

909.18.8 Special inspections for smoke control. Smoke control systems shall be tested by a special inspector in accordance with Chapter 17.

909.18.8.1 Scope of testing. Special inspections shall be conducted in accordance with the following:

1. During erection of ductwork and prior to concealment for the purposes of leakage testing and recording of device location.
2. Prior to occupancy and after sufficient completion for the purposes of pressure-difference testing, flow measurements, and detection and control verification.

909.18.8.2 Qualifications. Special inspectors for smoke control shall have a certification as air balancers and expertise in fire protection engineering or mechanical engineering.

909.18.8.3 Reports. A complete report of testing shall be prepared by the special inspector or approved agency. The report shall include identification of all devices by manufacturer, nameplate data, design values, measured values and identification tag or mark. The report shall be reviewed by the responsible engineer and, when satisfied that the design intent has been achieved, the engineer shall seal, sign and date the report.

909.18.8.3.1 Report filing. A copy of the final report and each inspection report shall be filed with the department and Fire Commissioner, and an identical copy shall be maintained in an approved location at the building.

909.18.9 Identification and documentation. Charts, drawings and other documents identifying and locating each component of the smoke control system, and describing its proper function and maintenance requirements, shall be maintained on file at the building as an attachment to the report required by Section 909.18.8.3. Devices shall have an approved identifying tag or mark on them consistent with the other required documentation and shall be dated indicating the last time they were successfully tested and by whom.

909.18.10 Reacceptance testing. The smoke control system shall require a reacceptance test after any modifications to the system or physical changes to the building that may affect system performance. Reacceptance testing shall be a retest of the entire system in accordance with Sections 909.18.1 through 909.18.9.

909.19 System acceptance. Buildings, or portions thereof, required by this code to comply with this section shall not be issued a certificate of occupancy until such time that the department determines that the provisions of this section have been fully satisfied and a written maintenance program is approved by the New York City Fire Department.

Exception: In buildings of phased construction, the department may issue a temporary certificate of occupancy provided that those portions of the building to be occupied meet the requirements of this section and that the remainder does not pose a significant hazard to the safety of the proposed occupants or adjacent buildings.

909.20 Smokeproof enclosures. Where required by Section 1023.11, a smoke proof enclosure shall be constructed in accordance with this section. Where access to the roof is required by the *New York City Fire Code*, such access shall be from the smoke proof enclosure where a smoke proof enclosure is required. Smokeproof enclosures shall consist of one of the following systems:

1. An enclosed interior exit stairway constructed in accordance with Section 1023 and accessed through an open exterior balcony.
2. An enclosed interior exit stairway constructed in accordance with Section 1023 and accessed through a naturally ventilated vestibule.
3. An enclosed interior exit stairway constructed in accordance with Section 1023 and accessed through a mechanically ventilated vestibule.
4. A pressurized interior exit stairway constructed in accordance with Section 1023.

909.20.1 Access. Access to the interior exit stairway or ramp shall be by way of a vestibule or an open exterior balcony, unless such stairway is pressurized in accordance with Section 909.20.5. The minimum dimension of the vestibule or open exterior balcony shall not be less than the required width of the corridor leading to the vestibule or open exterior balcony but shall not have a width of less than 44 inches (1117.6 mm) and shall not have a length of less than 72 inches (1828.8 mm) in the direction of egress travel.

909.20.2 Construction. The smoke proof enclosure shall be separated from the remainder of the building by not less than 2-hour fire barriers constructed in accordance with Section 707 or horizontal assemblies constructed in accordance with Section 711, or both. Openings are not permitted other than the required means of egress doors. The vestibule shall be separated from the stairway or ramp by not less than a 2-hour fire-resistance-rated fire barrier. The open exterior balcony shall be constructed in accordance with the fire-resistance-rating requirements for floor construction.

909.20.2.1 Door closers. Doors in a smoke proof enclosure shall be self- or automatic-closing by actuation of a smoke detector in accordance with Section 716.5.9.3 and shall be installed at the floor-side entrance to the smoke proof enclosure. The actuation of the smoke detector on any door shall activate the closing devices on all doors in the smoke proof enclosure at all levels. Smoke detectors shall be installed in accordance with Section 907.3.

909.20.3 Natural ventilation alternative. The provisions of Sections 909.20.3.1 through 909.20.3.3 shall apply to ventilation of smoke proof enclosures by natural means.

909.20.3.1 Balcony doors. Where access to the stairway or ramp is by way of an open exterior balcony, the door assembly into the enclosure shall be a fire door assembly in accordance with Section 716.5.

909.20.3.2 Vestibule doors. Where access to the stairway or ramp is by way of a vestibule, the door assembly into the vestibule shall be a fire door assembly complying with Section 716.5. The door assembly from the vestibule to the stairway shall have not less than a 90-minute fire protection rating complying with Section 716.5.

909.20.3.3 Vestibule ventilation. Each vestibule shall have a minimum net area of 16 square feet (1.5 m^2) of opening in a wall facing an outer court, yard or public way that is not less than 20 feet (6096 mm) in width.

909.20.4 Mechanical ventilation alternative. The provisions of Sections 909.20.4.1 through 909.20.4.5 shall apply to ventilation of smoke proof enclosures by mechanical means.

909.20.4.1 Vestibule doors. The door assembly from the building into the vestibule shall be a fire door assembly complying with Section 716.5.3. The door assembly from the vestibule to the stairway or ramp shall not have less than a 90-minute fire protection rating and shall meet the requirements for a smoke door assembly in accordance with Section 716.5.3. The door shall be installed in accordance with NFPA 105.

909.20.4.2 Vestibule ventilation. The vestibule shall be supplied with not less than one air change per minute and the exhaust shall not be less than 150 percent of supply. Supply air shall enter and exhaust air shall discharge from the vestibule through separate, tightly constructed ducts used only for that purpose. Supply air shall enter the vestibule within 6 inches (152.4 mm) of the floor level. The top of the exhaust register shall be located at the top of the smoke trap but not more than 6 inches (152.4 mm) down from the top of the trap, and shall be entirely within the smoke trap area. Doors in the open position shall not obstruct duct openings. Duct openings with controlling dampers are permitted where necessary to meet the design requirements, but dampers are not otherwise required.

909.20.4.2.1 Engineered ventilation system. Where a specially engineered system is used, the system shall exhaust a quantity of air equal to not less than 90 air changes per hour from any vestibule in the emergency operation mode and shall be sized to handle three vestibules simultaneously. Smoke detectors shall be located at the floor-side entrance to each vestibule and shall activate the system for the affected vestibule. Smoke detectors shall be installed in accordance with Section 907.3.

909.20.4.3 Smoke trap. The vestibule ceiling shall not be less than 20 inches (508 mm) higher than the door opening into the vestibule to serve as a smoke and heat trap and to provide an upward-moving air column. The height shall not be decreased unless approved and justified by design and test.

909.20.4.4 Stairway or ramp shaft air movement system. The stairway or ramp shaft shall be provided with a dampered relief opening and supplied with sufficient air to maintain a minimum positive pressure of 0.10 inch of water (24.9 Pa) in the shaft relative to the vestibule with all doors closed. The system shall maintain a maximum of 0.35 inch of water (87.2 Pa) in the shaft relative to the building measured with all stairway doors closed under maximum anticipated stack pressures.

909.20.4.5 Door opening force. Door opening force shall not exceed limits in Section 1010.1.3.

909.20.5 Stairway and ramp pressurization alternative. Where the building is equipped throughout with an automatic sprinkler system in accordance with Section 903.3.1.1, the vestibule is not required, provided each interior exit stairway or ramp is pressurized to not less than 0.10 inch of water (24.9 Pa) and not more than 0.35 inches of water (87.2 Pa) in the shaft relative to the building measured with all interior exit stairway and ramp doors closed under maximum anticipated conditions of stack effect and wind effect.

909.20.6 Ventilating equipment. The activation of ventilating equipment required by the alternatives in Sections 909.20.4 and 909.20.5 shall be by smoke detectors installed at each floor level at an approved location at the entrance to the smokeproof enclosure. When the closing device for the stairway and ramp shaft and vestibule doors is activated by smoke detection or power failure, the mechanical equipment shall activate and operate at the required performance levels. Smoke detectors shall be installed in accordance with Section 907.3.

909.20.6.1 Ventilation systems. Smokeproof enclosure ventilation systems shall be independent of other building ventilation systems. The equipment, control wiring, power wiring and ductwork shall comply with one of the following:

1. Equipment, control wiring, power wiring and ductwork shall be located exterior to the building and directly connected to the smokeproof enclosure or connected to the smokeproof enclosure by ductwork enclosed by not less than 2-hour fire barriers constructed in accordance with Section 707 or horizontal assemblies constructed in accordance with Section 711, or both.
2. Equipment, control wiring, power wiring and ductwork shall be located within the smokeproof enclosure with intake or exhaust directly from and to the outside or through ductwork enclosed by not less than 2-

hour fire barriers constructed in accordance with Section 707 or horizontal assemblies constructed in accordance with Section 711, or both.

3. Equipment, control wiring, power wiring and ductwork shall be located within the building if separated from the remainder of the building, including other mechanical equipment, by not less than 2-hour fire barriers constructed in accordance with Section 707 or horizontal assemblies constructed in accordance with Section 711, or both.

Exceptions:

1. Control wiring and power wiring utilizing a 2-hour rated cable or cable system in accordance with UL 2196.
2. Where encased with not less than 2 inches (50.8 mm) of concrete.
3. Control wiring and power wiring protected by a listed electrical circuit protective system with a fire-resistance rating of not less than 2 hours.

909.20.6.2 Standby power. Mechanical vestibule and stairway and ramp shaft ventilation systems and automatic fire detection systems shall be provided with standby power in accordance with Sections 403.4.8 and 2702.

909.20.6.3 Acceptance and testing. Before the mechanical equipment is approved, the system shall be tested in the presence of the department or approved agency to confirm that the system is operating in compliance with these requirements.

909.21 Elevator hoistway pressurization alternative. Where elevator hoistway pressurization is provided in lieu of required enclosed elevator lobbies, the pressurization system shall comply with Sections 909.21.1 through 909.21.11.

909.21.1 Pressurization requirements. Elevator hoistways shall be pressurized to maintain a minimum positive pressure of 0.10 inch of water (24.9 Pa) and a maximum positive pressure of 0.25 inch of water (62.2 Pa) with respect to adjacent occupied space on all floors. This pressure shall be measured at the midpoint of each hoistway door, with all elevator cars at the floor of recall and all hoistway doors on the floor of recall open and all other hoistway doors closed. The pressure differentials shall be measured between the hoistway and the adjacent elevator landing. The opening and closing of hoistway doors at each level must be demonstrated during this test. The supply air intake shall be from an outside, uncontaminated source located a minimum distance of 20 feet (6096 mm) from any air exhaust system or outlet.

Exceptions:

1. On floors containing only Group R occupancies, the pressure differential is permitted to be measured between the hoistway and a dwelling unit or sleeping unit.
2. Where an elevator opens into a lobby enclosed in accordance with Section 3007.6 or 3008.7, the pressure differential is permitted to be measured between the hoistway and the floor space immediately outside of the door(s) to the enclosed lobby.
3. The pressure differential is permitted to be measured relative to the outdoor atmosphere on floors other than the following:
   3.1. The fire floor.
   3.2. The two floors immediately below the fire floor.
   3.3. The floor immediately above the fire floor.
4. The minimum positive pressure of 0.10 inch of water (24.9 Pa) and a maximum positive pressure of 0.25 inch of water (62.2 Pa) with respect to occupied floors are not required at the floors of recall with the doors open.

909.21.1.1 Use of ventilation systems. Ventilation systems, other than hoistway supply air systems, are permitted to be used to exhaust air from adjacent spaces on the fire floor, two floors immediately below and one floor immediately above the fire floor to the building's exterior where necessary to maintain positive pressure relationships as required in Section 909.21.1 during operation of the elevator shaft pressurization system.

Where ventilation systems are being used as a component of the elevator hoistway pressurization system, they shall comply with Section 909.21.

909.21.2 Rational analysis. A rational analysis complying with Section 909.4 shall be submitted with the construction documents.

909.21.3 Ducts for system. Any duct system that is part of the pressurization system shall be protected with the same fire-resistance rating as required for the elevator shaft enclosure.

909.21.4 Fan system. The fan system provided for the pressurization system shall be as required by Sections 909.21.4.1 through 909.21.4.4.

909.21.4.1 Fire resistance. Where located within the building, the fan system that provides the pressurization shall be protected with the same fire-resistance rating required for the elevator shaft enclosure.

909.21.4.2 Smoke detection. The supply fan system shall be equipped with a smoke detector that will automatically shut down the supply fan system when smoke is detected within the system.

909.21.4.3 Separate systems. A separate fan system shall be used for each elevator hoistway.

909.21.4.4 Fan capacity. The supply fan capacity shall be specified by a registered design professional to meet the requirements of a designed pressurization system in accordance with the rational analysis required by Section 909.21.2.

909.21.5 Standby power. The pressurization system, including ventilation systems used per Section 909.21.1.1, shall be provided with standby power in accordance with Section 2702.

909.21.6 Activation of pressurization system. The elevator pressurization system shall be activated upon activation of either the building fire alarm system or the elevator landing smoke detectors. Where both a building fire alarm system and elevator landing smoke detectors are present, each shall be independently capable of activating the pressurization system.

909.21.7 Testing and special inspections. Special inspections for performance shall be required in accordance with Section 909.18.8. System acceptance testing shall be in accordance with Section 909.19.

909.21.8 Marking and identification. Detection and control systems shall be marked in accordance with Section 909.14.

909.21.9 Control diagrams. Control diagrams shall be provided in accordance with Section 909.15.

909.21.10 Firefighter's smoke control panel. A control panel complying with Section 909.16 shall be provided.

909.21.11 System response time. Hoistway pressurization systems shall comply with the requirements for smoke control system response time in Section 909.17.

SECTION BC 910
SMOKE AND HEAT REMOVAL

910.1 General. Where required by this code, smoke and heat vents or mechanical smoke removal systems shall conform to the requirements of this section.

910.2 Where required. Smoke and heat vents or a mechanical smoke removal system shall be installed as required by Sections 910.2.1 and 910.2.2.

Exceptions:

1. Frozen-food warehouses used solely for storage of Class I and II commodities where protected by an automatic sprinkler system in accordance with Section 903.3.1.1.

2. Smoke and heat removal shall not be required in areas of buildings equipped with early suppression fast-response (ESFR) sprinklers.

3. Smoke and heat removal shall not be required in areas of buildings equipped with control mode special application sprinklers with a response time index of 50 $(m \cdot s)^{1/2}$ or less that are listed to control a fire in stored commodities with 12 or fewer sprinklers.

910.2.1 Group F-1 or S-1. Smoke and heat vents installed in accordance with Section 910.3 or a mechanical smoke removal system installed in accordance with Section 910.4 shall be installed in buildings and portions thereof used as a Group F-1 or S-1 occupancy having more than 50,000 square feet (4645.2 m^2) of undivided area. In occupied portions of a building equipped throughout with an automatic sprinkler system in accordance with Section 903.3.1.1, where the upper surface of the story is not a roof assembly, a mechanical smoke removal system in accordance with Section 910.4 shall be installed.

Exception: Group S-1 aircraft repair hangars.

910.2.2 High-piled combustible storage. Smoke and heat removal for buildings and portions thereof containing high-piled combustible storage shall be installed in accordance with the *New York City Fire Code* and Section 413 of this code. Installation shall also be in conformance with Section 910.3 in unsprinklered buildings and portions thereof.

In buildings and portions thereof containing high-piled combustible storage equipped throughout with an automatic sprinkler system in accordance with Section 903.3.1.1, a smoke and heat removal system shall be installed in accordance with Section 910.3 or 910.4. In occupied portions of a building equipped throughout with an automatic sprinkler system in accordance with Section 903.3.1.1, where the upper surface of the story is not a roof assembly, a mechanical smoke removal system in accordance with Section 910.4 shall be installed.

910.3 Smoke and heat vents. The design and installation of smoke and heat vents shall be in accordance with Sections 910.3.1 through 910.3.3.

910.3.1 Listing and labeling. Smoke and heat vents shall be listed and labeled to indicate compliance with UL 793 or FM 4430.

910.3.2 Smoke and heat vent locations. Smoke and heat vents shall be located 20 feet (6096 mm) or more from adjacent lot lines and fire walls and 10 feet (3048 mm) or more from fire barriers. Vents shall be uniformly located within the roof in the areas of the building where the vents are required to be installed by Section 910.2 with consideration given to roof pitch, sprinkler location and structural members.

910.3.3 Smoke and heat vents area. The required aggregate area of smoke and heat vents shall be calculated as follows:

For buildings equipped throughout with an automatic sprinkler system in accordance with Section 903.3.1.1:

A_{VR} = V/9000 **(Equation 9-2)**

where:

A_{VR} = The required aggregate vent area (ft^2).

V = Volume (ft^3) of the area that requires smoke removal.

For unsprinklered buildings:

$A_{VR} = A_{FA}/50$ **(Equation 9-3)**

where:

A_{VR} = The required aggregate vent area (ft^2).

A_{FA} = The area of the floor in the area that requires smoke removal.

‡‡‡ **910.4 Mechanical smoke removal systems.** Mechanical smoke removal systems shall be designed and installed in accordance with Sections 910.4.1 through 910.4.7.

910.4.1 Automatic sprinklers required. The building shall be equipped throughout with an approved automatic sprinkler system in accordance with Section 903.3.1.1.

910.4.2 Exhaust fan construction. Exhaust fans that are part of a mechanical smoke removal system shall be rated for operation at 221°F (105°C). Exhaust fan motors shall be located outside of the exhaust fan air stream.

910.4.3 System design criteria. The mechanical smoke removal system shall be sized to exhaust the building at a minimum rate of two air changes per hour based upon the volume of the building or portion thereof without contents. The capacity of each exhaust fan shall not exceed 30,000 cubic feet per minute (14.2 m^3/sec).

910.4.3.1 Makeup air. No portion of the makeup air openings shall be higher than 6 feet (1828.8 mm) above the floor level. Operation of makeup air openings shall be manual or, if properly coordinated with the smoke removal system, automatic. The minimum gross area of makeup air inlets shall be 8 square feet per 1,000 cubic feet per minute (0.74 m^2 per 0.4719 m^3/s) of smoke exhaust.

910.4.4 Activation. The mechanical smoke removal system shall be started by manual controls.

910.4.5 Manual control location. Manual controls shall be located so as to be accessible to the Fire Department from an exterior door of the building and protected against interior fire exposure by not less than 1-hour fire barriers constructed in accordance with Section 707 or horizontal assemblies constructed in accordance with Section 711, or both. The location of manual controls is subject to the approval of the Fire Commissioner.

910.4.6 Control wiring. Wiring for operation and control of mechanical smoke removal systems shall be connected ahead of the main disconnect in accordance with the *New York City Electrical Code* and be protected against interior fire exposure to temperatures in excess of 1,000°F (537.8°C) for a period of not less than 15 minutes.

910.4.7 Controls. Where building air-handling and mechanical smoke removal systems are combined or where independent building air-handling systems are provided, fans shall automatically shut down in accordance with the *New York City Mechanical Code*. The manual controls provided for the smoke removal system shall have the capability to override the automatic shutdown of fans that are part of the smoke removal system.

910.5 Maintenance. Smoke and heat vents and mechanical smoke removal systems shall be maintained in accordance with the *New York City Fire Code*.

SECTION BC 911
FIRE COMMAND CENTER

911.1 General. Where required by other sections of this code and in buildings classified as high-rise buildings by this code, a fire command center for Fire Department operations shall be provided and shall comply with Sections 911.1.1 through 911.1.6.

911.1.1 Location and access. The fire command center location shall be in the lobby of the building on the main entrance floor near the Fire Department designated response point.

911.1.2 Reserved.

911.1.3 Reserved.

911.1.4 Reserved.

911.1.5 Storage. Storage unrelated to operation of the fire command center shall be prohibited.

911.1.6 Required features. The fire command center shall comply with the *New York City Fire Code* and NFPA 72 and shall include the following features as applicable in their respective control units or panels:

1. Fire alarm control unit:
    - 1.1. The emergency voice/alarm communication system control unit.
    - 1.2. The two-way voice‡ communications system.
    - 1.3. Fire detection and alarm system annunciator.
    - 1.4. Controls for unlocking stairway doors and locked elevator vestibule doors simultaneously.
    - 1.5. Sprinkler valve and water-flow detector display panels.
    - 1.6. Emergency and standby power status indicators (generator running, generator failure to start).
    - 1.7. Fire pump status indicators.
    - 1.8. Elevator fire recall switch in accordance with ASME A17.1.
2. Auxiliary Radio Communication System Panel.
3. Elevator control panel:

3.1. Annunciator visually indicating the location of the elevators and whether they are operational.

3.2. Elevator emergency or standby power selector switch(es), where emergency or standby power is provided.

4. The firefighter's control panel required by Section 909.16 for smoke control systems installed in the building.
5. Monitoring/Control for Post-Fire Smoke Purge:

5.1 Status indicators and controls for air distribution systems.

5.2 Manual controls of post-fire smoke purge system in accordance with Section 917.2.3.

6. A telephone for Fire Department use with controlled access to the public telephone system.
7. Public address system, where specifically required by other sections of this code.
8. Manual controls for the release of doors that are automatic-closing by the actuation of smoke detectors or activation of the fire alarm in accordance with the exception to Section 713.7.

911.2 Secondary fire command center. Where required in locations described in Appendix G of this code, a secondary fire command center shall be provided subject to the approval of the Fire Department. Design and installation requirements shall be in accordance with NFPA 72.

SECTION BC 912
FIRE DEPARTMENT CONNECTIONS

912.1 Installation. Fire Department connections shall be installed in accordance with the NFPA standard applicable to the system design and shall comply with Sections 912.2 through 912.6.

912.2 Location. With respect to hydrants, driveways, buildings and landscaping, Fire Department connections shall be so located that fire apparatus and hose connected to supply the system will not obstruct access to the buildings for other fire apparatus.

912.2.1 Fire Department connections. The location of Fire Department connections shall be as follows:

912.2.1.1 Facing each street or public space.‡ One Fire Department connection shall be provided for each 300 feet (91 140 mm) of exterior building wall or fraction thereof facing upon each street or public space.

912.2.1.2 Facing two parallel streets or public spaces. ‡ Where buildings face upon two parallel streets or public spaces without an intersecting street or public space, one connection shall be provided for each 300 feet (91 140 mm) of exterior building wall or fraction thereof facing upon each such parallel street or public space.

912.2.1.3 Facing two intersecting streets or public spaces. ‡ Where a building faces upon two intersecting streets or public spaces and the total length of the exterior building walls facing upon such streets or public spaces does not exceed 300 feet (91 140 mm), only one Fire Department connection need be installed provided the Fire Department connection is located within 15 feet (4572 mm) of the corner and on the street with the longest building frontage.

912.2.1.4 Facing three streets or public spaces. ‡ Where a building faces on three streets or public spaces, one Fire Department connection shall be provided for each 300 feet (91 140 mm) of building wall or fraction thereof facing upon such streets or public spaces provided that at least one Fire Department connection is installed on each of the parallel streets or public spaces, and further provided that the Fire Department connections shall be located so that the distance between them does not exceed 300 feet (91 140 mm).

912.2.1.5 Facing four streets or public spaces. ‡ Where a building faces upon four streets or public spaces, at least one Fire Department connection shall be provided on each street front or public space; however, only one Fire Department connection need be provided at the corner of two intersecting streets or public spaces if the Fire Department connection is located within 15 feet (4572 mm) of the corner and on the street with the longest building frontage or public space, and if the distances between Fire Department connections, in all cases, do not exceed 300 feet (91 140 mm).

912.2.1.6 Obstruction by another building. ‡ In any case where the exterior building walls of a building facing a street or public space are obstructed in part by another building, one Fire Department connection shall be provided for each clear 300 feet (91 140 mm) of exterior building wall or fraction thereof facing upon such street or public space.

912.2.2 Existing buildings. On existing buildings, wherever the Fire Department connection is not visible to approaching fire apparatus, the Fire Department connection shall be indicated by an approved sign mounted on the street front or on the side of the building. Such sign shall have the letters "FDC" not less than 6 inches (152.4 mm) high and words in letters not less than 2 inches (50.8 mm) high or an arrow to indicate the location. Such signs shall be subject to the approval of the Fire Department.

912.3 Fire hose threads. Fire hose threads used in connection with standpipe, sprinkler and combination standpipe-sprinkler systems shall be approved and shall be compatible with New York City Fire Department hose threads.

912.4 Access. Immediate access to Fire Department connections shall be maintained at all times and without obstruction by fences, bushes, trees, walls or any other object.

Exception: Fences, where provided with an access gate and a means of emergency operation that shall be maintained operational at all times in accordance with the *New York City Fire Code*.

912.4.1 Locking Fire Department connection caps. The Fire Department may require locking caps on Fire Department connections in accordance with the *New York City Fire Code*.

912.4.2 Clear space around connections. A working space of not less than 36 inches (914.4 mm) in width, 36 inches (914.4 mm) in depth and 78 inches (1981.2 mm) in height shall be provided and maintained in front of and to the sides of wall-mounted Fire Department connections and around the circumference of free-standing Fire Department connections, except as otherwise required or approved by the Fire Commissioner.

912.4.3 Physical protection. Where Fire Department connections are subject to impact by a motor vehicle, vehicle impact protection shall be provided in accordance with the *New York City Fire Code*.

912.5 Signs. Fire Department connections shall be provided with signage in accordance with Section 912 of the *New York City Fire Code*.

912.6 Backflow protection. The potable water supply to automatic sprinkler and standpipe systems shall be protected against backflow as required by the *New York City Plumbing Code*.

SECTION BC 913
FIRE PUMPS

913.1 General. Where provided or required, fire pumps shall be installed in accordance with this section, NFPA 20 and other applicable sections of this code.

913.2 Protection against interruption of service. The fire pump, driver and controller shall be protected in accordance with NFPA 20 against possible interruption of service.

913.2.1 Protection of fire pump rooms. Fire pumps shall be located in rooms that are separated from all other areas of the building by 2-hour fire barriers constructed in accordance with Section 707 or 2-hour horizontal assemblies constructed in accordance with Section 711, or both.

Exceptions:

1. In other than high-rise buildings, separation by 1-hour fire barriers constructed in accordance with Section 707 or 1-hour horizontal assemblies constructed in accordance with Section 711, or both, shall be permitted in buildings equipped throughout with an automatic sprinkler system in accordance with Section 903.3.1.1 or 903.3.1.2.
2. Separation is not required for fire pumps physically separated in accordance with NFPA 20.
3. Separation is not required for a fire pump, other than an automatic standpipe fire pump, where such fire pump is located in a mechanical equipment room, as defined by the *New York City Mechanical Code*, enclosed by 2-hour fire barriers constructed in accordance with Section 707 or 2-hour horizontal assemblies constructed in accordance with Section 711, or both. Refrigerants, gas piping, gas consumption

devices, gas meters or any other gas equipment and fuel storage or fuel consuming appliances shall not be installed in any space housing a fire pump.

913.2.2 Circuits supplying fire pumps. Cables used for survivability of circuits supplying fire pumps shall be in accordance with the *New York City Electrical Code*. Electrical circuit protective systems shall be installed in accordance with their listing requirements, and the *New York City Electrical Code*.

913.3 Temperature of pump room. Suitable means shall be provided for maintaining the temperature of a pump room or pump house, where required, above 40°F (4.4°C).

913.3.1 Engine manufacturer's recommendation. Temperature of the pump room, pump house or area where engines are installed shall never be less than the minimum recommended by the engine manufacturer. The engine manufacturer's recommendations for oil heaters shall be followed.

913.4 Valve supervision. Where provided, the fire pump suction, discharge and bypass valves, and isolation valves on the backflow prevention device or assembly shall be supervised open as required by Section 907.

913.4.1 Test outlet control valve supervision. Fire pump test outlet control valves shall be supervised in the closed position. Individual hose valves on the test header are not required to be monitored.

‡‡‡ **913.5 Acceptance test.** Acceptance testing shall be done in accordance with the requirements of Section 1705.30 of this code, the *New York City Fire Code* and NFPA 20. Refurbished or repaired fire pumps shall be tested in accordance with Section 1705.30 of this code, the *New York City Fire Code* and NFPA 20. A notification shall be given to the department prior to performance of the test.

SECTION BC 914
EMERGENCY RESPONDER SAFETY FEATURES

914.1 Shaftway markings. Vertical shafts shall be identified as required by the *New York City Fire Code* and Sections 914.1.1 and 914.1.2 of this code.

914.1.1 Exterior access to shaftways. Outside openings accessible to the Fire Department and that open directly on a hoistway or shaftway communicating between two or more floors in a building shall be plainly marked with the word "SHAFTWAY" in red letters not less than 6 inches (152.4 mm) high on a white background. Such warning signs shall be placed so as to be readily discernible from the outside of the building.

914.1.2 Interior access to shaftways. Door or window openings to a hoistway or shaftway from the interior of the building shall be plainly marked with the word "SHAFTWAY" in red letters not less than 6 inches (152.4 mm) high on a white background. Such warning signs shall be placed so as to be readily discernible.

Exception: Markings shall not be required on shaftway openings that are readily discernible as openings onto a shaftway by the construction or arrangement.

914.2 Equipment room identification. Fire protection equipment shall be identified in an approved manner. Rooms containing controls for air-conditioning systems, sprinkler risers and valves or other fire detection, suppression or control elements shall be identified for the use of the Fire Department. Approved signs required to identify fire protection equipment and equipment location shall be constructed of durable materials, permanently installed and readily visible.

SECTION BC 915
CARBON MONOXIDE DETECTION

915.1 Carbon monoxide alarms and detectors. Carbon monoxide alarms and detectors shall be provided and installed in accordance with Sections 915.1.1 through 915.7.

915.1.1 Group I-1 and R occupancies. Listed carbon monoxide alarms or detectors shall be installed as follows:

1. Group I-1, Group R-1, and Group R-2 where the main use or dominant occupancy of a building is classified as Group R-2 student apartments. Carbon monoxide detectors and audible notification appliances shall be

installed in affected dwelling units in accordance with Section 915.1.1.1 and shall be annunciated by dwelling unit at a constantly attended location from which the fire alarm system is capable of being manually activated.

2. Groups R-2 (other than occupancies covered by Item 1) and R-3. Carbon monoxide alarms shall be installed in affected dwelling units in accordance with Section 915.1.1.1.

915.1.1.1 Affected dwelling units. Carbon monoxide alarms or detectors shall be required within the following dwelling units:

1. Units on the same story where carbon monoxide-producing equipment or enclosed parking is located.
2. Units on the stories immediately above and below the floor where carbon monoxide-producing equipment or enclosed parking is located.
3. Units in a building containing a carbon monoxide-producing furnace, boiler, or water heater as part of a central system.
4. Units in a building served by a carbon monoxide-producing furnace, boiler, or water heater as part of a central system that is located in an adjoining or attached building.

915.1.1.1.1 Required locations within dwelling units. Carbon monoxide alarms or detectors shall be located within dwelling units as follows:

1. Outside of any room used for sleeping purposes, within 15 feet (4572 mm) of the entrance to such room.
2. In any room used for sleeping purposes.
3. On any story within a dwelling unit, including below-grade stories and penthouses of any area, but not including crawl spaces and uninhabitable attics.

‡‡‡ **915.1.1.1.2 Exhaust of carbon monoxide in Group R-3 occupancy (one- and two-family dwellings and townhouses).** Means of exhausting carbon monoxide from garages shall be provided when a carbon monoxide alarm or detector is activated in a Group R-3 occupancy, provided such garage is attached within the Group R-3 occupancy. Such exhaust system shall be arranged to operate automatically upon detection of a concentration of carbon monoxide of 35 parts per million (ppm) or greater by approved automatic detection device. The system shall be capable of producing an exhaust rate of 1.5 cfm per square foot of floor area of the garage. Removal of sensor, interruption of power or cut wires shall cause the relay circuit to open and start fan. The relay contact shall close and the fan may shut off when the carbon monoxide level is below 35 ppm. Carbon monoxide exhausting means shall be connected to a separate circuit and provided with a lock and identified at the power source. Such circuit shall not be connected to a power source through an arc-fault or Ground Fault Circuit Interrupter (GFCI) devices. Additionally, when the carbon monoxide exhausting means is connected to the plug-in-type overcurrent protection device, such device shall be secured in place by an additional fastener.

915.1.2 Buildings that are equipped with a fire alarm system and that contain Group A-1, A-2, A-3, Group B or Group M occupancies. Listed carbon monoxide detectors shall be installed in buildings that are equipped with a fire alarm system and that contain Group A-1, A-2 or A-3, Group B or Group M occupancies. Such carbon monoxide detectors installed pursuant to this section shall have built-in sounder bases, shall transmit a carbon monoxide alarm signal to a central supervising station and shall be permitted to initiate an audible and visual supervisory signal at a constantly attended location. The department shall adopt rules and/or reference standards governing the installation and location of carbon monoxide detectors provided such detectors shall be required within rooms containing carbon monoxide-producing equipment and addressing the installation of such detectors or any alternative means of compliance in existing buildings.

915.1.2.1 Retroactive provisions for existing buildings. Notwithstanding any other provision of law, listed carbon monoxide detectors shall be installed in existing buildings that are equipped with a fire alarm system and that contain group A-1, A-2, A-3, Group B or Group M occupancies in accordance with Section 915.1.2 by July 1, 2021.

915.2 Group E, I-2 and I-4 occupancies. Listed carbon monoxide detectors with built-in sounder bases shall transmit a carbon monoxide alarm signal to a central supervising station and shall be permitted to initiate an audible and visual supervisory signal at a constantly attended location.

1. Carbon monoxide detectors with built-in sounder bases shall be installed within any room containing carbon monoxide-producing equipment.
2. Carbon monoxide detectors with built-in sounder bases shall be installed in corridors on the story where carbon monoxide-producing equipment unit is located, as well as one story above and one story below.
3. Carbon monoxide detectors with built-in sounder bases shall be installed in all corridors on the story where enclosed parking or a loading dock is located, as well as one story above and one story below.

915.3 Installation requirement for all occupancies. Where a fire alarm system is required or provided, carbon monoxide detectors with sounder bases shall be installed within any room containing carbon monoxide-producing equipment.

915.3.1 Equipment shutdown. Activation of a carbon monoxide detector located at the source of carbon monoxide-producing equipment shall shut down that source.

Exception: This provision does not apply where the source is a generator.

915.4 Detection equipment. Carbon monoxide detection required by Sections 915.1 through 915.2 shall be provided by carbon monoxide alarms complying with Section 915.5 or carbon monoxide detection systems complying with Section 915.6, unless otherwise specified in this code.

915.5 Carbon monoxide alarms. Carbon monoxide alarms shall comply with Sections 915.5.1 through 915.5.4.

915.5.1 Power source. Carbon monoxide alarms shall receive their primary power from the building wiring where such wiring is served from a commercial source, and when primary power is interrupted, shall receive power from a battery. Wiring shall be permanent and without a disconnecting switch other than that required for overcurrent protection.

Exception: Where installed in buildings without commercial power, battery-powered carbon monoxide alarms shall be an acceptable alternative.

915.5.2 Listings. Carbon monoxide alarms shall be listed in accordance with UL 2034.

915.5.3 Combination alarms. Combination carbon monoxide/smoke alarms shall be an acceptable alternative to carbon monoxide alarms. Combination carbon monoxide/smoke alarms shall be listed in accordance with UL 2034 and UL 217.

915.5.4 Interconnection of alarms. When two or more alarms are installed, they shall be interconnected in accordance with NFPA 720.

915.6 Carbon monoxide detection systems. Carbon monoxide detection systems shall be an acceptable alternative to carbon monoxide alarms and shall comply with Sections 915.6.1 through 915.6.4.

915.6.1 General. Carbon monoxide detection systems shall comply with NFPA 720. Carbon monoxide detectors shall be listed in accordance with UL 2075.

915.6.2 Locations. Carbon monoxide detectors shall be installed in the locations specified in Section 915.2. These locations supersede the locations specified in NFPA 720.

915.6.3 Combination detectors. Combination carbon monoxide/smoke detectors installed in carbon monoxide detection systems shall be an acceptable alternative to carbon monoxide detectors, provided they are listed in accordance with UL 2075 and UL 268.

915.6.4 Wiring and power supply. Wiring and power supply for carbon monoxide detectors shall comply with Sections 907.6.1 and 907.6.2, respectively.

915.7 Inspection, testing and maintenance. Carbon monoxide alarms and carbon monoxide detection systems shall be inspected, tested and maintained in accordance with NFPA 720 and the *New York City Fire Code*.

SECTION BC 916
FIRE DEPARTMENT IN-BUILDING AUXILIARY RADIO COMMUNICATION SYSTEM (ARCS)

916.1 General. This section covers the design, installation and performance criteria of Fire Department In-Building Auxiliary Radio Communication System (ARCS). Such systems shall be designed and installed by the *New York City Fire Code*, Sections 403 and 907 of this code, or where installed voluntarily, such systems shall be designed and installed in accordance with this section, NFPA 72, the *New York City Electrical Code* and in accordance with Fire Department requirements.

916.1.1 Construction documents. Construction documents for ARCS shall be submitted for approval to the Fire Department prior to system installation. Construction documents shall include, but need not be limited to, all of the following:

1. Type of radio equipment and antenna.
2. Riser diagram and floor plans showing location of elements of the ARCS, including but not limited to building fire command center or fire alarm control panel, dedicated radio console, base station and all other critical system components such as antennas, amplifiers, cables as applicable.
3. Legend of all ARCS symbols and abbreviations used.
4. Location of primary and secondary power source.
5. Specification and listing details for all equipment, devices and cables.

916.1.2 Acceptance testing, maintenance and operational testing. Acceptance testing, maintenance and operational testing of the ARCS shall be performed in accordance with the *New York City Fire Code* and rules promulgated by the Fire Department.

916.2 Instructions. Operating, testing and maintenance instructions and record drawings ("as-builts") and detailed specifications of all the components shall be provided at an approved location.

916.3 Where required. ARCS, which shall be in accordance with this section, shall be required in the following:

1. High-rise buildings constructed in accordance with Section 403.
2. Underground buildings constructed in accordance with Section 405.
3. Buildings having a total gross area exceeding 250,000 square feet (23 225.8 m^2).

Exceptions:

1. Group R-2 buildings that meet all of the following requirements:
   a. The highest occupied floor is less than 125 feet (38 100 mm) above the lowest level of Fire Department vehicle access;
   b. The building has no more than 1 story below grade; and
   c. The floor area of the building does not exceed 250,000 square feet (23 225.8 m^2).
2. Where it is determined by the Fire Department that a radio communication system is not required.

SECTION BC 917
POST-FIRE SMOKE PURGE SYSTEMS

917.1 Scope and purpose. The purpose of this section is to establish minimum requirements for the design and installation of post-fire smoke purge systems, which are intended for the timely restoration of operations and overhaul activities once a fire is extinguished. Post-fire smoke purge systems are not intended or designed as life safety systems and are not required to meet the provisions of Section 909. Post-fire smoke purge systems shall be required in:

1. High-rise buildings subject to Section 403.
2. Buildings with any story exceeding 50,000 square feet (4645.2 m^2) in floor area.

3. Building with spaces exceeding 100 feet (30 480 mm) from natural ventilation openings. Natural ventilation openings shall consist of operable windows and doors of at least 5 percent of the floor area or roof vents per Section 910.
4. High-piled stock or rack storage in accordance with the *New York City Fire Code*.

Exceptions: A post-fire smoke purge system is not required in Group R-2 occupancies where either of the following conditions exists:

1. **Openable windows.** A post-fire smoke purge system is not required where every habitable room located in dwelling units is provided with windows complying with Chapter 12 and all of the following:

   1.1. **Minimum window area.** Each required window shall provide at least 12 square feet (1.1 m^2) of glazed area. The total area of all such windows shall not be less than 10 percent of the floor area of the room or space served.

   1.2. **Minimum openable area.** Each required window shall provide a minimum of 6 square feet (0.56 m^2) of openable area. The total area of all such openings shall not be less than 5 percent of the floor area of the room or space served. In addition, each required openable area shall be:

      1.2.1. Located wholly at least 30 inches (762 mm) above the finished floor; and

      1.2.2. Fully openable to the minimum 6 square feet (0.56 m^2), at all times and without limiting stops or devices. Such openings may be achieved through the use of double-hung, sliding, or similar types of windows. However, in the event of the use of casement-, hopper-, pivot-, or awning-type windows, such windows shall satisfy the requirements of this section only when they open to at least 75 degrees (1.22 rad).

   1.3 **Window guards.** This exception shall not apply where the type of window guards installed in compliance with Section 27-2043.1 of the *Administrative Code* and Section 131.15 of the *New York City Health Code* requires the installation of limiting devices or stops.

2. **Smokeproof enclosures.** A post-fire smoke purge system is not required where all exits are constructed as smokeproof enclosures in accordance with Section 1023.11.

917.2 Post-fire smoke purge systems in occupancy groups other than R-2. Post-fire smoke purge systems in occupancy groups other than R-2 shall comply with Sections 917.2.1 through 917.2.4.‡

917.2.1 General design requirements. Post-fire smoke purge systems are permitted to use dedicated equipment, the normal building HVAC system or other openings and shall have the capability to exhaust smoke from occupied spaces. Smoke removal may be by either mechanical or natural ventilation, but shall be capable of removing cold smoke. Smoke removed from a space must be discharged to a safe location outside the building and shall not be recirculated into the building in accordance with the *New York City Mechanical Code*.

917.2.2 Exhaust capability. The system shall have an air supply and smoke exhaust capability that will provide a minimum of 6 air changes per hour or 1 cubic foot per minute per square foot (cfm/ft^2) (0.00508 $m^3/s \bullet m^2$), whichever is greater. The system need not exhaust from all areas at the same time, but is permitted to be zoned based on the largest fire area served. For the purpose of calculating system size, the height of a compartment shall be considered to run from slab to slab and include the volume above suspended ceilings. Provisions shall be made for sufficient make-up air. The provisions may include operable windows, doors, building leakage, or mechanical systems. In buildings having occupied floors located less than 75 feet (22 860 mm) above the lowest level of Fire Department vehicle access, breakable windows may be utilized.

917.2.3 Operation. The post-fire smoke purge system shall be operated by manual controls that are part of the fire command center, in accordance with Section 911, or fire alarm control unit when a fire command center is not required. Such control center or panel shall display a graphic indicating the portions of the building served by each post-fire smoke purge system. When a system is zoned into areas of operation less than the entire building, each zone shall have an individual control. Fire Department manual controls of post-fire smoke purge systems shall not override the manual or automatic operation of the smoke control system. Such Fire Department manual controls shall override the fire shutdown signal from the fire alarm system.

917.2.4 Interior exit stairways or ramps or exit passageways in occupancies other than Group R-2. Interior exit stairways or ramps or exit passageways shall not be used as a portion of the post-fire smoke purge system in occupancies other than Group R-2. Doors in interior exit stairways or ramps or exit passageways shall not be permitted to be used as a portion of the post-fire smoke purge system. Air transfer and duct openings associated with the post-fire smoke purge system shall not be permitted in the interior exit stairway or ramp or exit passageway.

917.3 Post-fire smoke purge systems in occupancy Group R-2. Post-fire smoke purge systems in Group R-2 occupancies shall comply with either Section 917.3.1 or 917.3.2. Smoke removed must be discharged to a safe location outside the building and shall not be recirculated into the building in accordance with the *New York City Mechanical Code*.

917.3.1 Stair ventilation. The top of all enclosed exit stairs shall be provided with a reversible fan system capable of introducing fresh air or exhausting air at a rate of 6 air changes per hour or 1 cubic foot per minute per square foot (cfm/ft^2) (0.00508 m^3/s • m^2), whichever is greater, based on the area of the largest floor. Such system shall be operated by manual controls that are part of the fire command center, as per Section 911, or fire alarm panel when a fire command center is not required. Such control center or panel shall display a graphic indicating the portions of the building served by each post-fire smoke purge system. The operation of such system shall be controlled by Fire Department personnel by manually opening stair doors at the appropriate story.

917.3.2 Corridor ventilation. The ducts and fans that provide fresh air supply to the public corridors in accordance with the *New York City Mechanical Code* shall be provided with reversible fans and duct sizes capable of introducing fresh air to or exhausting air from the corridor at a rate of 6 air changes per hour or 1 cubic foot per minute per square foot (cfm/ft^2) (0.00508 m^3/s • m^2), whichever is greater, based on the area of the largest apartment plus the area of the public corridor. Such system shall be operated by manual controls that are part of the fire command center, as per Section 911 of this code, or fire alarm panel when a fire command center is not required. Each floor to be ventilated shall be by individual controls. Such control center or panel shall display a graphic indicating the portions of the building served by each postfire smoke purge system.

917.4 Maintenance. The building owner shall maintain post-fire smoke exhaust systems in good operational condition. Records of testing shall be maintained on the premises for inspection by the department and Fire Department personnel.

SECTION BC 918
GAS DETECTION SYSTEMS

918.1 Gas detection systems. Gas detection systems, including systems designed to detect flammable, toxic, asphyxiants and other gases, required by the *New York City Construction Codes* or the *New York City Fire Code*, shall comply with Sections 918.2 through 918.10 of this code.

918.2 Design and installation. Gas detection systems shall be designed and installed in accordance with this code, the *New York City Electrical Code*, and manufacturer's instructions. Such systems shall be inspected, tested and maintained in accordance with the *New York City Fire Code*.

918.2.1 Construction documents. Construction documents for gas detection systems shall be submitted for review and approval to the Fire Department prior to system installation. Construction documents shall include, but not be limited to, the following:

1. Gas detection system devices and equipment.
2. Engineering evaluation.
3. Items required by Section 907.1.1, as applicable.

918.3 Equipment. Gas detection system devices and equipment shall be designed for use with the particular gas being detected.

918.4 Reserved.

918.5 Emergency power. Gas detection systems shall be provided with emergency power in accordance with the *New York City Fire Code* and Chapter 27 of this code.

918.6 Sensor locations. Where leaking gases are expected to accumulate, sensors shall be installed in approved locations in accordance with the *New York City Fire Code* and NFPA 72.

918.7 Gas sampling. Gas detection systems shall perform continuous gas sampling. Sample analysis shall be processed immediately after sampling, except as follows:

1. For HPM gases, sample analysis shall be performed at intervals not exceeding 30 minutes.
2. For toxic gases, sample analysis shall be performed at intervals not exceeding 5 minutes in accordance with the Section 6004 of the *New York City Fire Code*.
3. Where a less frequent or delayed sampling interval is approved by the Fire Commissioner.

918.8 System activation. Gas detection systems shall activate and initiate an alarm system in compliance with the requirements of Sections 918.8.1 and 918.8.2.

918.8.1 Activation thresholds. A gas detection system alarm shall be initiated when a sensor detects a concentration of gas exceeding the following thresholds unless otherwise specified in this code or the *New York City Fire Code*:

1. For flammable gases, a gas concentration exceeding 25 percent of the lower flammability limit (LFL).
2. For nonflammable gases, a gas concentration exceeding one-half of the immediately dangerous to life and health (IDLH) concentration.

918.8.2 Alarm signals. The gas detection system shall initiate a local alarm and transmit a signal to a constantly attended control station when a short-term hazard condition is detected. The alarm shall be both visible and audible and shall provide warning both inside and outside the area where gas is detected. The audible alarm shall be distinct from all other alarms.

918.8.3 Shutoff of gas supply. The gas detection system shall automatically close the shutoff valve at the source of gas supply piping and tubing related to the system being monitored for whichever gas is detected.

Exception: Automatic shutdown is not required for reactors utilized for the production of highly toxic or toxic compressed gases where such reactors comply with all of the following:

1. Operated at pressures less than 15 pounds per square inch gauge (psig) (103.4 kPa).
2. Constantly attended.
3. Provided with readily accessible emergency shutoff valves.

918.8.4 Valve closure. The automatic closure of shutoff valves shall be in accordance with the following:

1. When the gas-detection sampling point initiating the gas detection system alarm is within a gas cabinet or exhausted enclosure, the shutoff valve in the gas cabinet or exhausted enclosure for the specific gas detected shall automatically close.
2. Where the gas-detection sampling point initiating the gas detection system alarm is within a gas room and compressed gas containers are not in gas cabinets or exhausted enclosures, the shutoff valves on all gas lines for the specific gas detected shall automatically close.
3. Where the gas-detection sampling point initiating the gas detection system alarm is within a piping distribution manifold enclosure, the shutoff valve for the compressed container of specific gas detected supplying the manifold shall automatically close.

   Exception: When the gas-detection sampling point initiating the gas detection system alarm is at a use location or within a gas valve enclosure of a branch line downstream of a piping distribution manifold, the shutoff valve in the gas valve enclosure for the branch line located in the piping distribution manifold enclosure shall automatically close.

918.9 Reserved.

918.10 Fire alarm system connections. Gas detection systems shall be connected to and monitored at the Fire Command Center or fire alarm control unit where required or provided. Where a Fire Command Center or fire alarm control unit is not required or provided, an approved gas detection system control panel or annunciator shall be installed at the building main front entrance. A gas detection system shall be monitored by a central supervising station. Gas detection supervisory and alarm signals shall be transmitted as separate and distinct signals to the central supervising station.